

AUTOMATED AND DISTRIBUTED STATISTICAL ANALYSIS OF ECONOMIC AGENT-BASED MODELS

Andrea Vandin



Sant'Anna

School of Advanced Studies – Pisa

Institute of Economics



Department
of Excellence
2018 - 2022

2023-2027!

EMbeDS

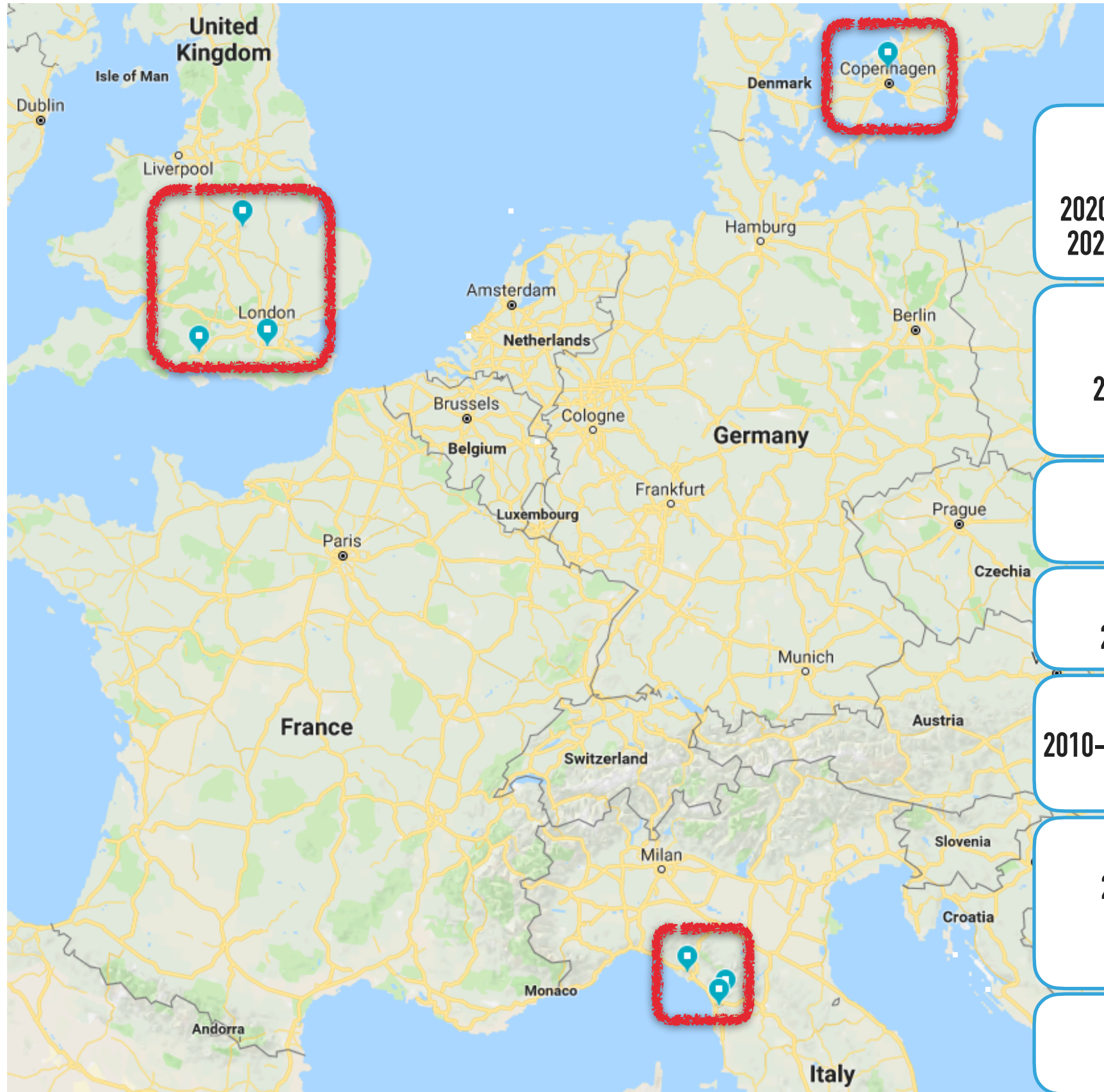
Economics and Management
in the era of Data Science



Danmarks
Tekniske
Universitet

1. About me & EMbeDS & Institute of Economics
2. Automated Statistical Analysis of Economic Agent-Based Models by Statistical Model Checking
3. PM meets SMC: Process-oriented Behavioral White-box Validation of ABMs

ABOUT ME



PISA

2023-??? Associate Professor @ Sant'Anna
2020-2022 Tenure-track Assistant Professor @ Sant'Anna
2020- NOW Adjunct : Associate Professor @ DTU

COPENHAGEN

2019 Associate Professor @ DTU
2017-2019 Tenure-track Assistant Professor @ DTU
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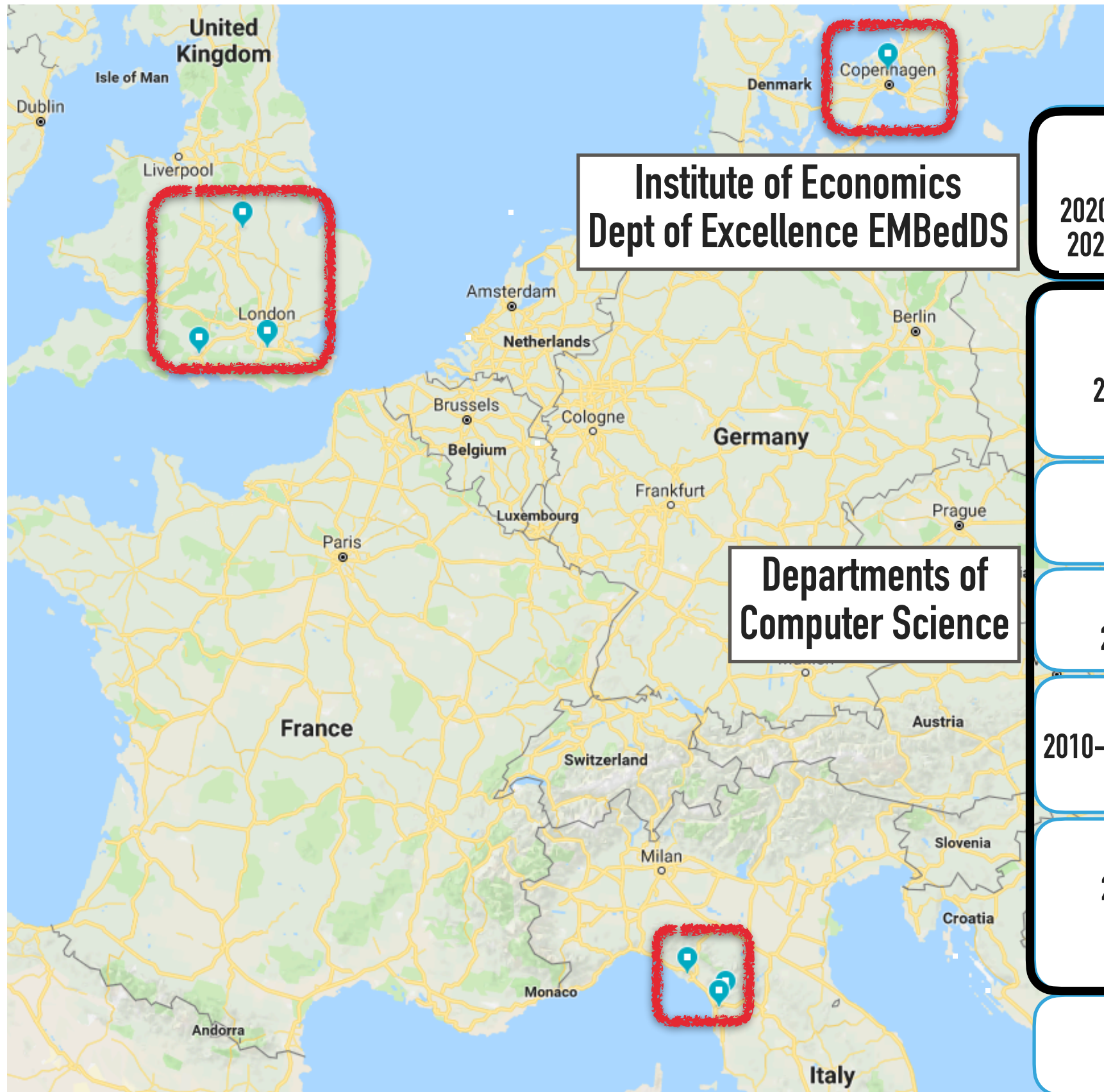
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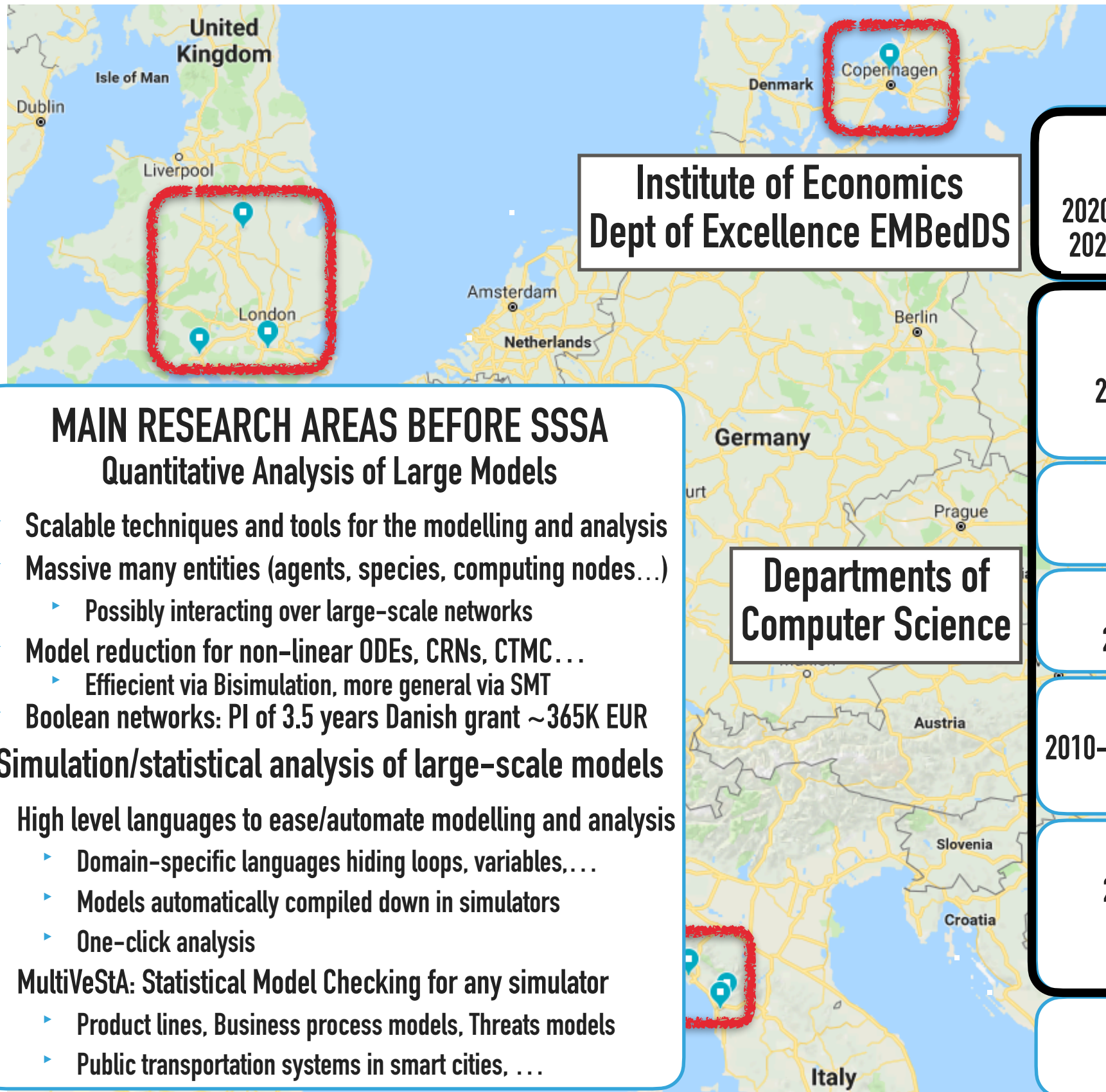
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MAIN RESEARCH AREAS BEFORE SSSA

Quantitative Analysis of Large Models

- ▶ Scalable techniques and tools for the modelling and analysis
- ▶ Massive many entities (agents, species, computing nodes...)
 - ▶ Possibly interacting over large-scale networks
- ▶ Model reduction for non-linear ODEs, CRNs, CTMC...
 - ▶ Efficient via Bisimulation, more general via SMT
- ▶ Boolean networks: PI of 3.5 years Danish grant ~365K EUR

Simulation/statistical analysis of large-scale models

- ▶ High level languages to ease/automate modelling and analysis
 - ▶ Domain-specific languages hiding loops, variables,...
 - ▶ Models automatically compiled down in simulators
 - ▶ One-click analysis
- ▶ MultiVeStA: Statistical Model Checking for any simulator
 - ▶ Product lines, Business process models, Threats models
 - ▶ Public transportation systems in smart cities, ...

**Institute of Economics
Dept of Excellence EMBedDS**

**Departments of
Computer Science**

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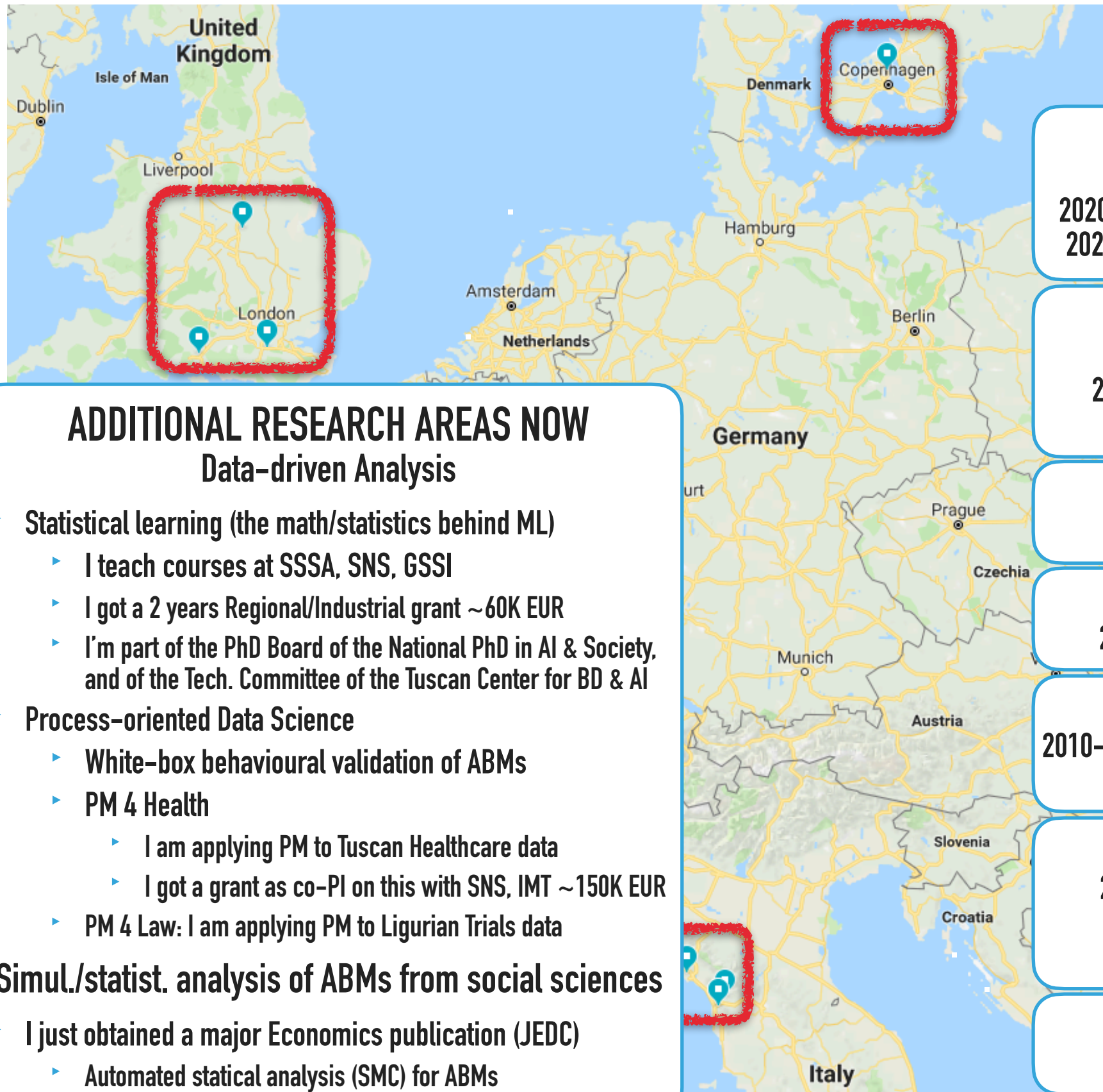
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ADDITIONAL RESEARCH AREAS NOW

Data-driven Analysis

- ▶ Statistical learning (the math/statistics behind ML)
 - ▶ I teach courses at SSSA, SNS, GSSI
 - ▶ I got a 2 years Regional/Industrial grant ~60K EUR
 - ▶ I'm part of the PhD Board of the National PhD in AI & Society, and of the Tech. Committee of the Tuscan Center for BD & AI
- ▶ Process-oriented Data Science
 - ▶ White-box behavioural validation of ABMs
 - ▶ PM 4 Health
 - ▶ I am applying PM to Tuscan Healthcare data
 - ▶ I got a grant as co-PI on this with SNS, IMT ~150K EUR
 - ▶ PM 4 Law: I am applying PM to Ligurian Trials data

Simul./statist. analysis of ABMs from social sciences

- ▶ I just obtained a major Economics publication (JEDC)
 - ▶ Automated statical analysis (SMC) for ABMs

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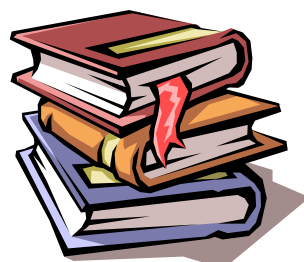
Large funding as 'Department of Excellence' 2018-2022

- Led by Prof. Francesca Chiaromonte
 - Statistician, also working at PennState University, US
- Involves a wide range of profiles: economics, management, law, statistics, computer science
- I am, so far, the only computer scientist, the first ever hired in the 'social sciences' part of Sant'Anna
 - First INF/01 ever at Sant'Anna

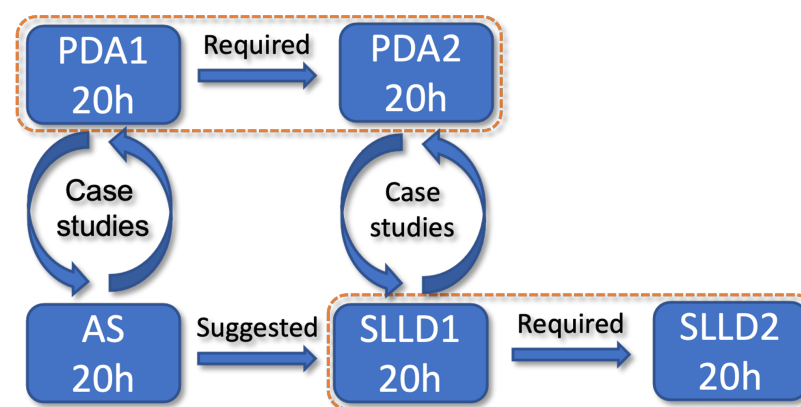
The mission of EMbeDS is to

- Foster data-driven statistical and computational approaches in Economics, Management, Law, and in the Social Sciences in general

Research



Teaching



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ABOUT INSTITUTE OF ECONOMICS

Research Areas



Complexity Economics >



The dynamics of industries and markets >



Innovation, technical change and economic history >



Data Science for economics and social science >

Data Science for economics and social science

Statistical learning; analysis and methods for big & complex data; data mining; patterns of causality in economic data; statistical model checking; calibration and validation of economic models

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AUTOMATED STATISTICAL ANALYSIS OF ECONOMIC AGENT-BASED MODELS BY STATISTICAL MODEL CHECKING

Joint work with

Daniele Giachini, Francesco Lamperti, Francesca Chiaromonte

Tool, models, and draft available at:

<https://bit.ly/MultiVeStATool>

Published at JEDC in 2022

Journal of Economic Dynamics and Control

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Journal of Economic Dynamics and Control

2013

■ [c8]    Stefano Sebastio, Andrea Vandin 

MultiVeStA: statistical model checking for discrete event simulators. VALUETOOLS 2013: 310-315

But also

Stefano Sebastio, INRIA

Stephen Gilmore, University of Edinburgh

Daniël Reijsbergen, Singapore University of Technology and Design

But also

Alberto Lluch Lafuente, DTU. Maurice ter Beek, Mieke Massink, Diego Latella, Vincenzo Ciancia, CNR. Axel Legay, ULouvain, Flavio Corradini, Fabrizio Fornari, Andrea Polini, Barbara Re, Francesco Tiezzi, Michele Loreti, University of Camerino. Rocco De Nicola, Mirco Tribatone IMT. Danilo Pianini, UniMore. Fabio Gadducci, Roberto Bruni, Andrea Corradini, University of Pisa. Martin Wissing, Lenz Benzner, LMU. Vashti Galpin University of Edinburgh.

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1. Motivation and context

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Examples of Considered models/simulations

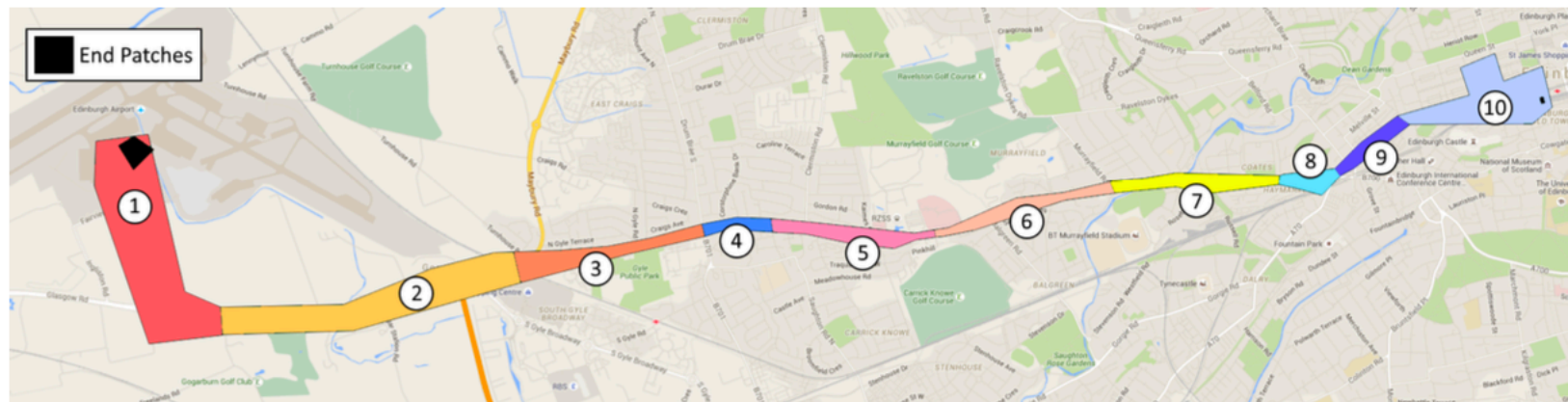
MultiVeStA has been successfully applied in a wide range of domains including, e.g.,

- economical agent-based models (see the replicability page in the right-menu),
- [engineering of highly-configurable systems](#),
- public transportation systems in smart cities [example 1](#), [example 2](#), [example 3](#)
- [biological systems](#)
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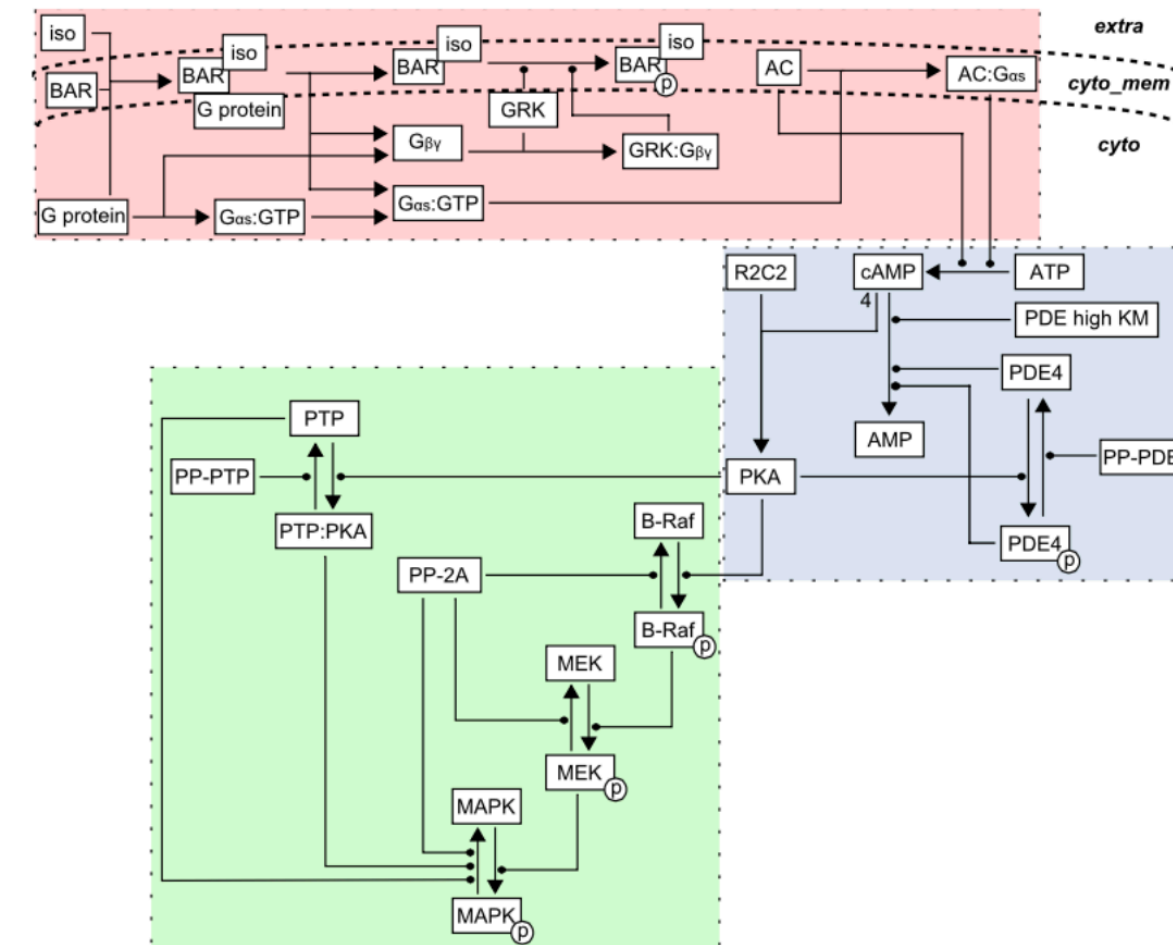


Fig. 1. Schematic representation of the cAMP/PKA/MAPK pathway (from [15]).

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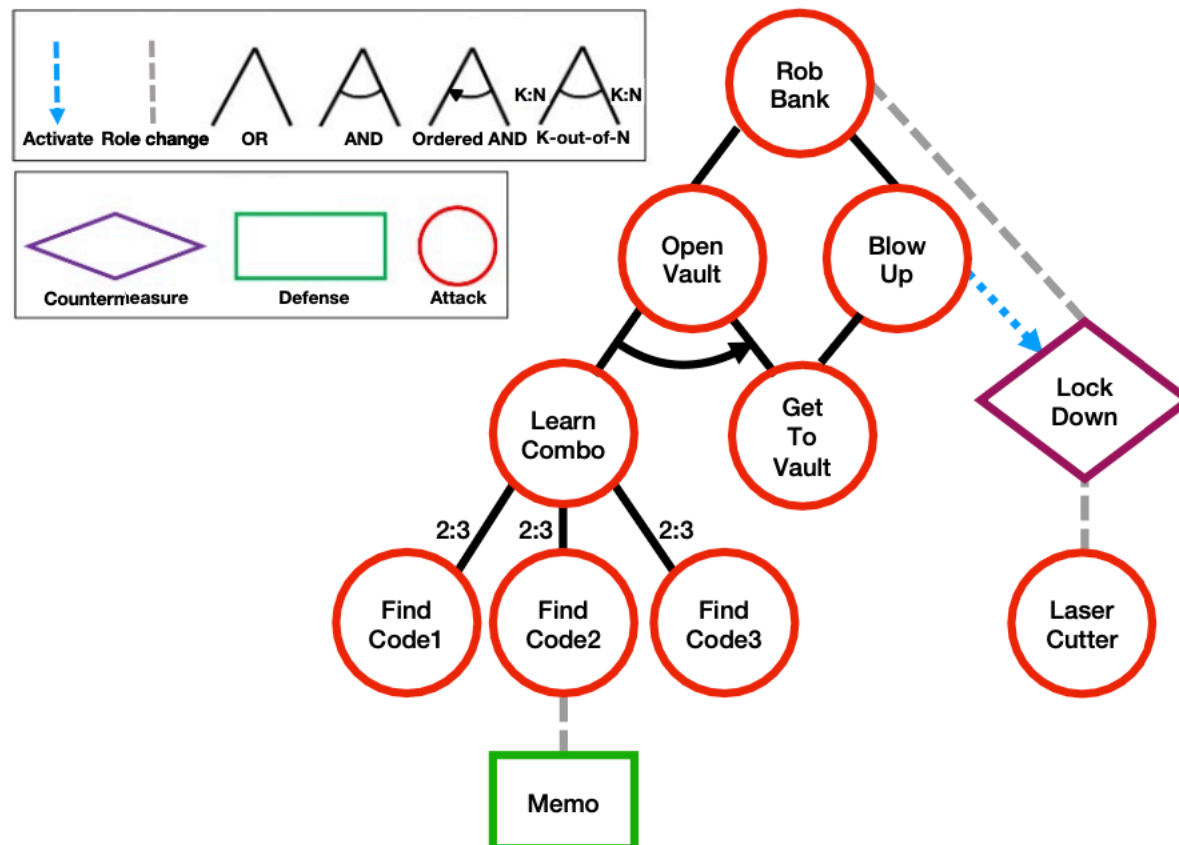
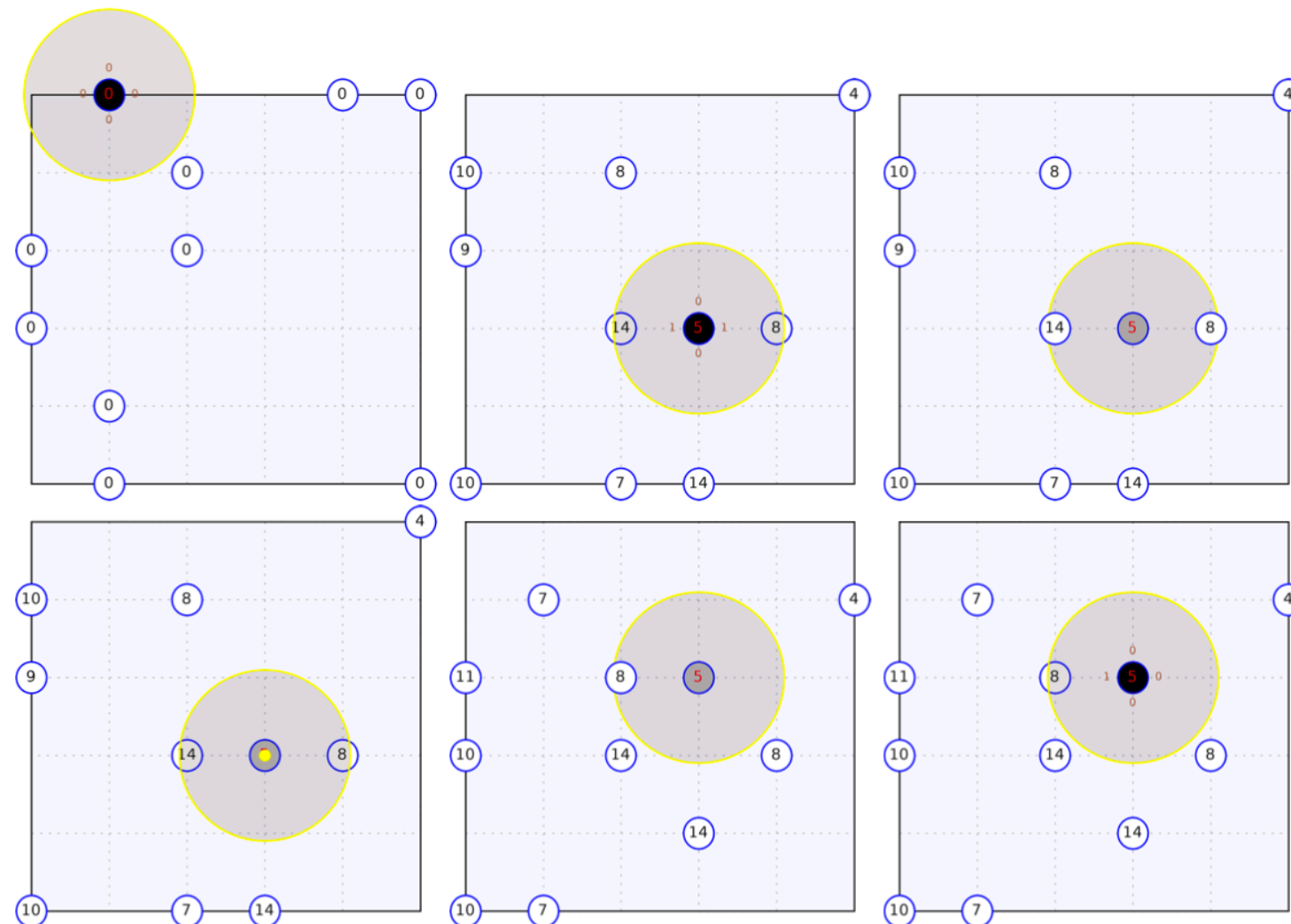


Fig. 1. An attack-defense diagram

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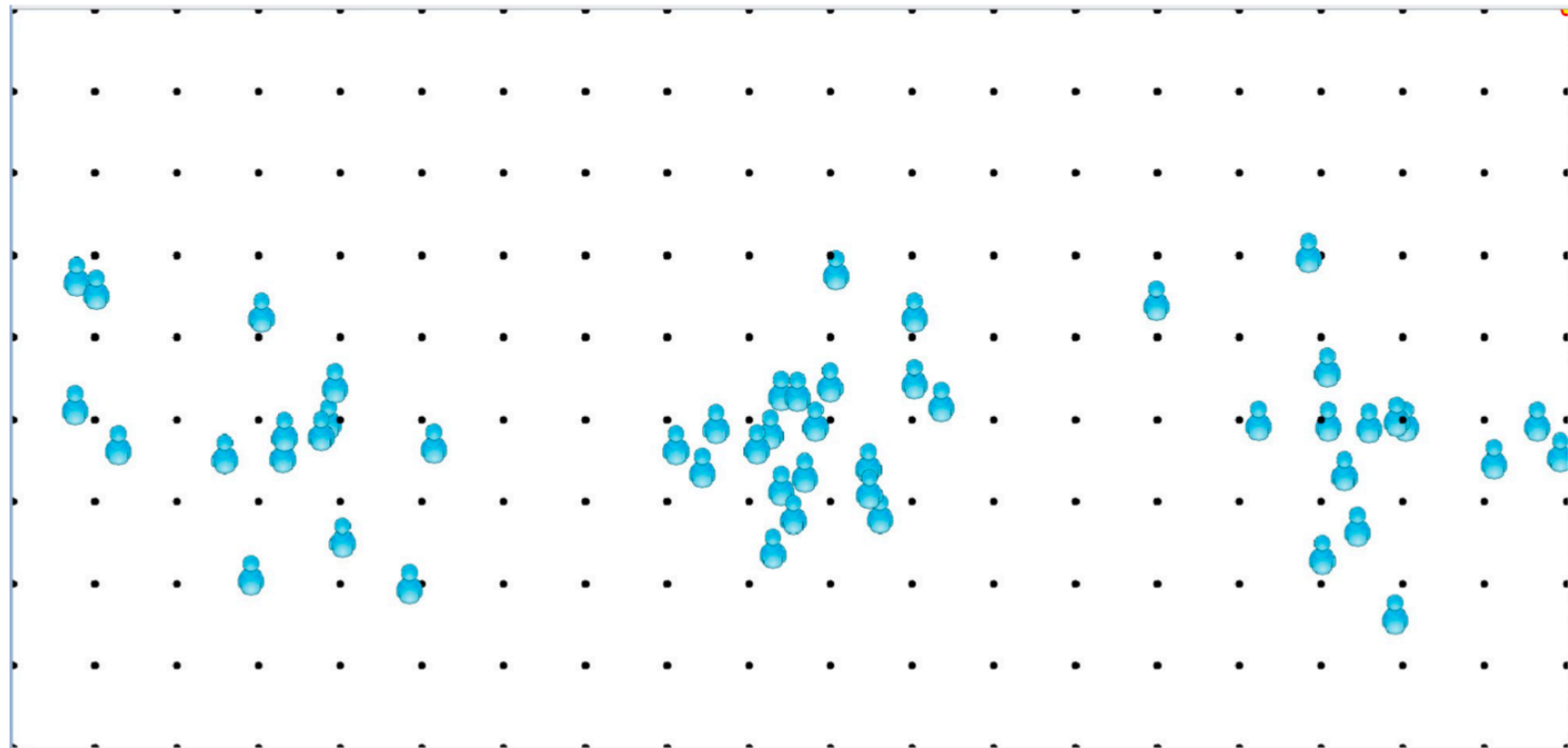
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What is an Economic Agent-Based Model? My personal view

NON ABMs

- 'Mainstream economists' tend to reason in terms of models that
 - Are given as a unique monolithic model
 - Do not have focus on their single components, but on the overall dynamics of the model
 - *What* the system does, rather than *how* the system does
 - Have explicit representations of the laws governing the economic system
 - **Can typically be analysed analytically**

ABMs

- Some economists are interested in modeling an economic system in terms of its components
 - The **agents** that operate in it: firms, households, banks...
- The modeller does not specify explicitly the laws governing the model.
- It describes explicitly
 - The behaviour of every agent
 - The interactions among the agents
- **The laws governing the model then *emerge* from these behaviours and interactions**
- These types of models are named ABMs.
 - These are typically too difficult to be solved analytically
 - We need to do simulations
 - My message: **we need to do simulations well!**
 - A variant of model checking, statistical model checking, can help on this

What is SMC?

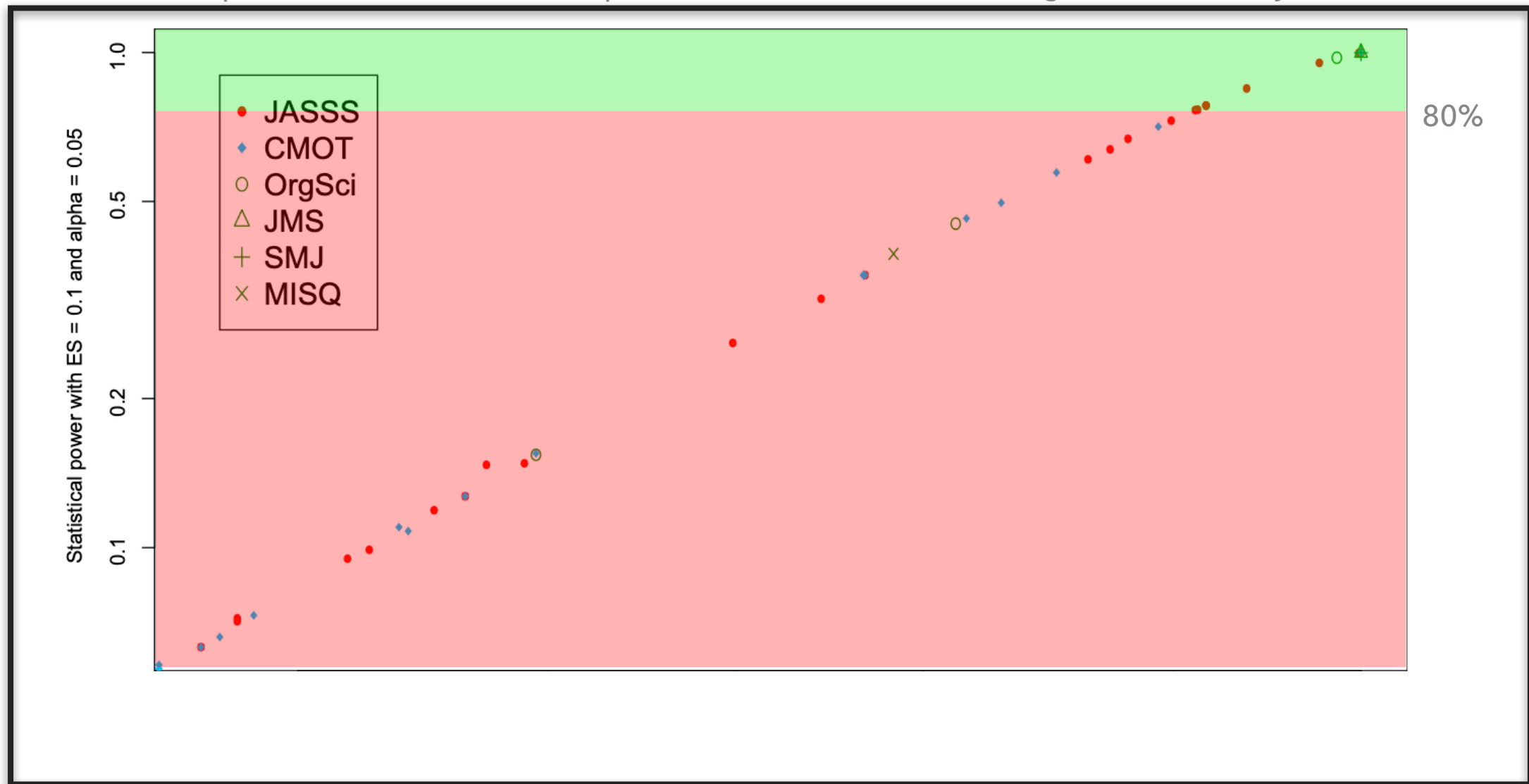
The Model Checking problems

Techniques well-known within the Theoretical Computer Science (Formal Methods) community

- ▶ **Model Checking (MC):**
 - ▶ To decide whether a **non-deterministic** model satisfies a temporal logic property
- ▶ **Probabilistic MC (PMC):**
 - ▶ To decide whether a **stochastic** model satisfies a temporal logic property with a probability greater than a certain threshold
- ▶ **Statistical MC (SMC):**
 - ▶ Simulation-based technique to **statistically approximate the PMC problem**
 - ▶ Only requires independent and identically distributed samplings (simulations)
 - ▶ Highly parallelizable
 - ▶ Many tools supporting it. E.g.
 - ▶ MultiVeStA, PRISM, UPPAAL, APMC, COSMOS, YMER, SAM, BIP,(P)VeStA...
 - ▶ Two main approaches: Probability estimation VS Hypothesis testing
 - ▶ Probability estimation → Real-valued property estimation

'Quality' of Statistical Analysis on 55 ABM from Management & Organisational Research

Adapted from Secchi, Seri, Computational and Mathematical Organization Theory, 2017



- ▶ The importance of designing well simulation-based analysis.
 - ▶ Power analysis on 'are the expected outcomes of different configurations of parameters the same'?
- ▶ **Power** is $1 - P(\text{Type II error})$
 - ▶ Roughly, **P(test='outcomes are different' | outcomes are different)**
 - ▶ "The value that seems to be more commonly accepted is 80%"
- ▶ "We need to encourage researchers to be more precise in the determination of the number of runs"

A systematic review of statistical power in software engineering experiments

Tore Dybå^{a,b,*}, Vigdis By Kampenes^a, Dag I.K. Sjøberg^a

^a Simula Research Laboratory, P.O. Box 134, NO-1325 Lysaker, Norway

^b SINTEF ICT, NO-7465 Trondheim, Norway

Received 11 May 2005; revised 24 August 2005; accepted 31 August 2005

Available online 3 November 2005

Abstract

Statistical power is an inherent part of empirical studies that employ significance testing and is essential for the planning of studies, for the interpretation of study results, and for the validity of study conclusions. This paper reports a quantitative assessment of the statistical power of empirical software engineering research based on the 103 papers on controlled experiments (of a total of 5,453 papers) published in nine major software engineering journals and three conference proceedings in the decade 1993–2002. The results show that the statistical power of software engineering experiments falls substantially below accepted norms as well as the levels found in the related discipline of information systems research. Given this study's findings, additional attention must be directed to the adequacy of sample sizes and research designs to ensure acceptable levels of statistical power. Furthermore, the current reporting of significance tests should be enhanced by also reporting effect sizes and confidence intervals.

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1. Motivation and context

2. Transient analysis

3. PM meets SMC: Process-oriented Behavioral White-box Validation of ABMs

Transient Analysis by autoIR: Large Macro ABM

Large-scale macro financial ABM from Caiani et al, JEDC, 2016

- ▶ An economy with households, consumption/capital firms, commercial banks, government, central bank
- ▶ Thousands of agents
- ▶ Implemented in JMAB: Java framework for macro stock-flow consistent ABM models.
 - ▶ Side product: any model implemented in JMAB is now natively integrated with MultiVeStA

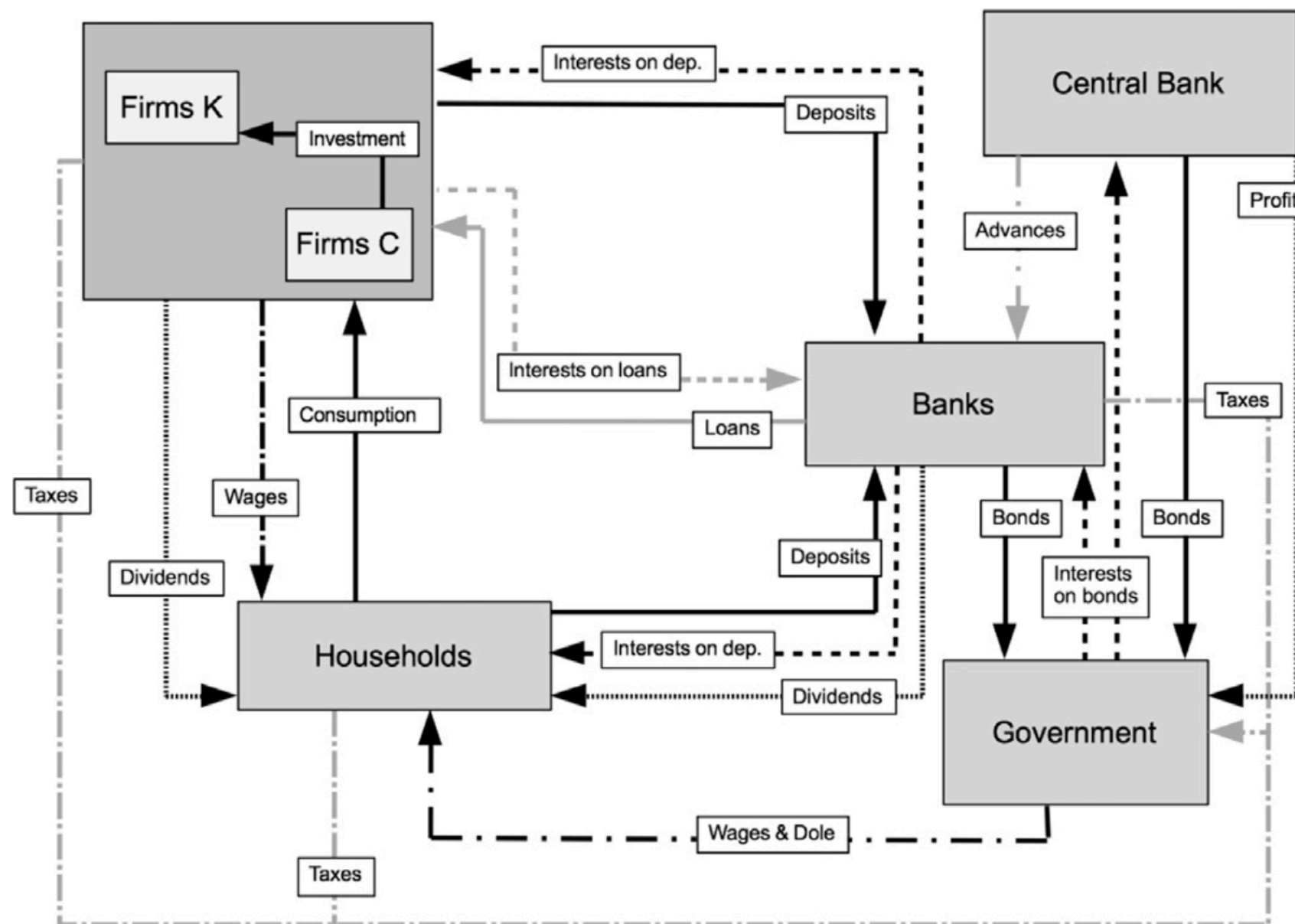


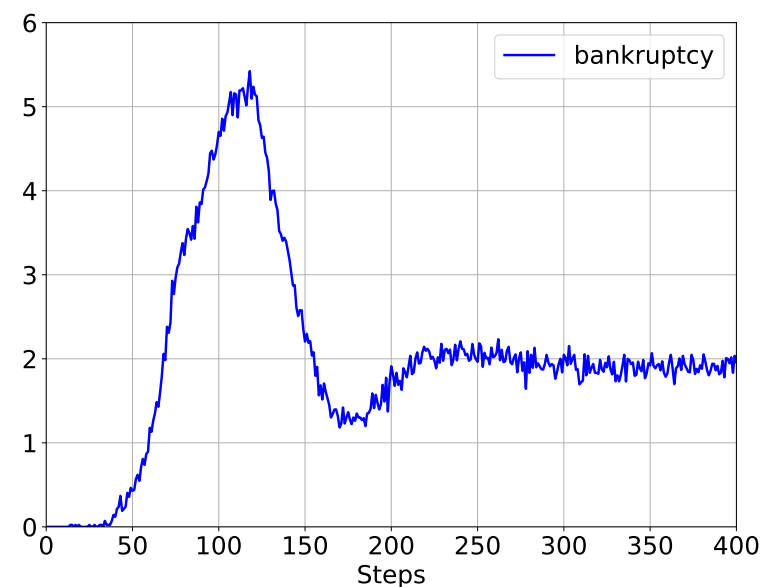
Fig. 1. Flow diagram of the model. Arrows point from paying sectors to receiving sectors.

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Evolution of bankruptcies

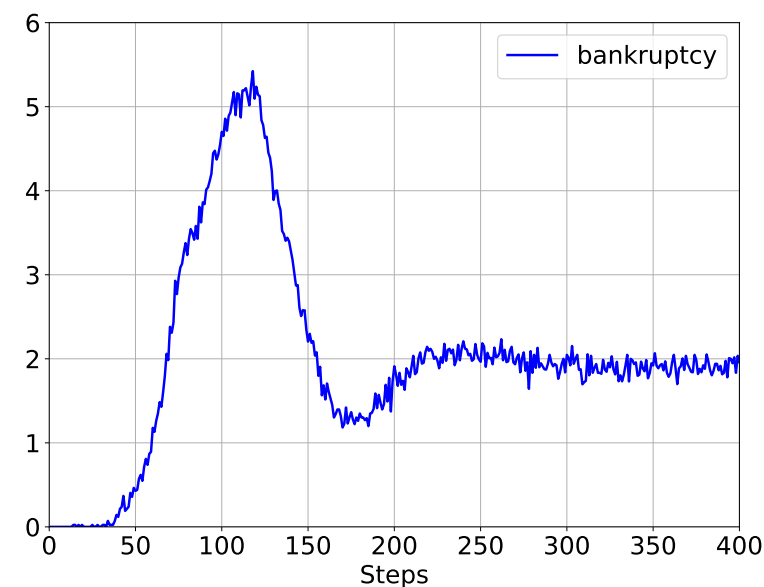


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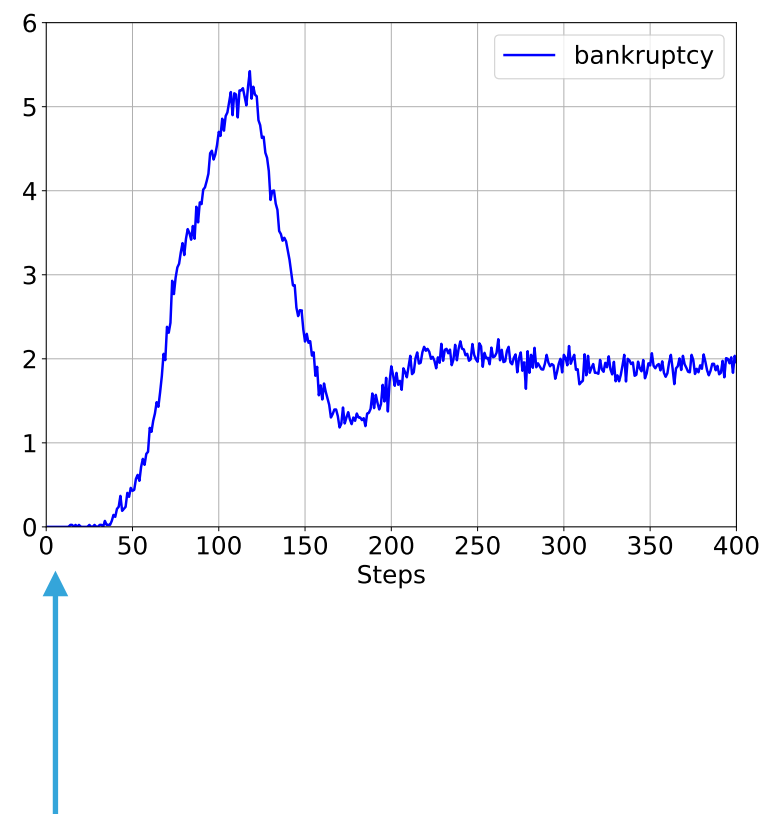
y_1

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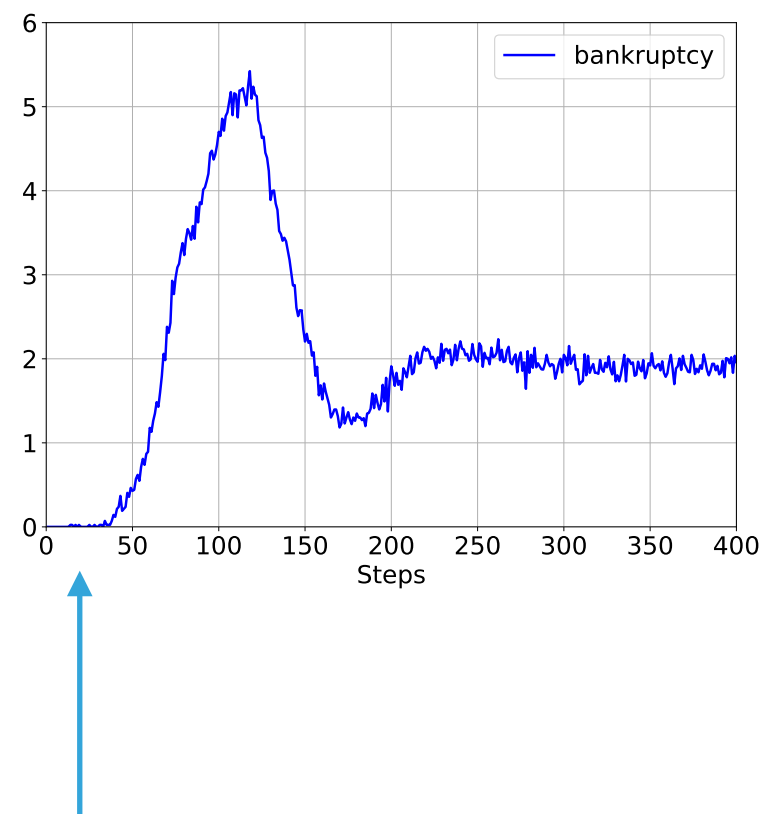
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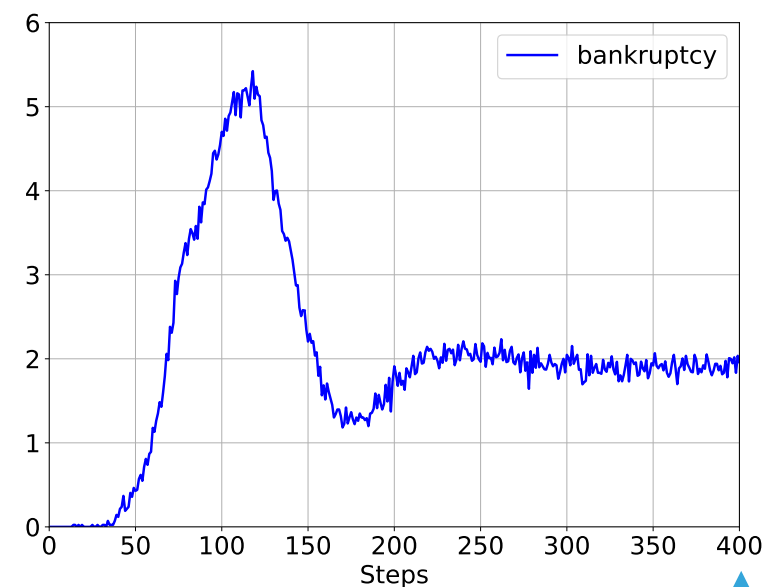
y_1 y_2

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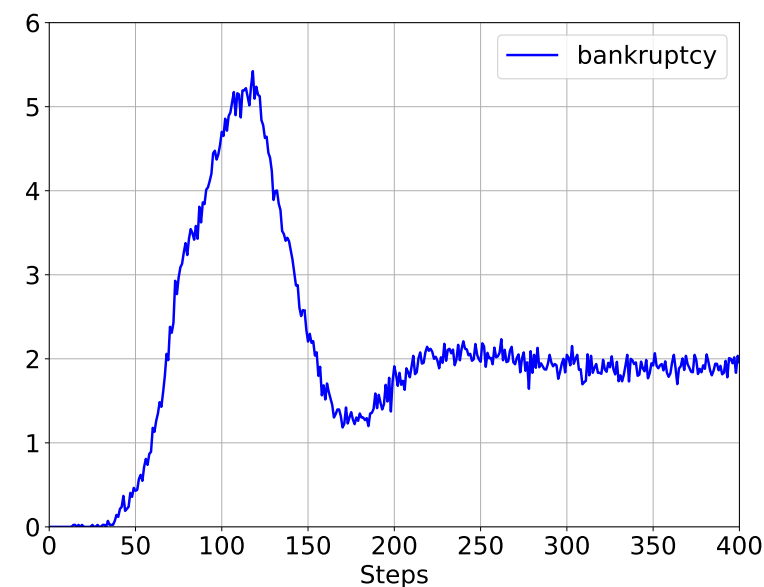
$y_1 \quad y_2 \quad \dots \quad y_{400}$

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$y_{1,1}$ $y_{1,2}$... $y_{1,400}$

$y_{2,1}$ $y_{2,2}$... $y_{2,400}$

$\vdots$ $\vdots$ $\vdots$ $\vdots$

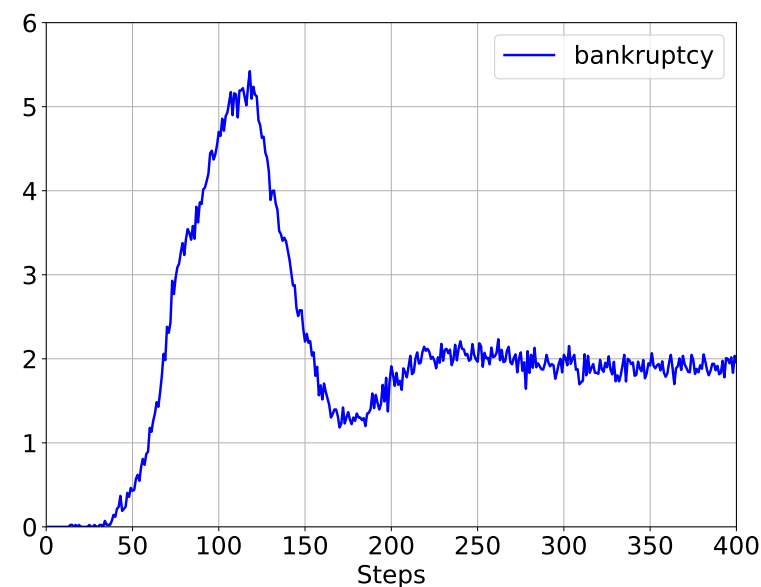
$y_{n,1}$ $y_{n,2}$... $y_{n,400}$

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$y_{2,1}$	$y_{2,2}$	$\dots$	$y_{2,400}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$y_{n,1}$	$y_{n,2}$	$\dots$	$y_{n,400}$

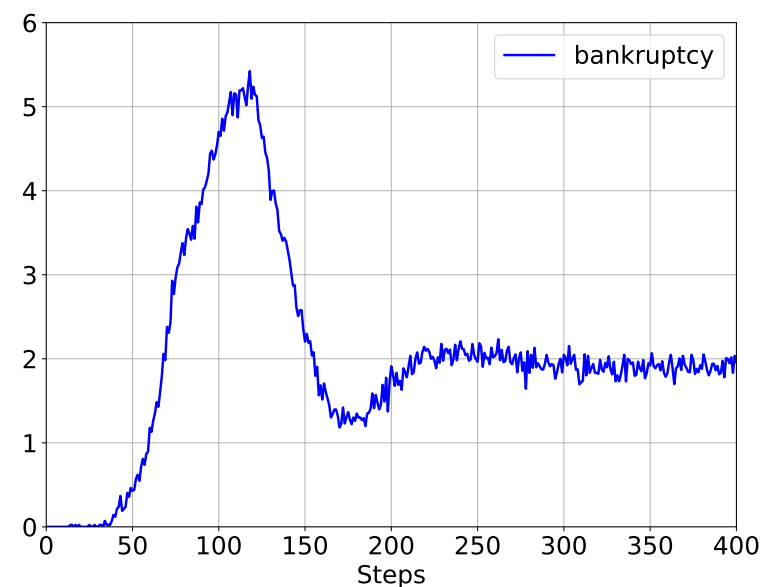
$$\sum_{i=1}^n \frac{y_{i,2}}{n} = \bar{Y}_2 \approx E[Y_2]$$

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$\vdots$	$\vdots$	$\vdots$	$\vdots$
$y_{n,1}$	$y_{n,2}$	$\dots$	$y_{n,400}$

$$\sum_{i=1}^n \frac{y_{i,2}}{n} = \bar{Y}_2 \approx E[Y_2]$$

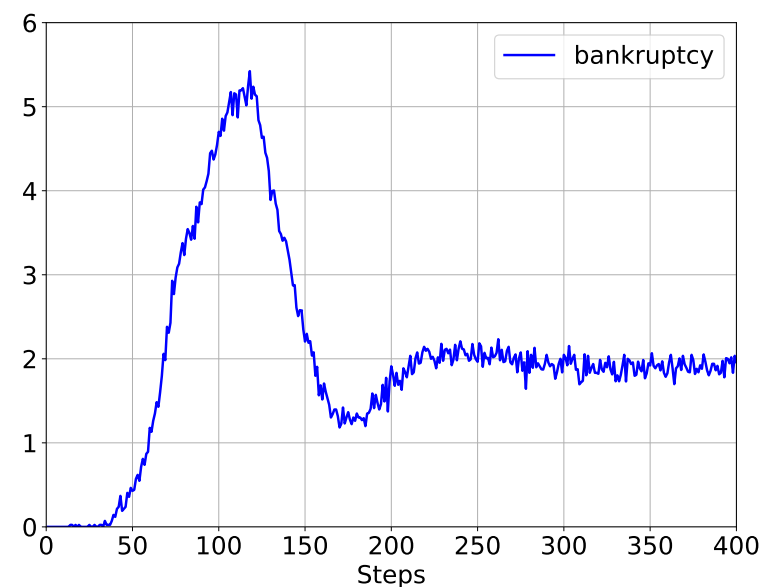
What is the correct value of n?

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$\vdots$	$\vdots$	$\vdots$	$\vdots$
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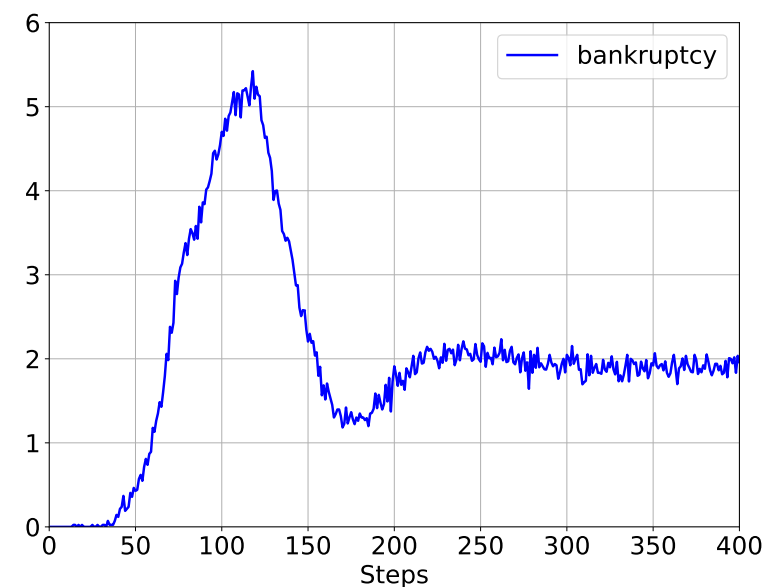
What is the correct value of n?
Typical answer: 100

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$y_{n,1}$	$y_{n,2}$	$\dots$	$y_{n,400}$

$$\sum_{i=1}^n \frac{y_{i,2}}{n} = \bar{Y}_2 \approx E[Y_2]$$

Our answer:

What is the correct value of n?
Typical answer: 100

The question itself is ill-posed
Each property and time step might require a different n

Transient Analysis by autoIR: Large Macro ABM

Large-scale macro financial ABM from Caiani et al, JEDC, 2016

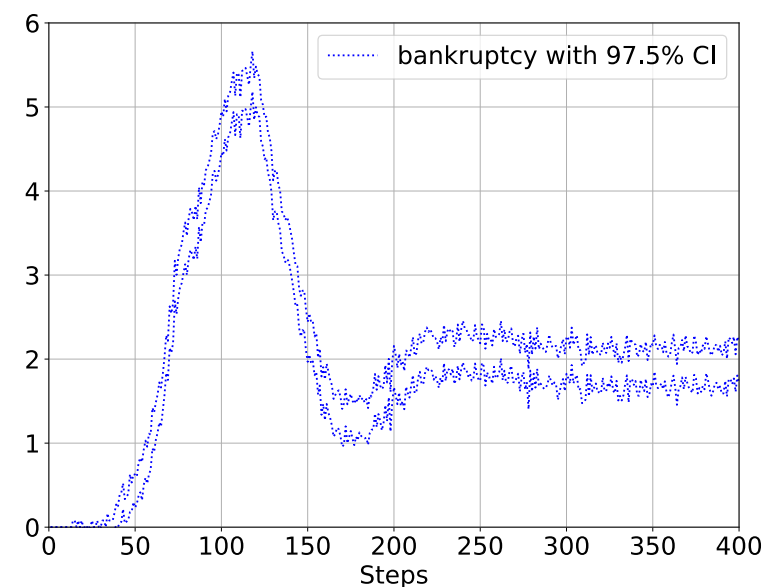
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$$\bar{Y}_t \pm t_{n-1, 1-\frac{\alpha}{2}} \cdot \sqrt{\frac{s_t^2}{n}}$$

$y_{1,1}$	$y_{1,2}$	...	$y_{1,400}$
$y_{2,1}$	$y_{2,2}$	...	$y_{2,400}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$y_{n,1}$	$y_{n,2}$	...	$y_{n,400}$

$$\sum_{i=1}^n \frac{y_{i,2}}{n} = \bar{Y}_2 \approx E[Y_2]$$

Evolution of bankruptcies



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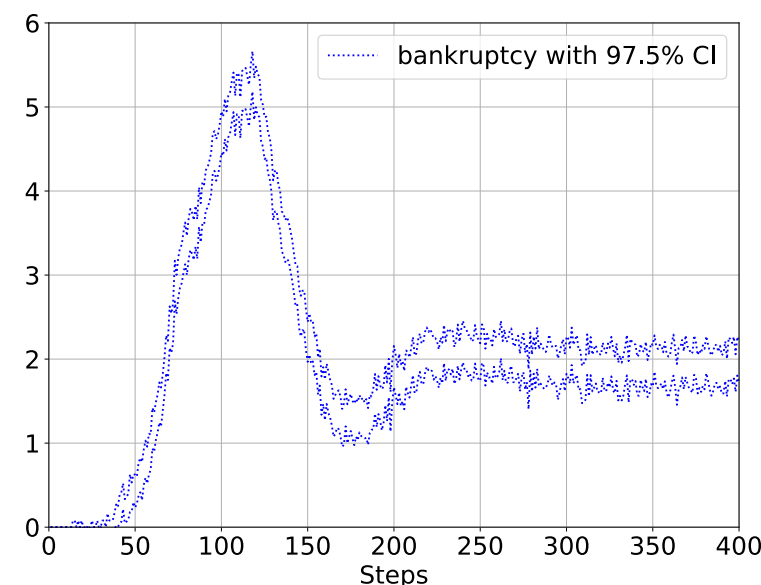
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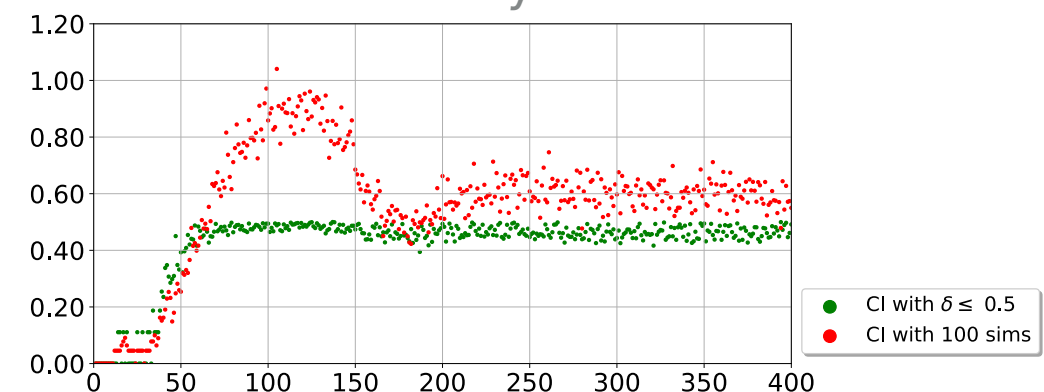
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Confidence Intervals width MultiVeStA VS by 100 sims



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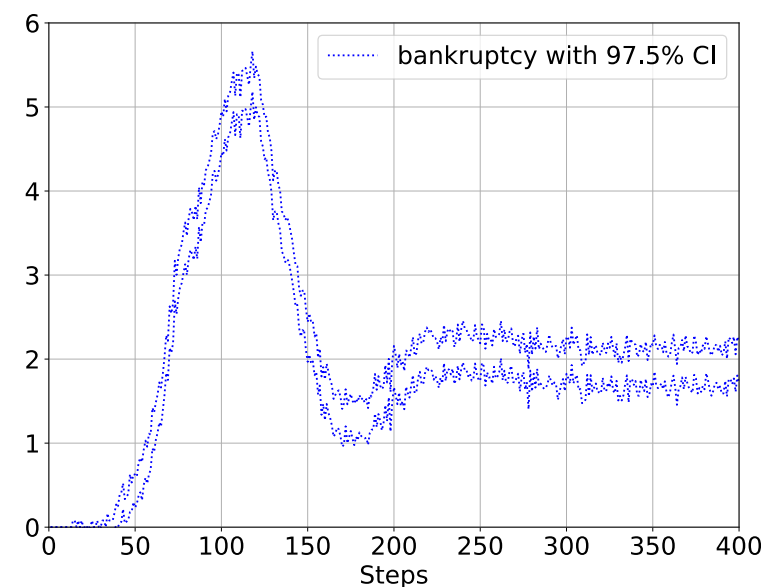
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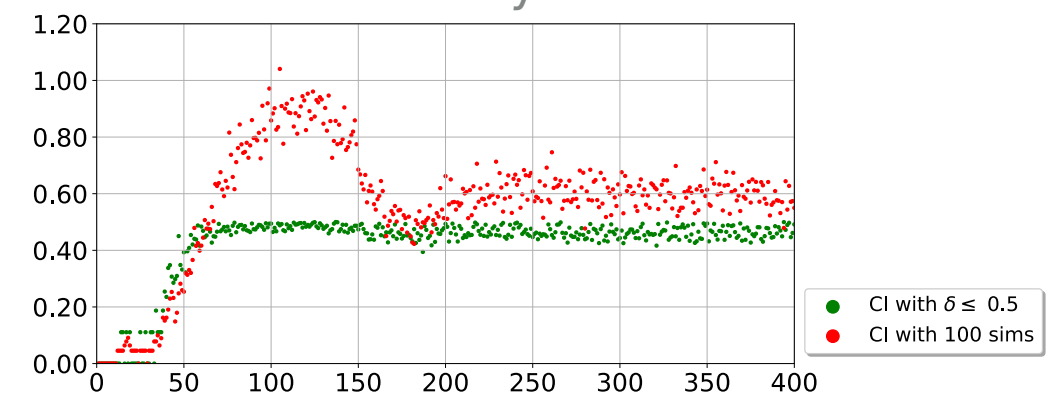
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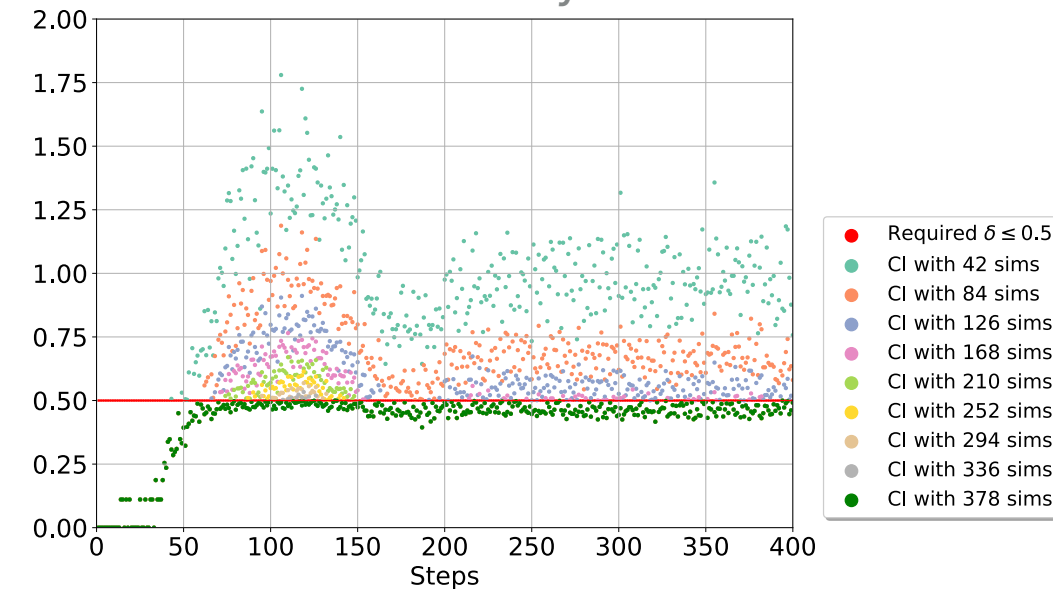
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Intermediate CIs width by MultiVeStA



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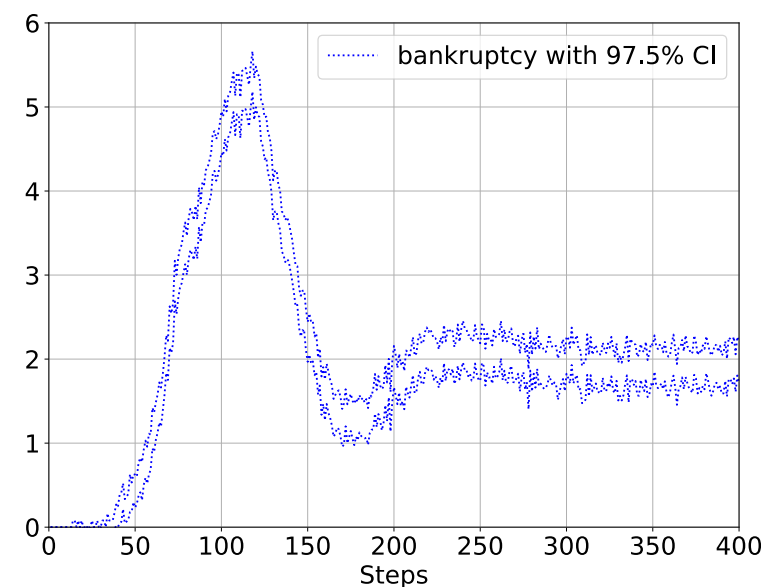
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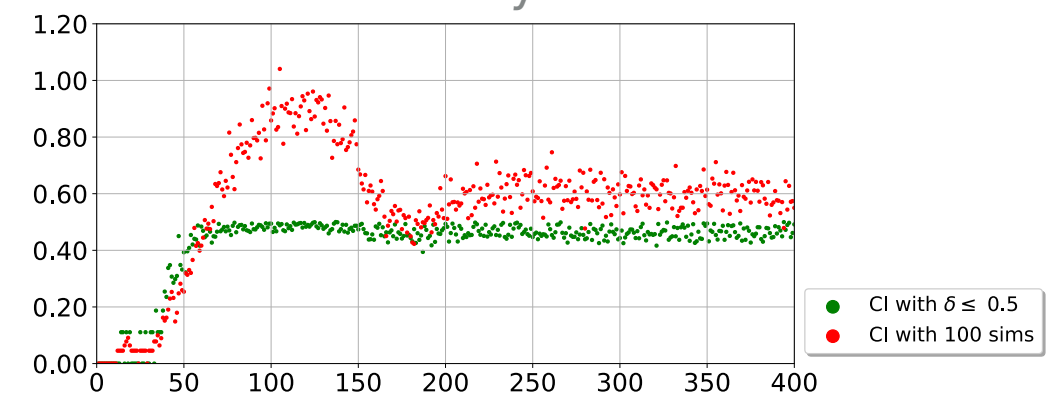
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Evolution of bankruptcies

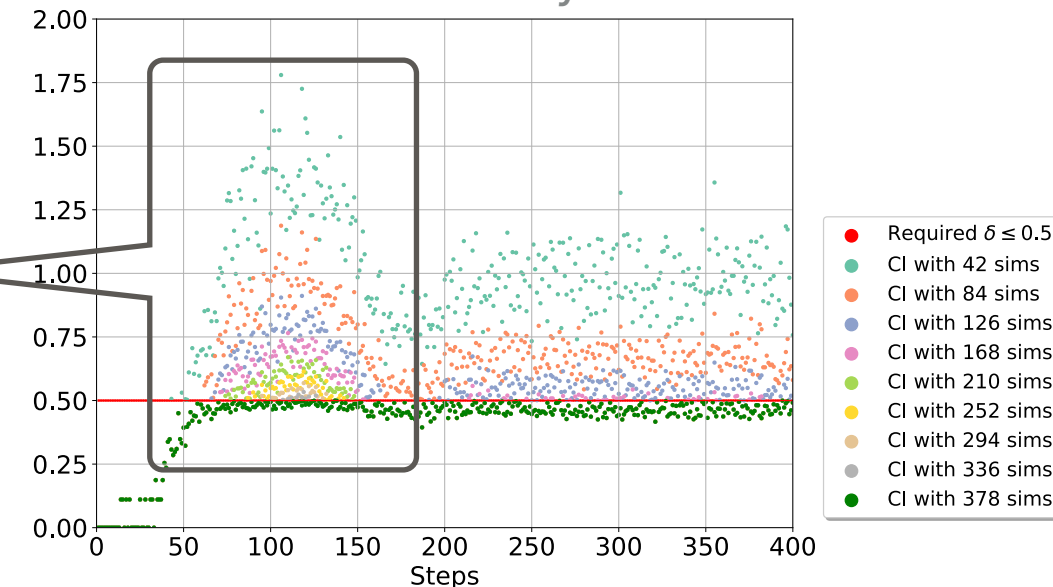


More simulations required here
We save a lot of time halting the last simulations to 150 steps

Confidence Intervals width MultiVeStA VS by 100 sims



Intermediate CIs width by MultiVeStA



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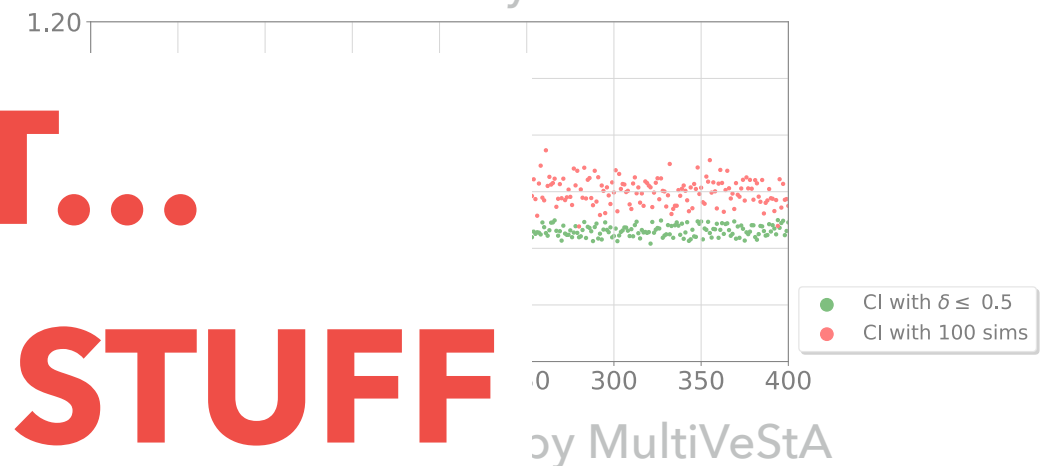
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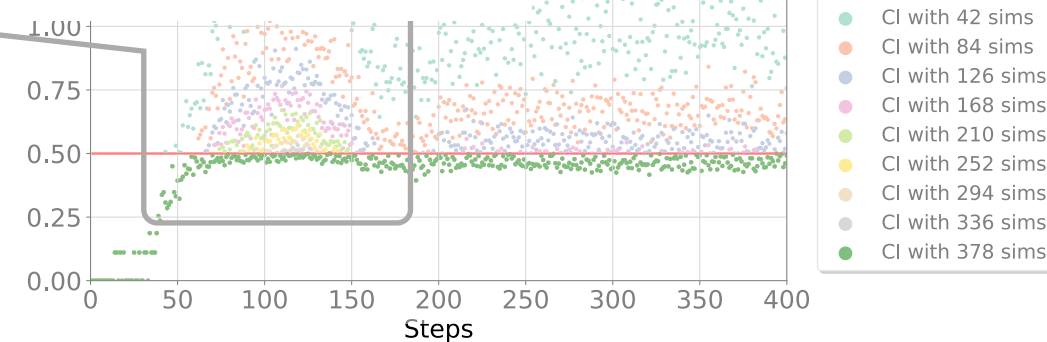


Confidence Intervals width
MultiVeStA VS by 100 sims



OK, BUT...
IS ALL THIS STUFF
IMPORTANT ???

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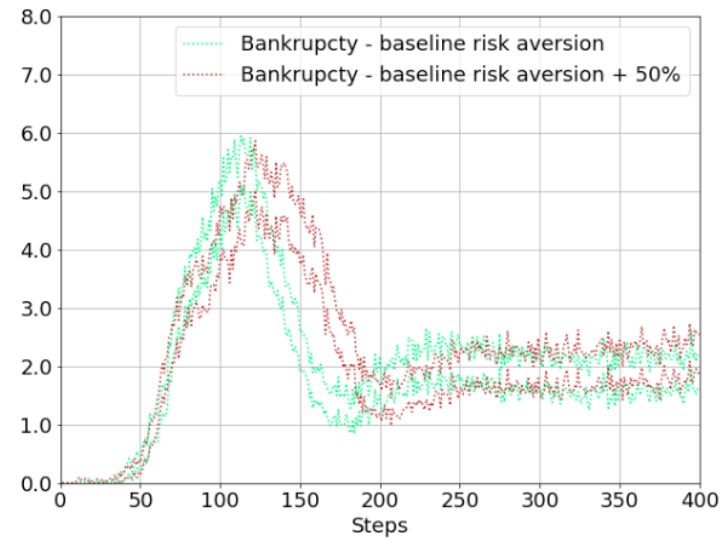
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Statistically Meaningful Counterfactual Analysis

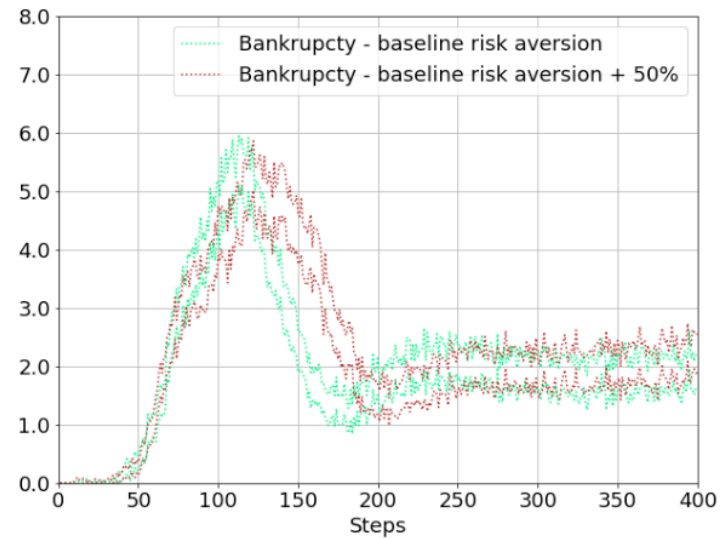
97.5% CI
100 Simulations



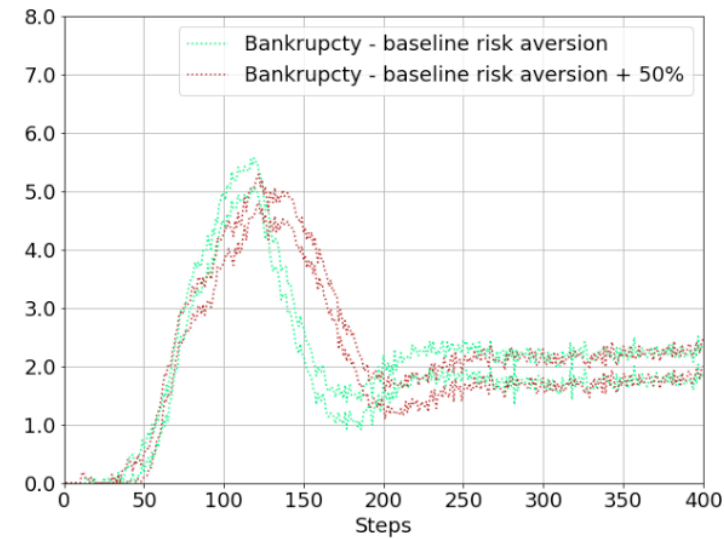
(a) CIs width for $\alpha = 0.025$ and $N = 100$ simulations

Statistically Meaningful Counterfactual Analysis

97.5% CI
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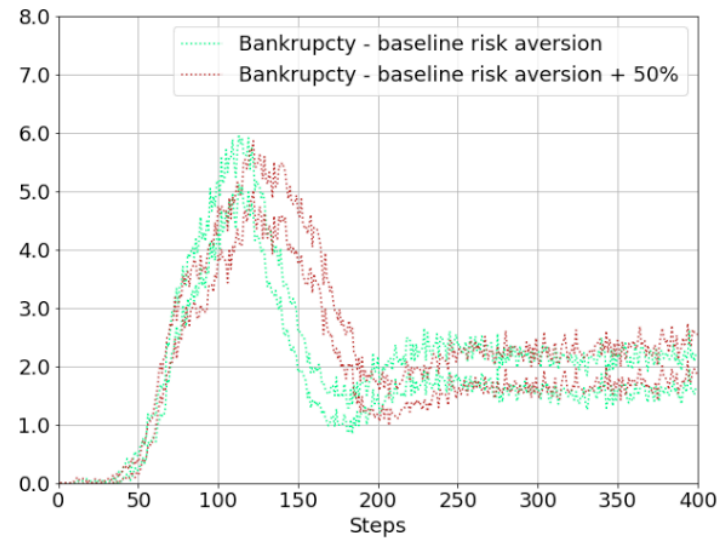


(b) CIs width for $\alpha = 0.025$ and $\delta = 0.5$

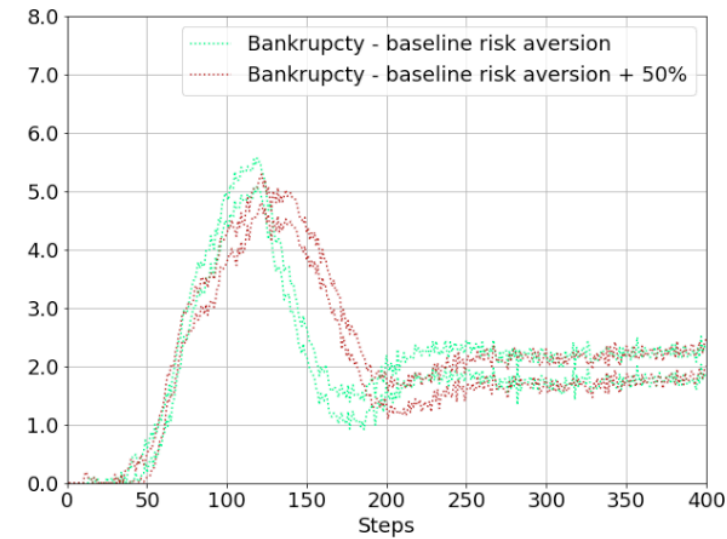
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MultiVeStA
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Statistically Meaningful Counterfactual Analysis

97.5% CI
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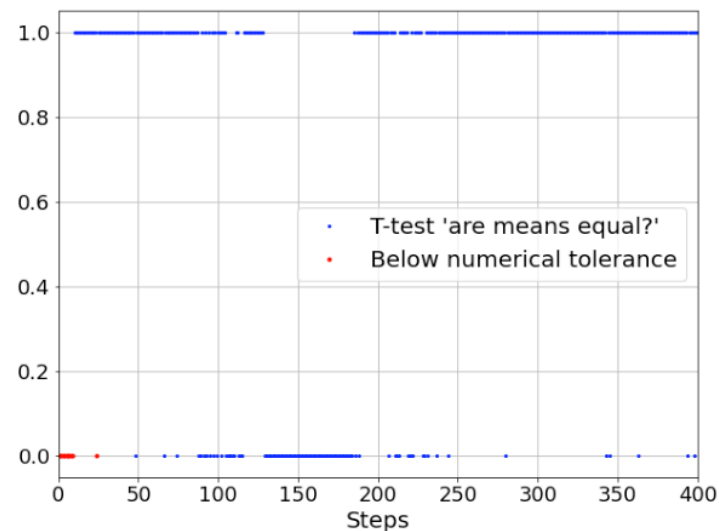
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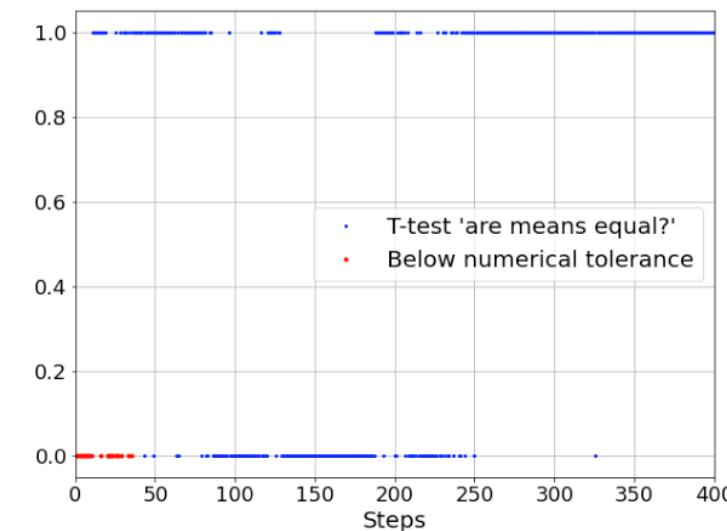
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97.5% CI
MultiVeStA
 $\delta = 0.5$

Welch's t-test with
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[Welch1947]



(c) T-test are means in (a) point-wise equal for significance $\alpha = 0.025$

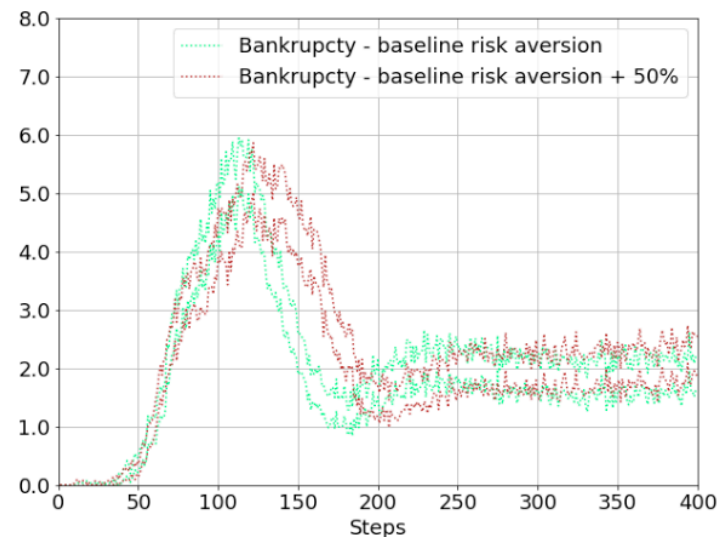


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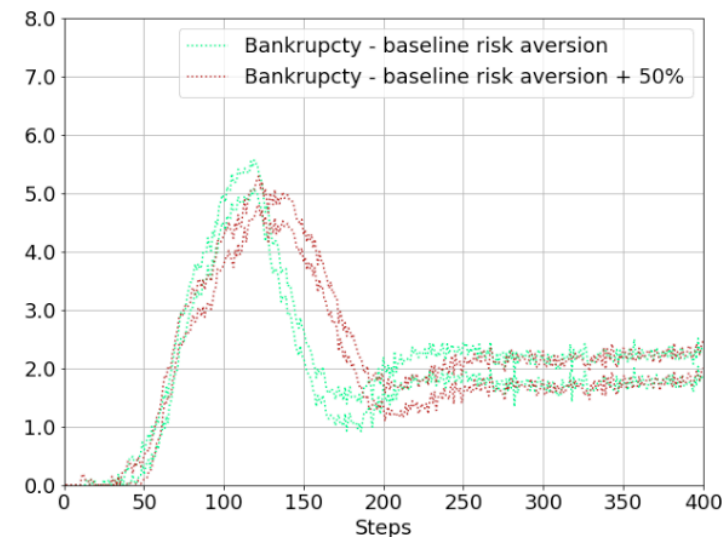
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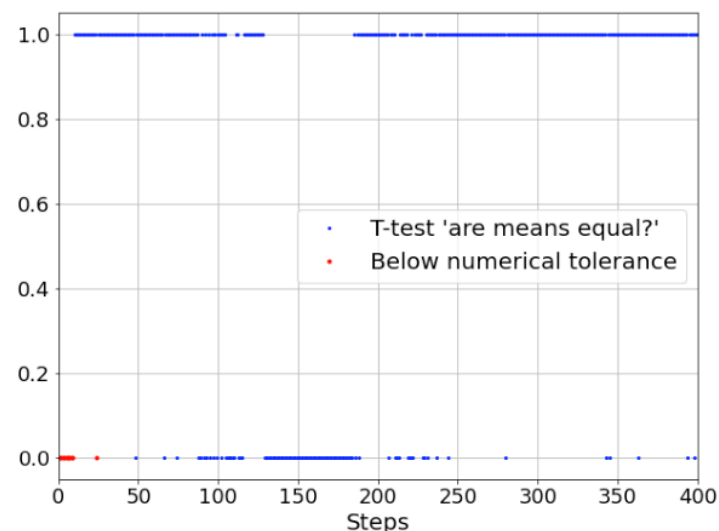
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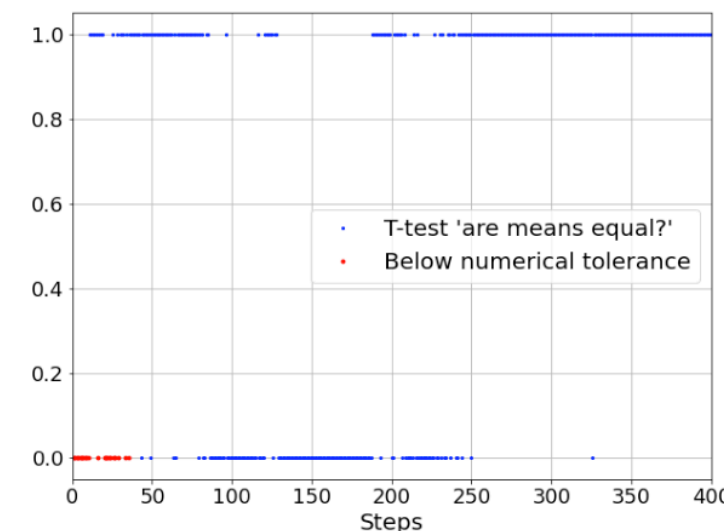
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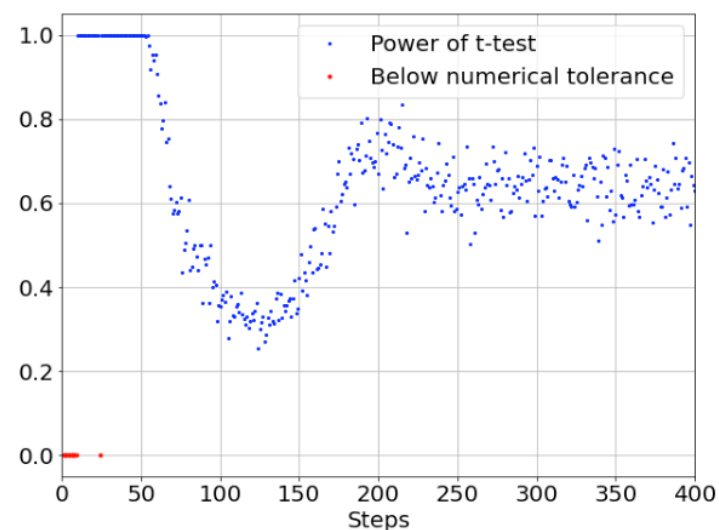
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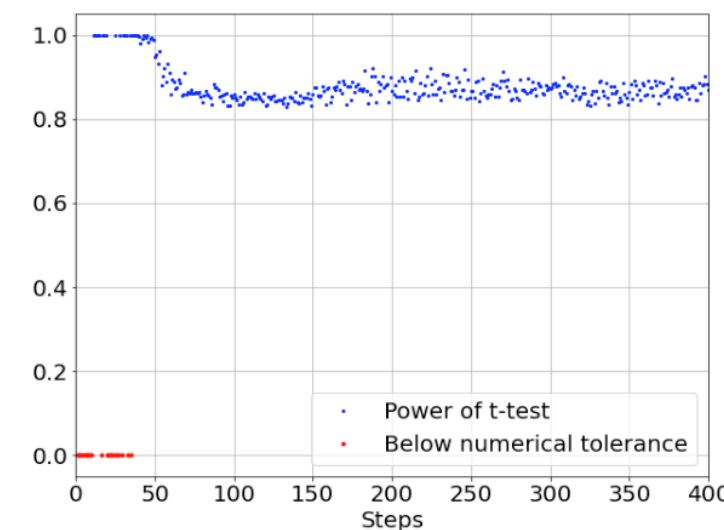
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Power of the test
 $P(\text{Test}=0 \mid \text{Real}=0)$
 $1 - P(\text{Type II error})$
[Chow2002]



(e) Power of t-test in (c) for difference $\varepsilon = 0.5$

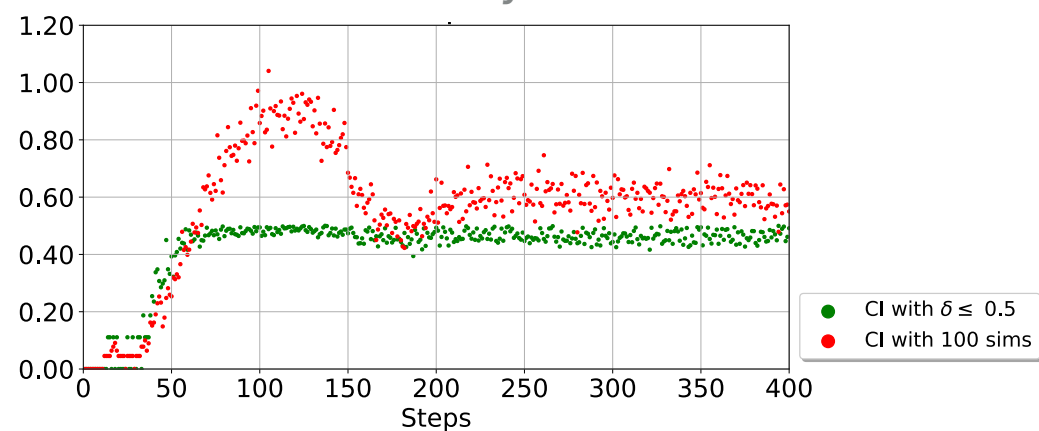


(f) Power of t-test in (d) for difference $\varepsilon = 0.5$

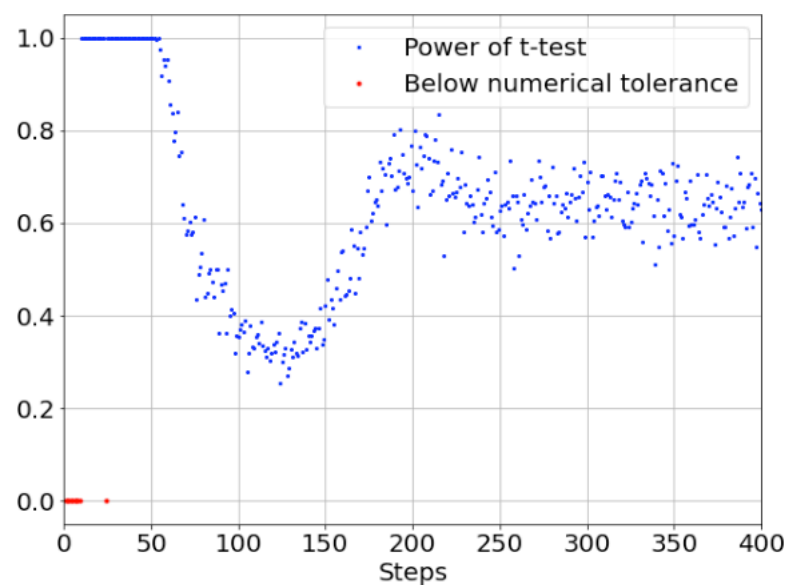
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Width of the Confidence Intervals and T-Test Power

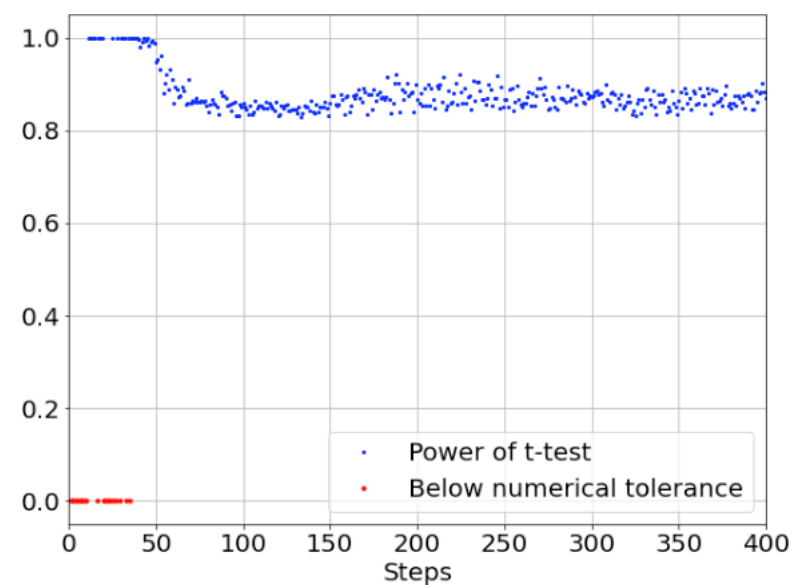
Confidence Intervals width
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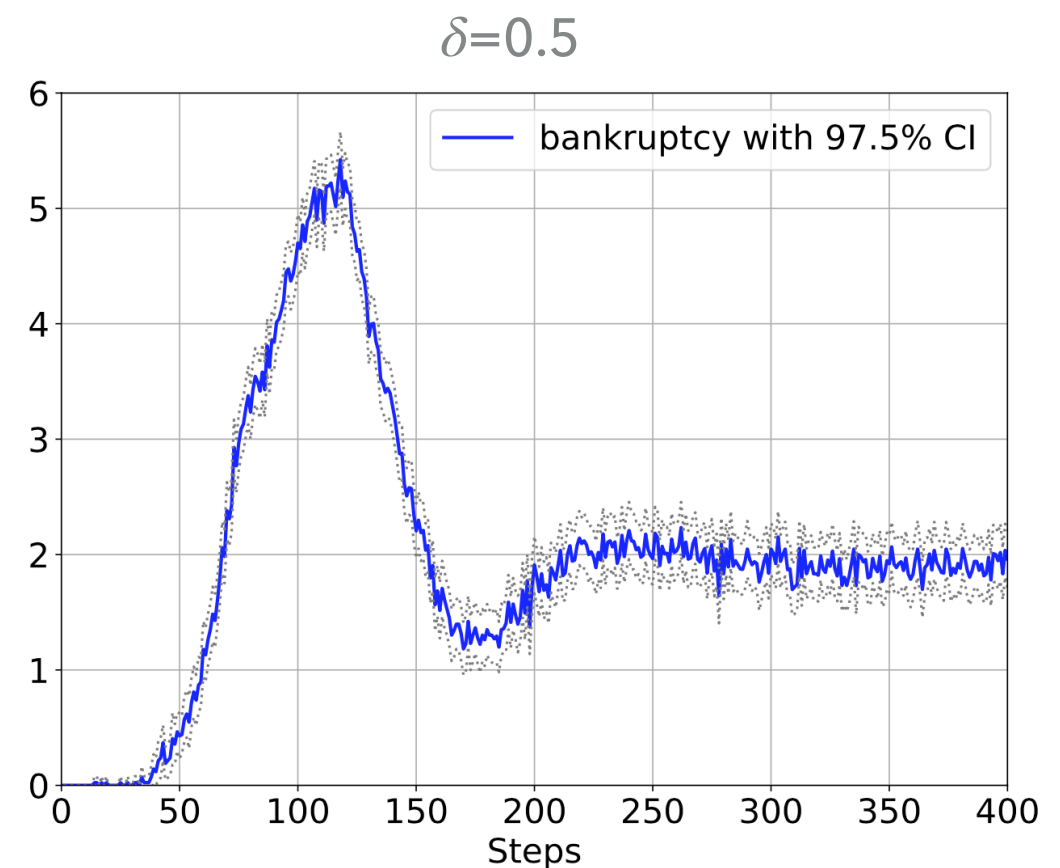
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A query to study the evolution of bankruptcies

```
obsAtStep(t,obs) = if (s.eval("steps") == t)
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                    else next(obsAtStep(t,obs))
                    fi ;
eval autoIR(E[ obsAtStep(t,"bankruptcy") ], ,t,1,1,400) ;
```



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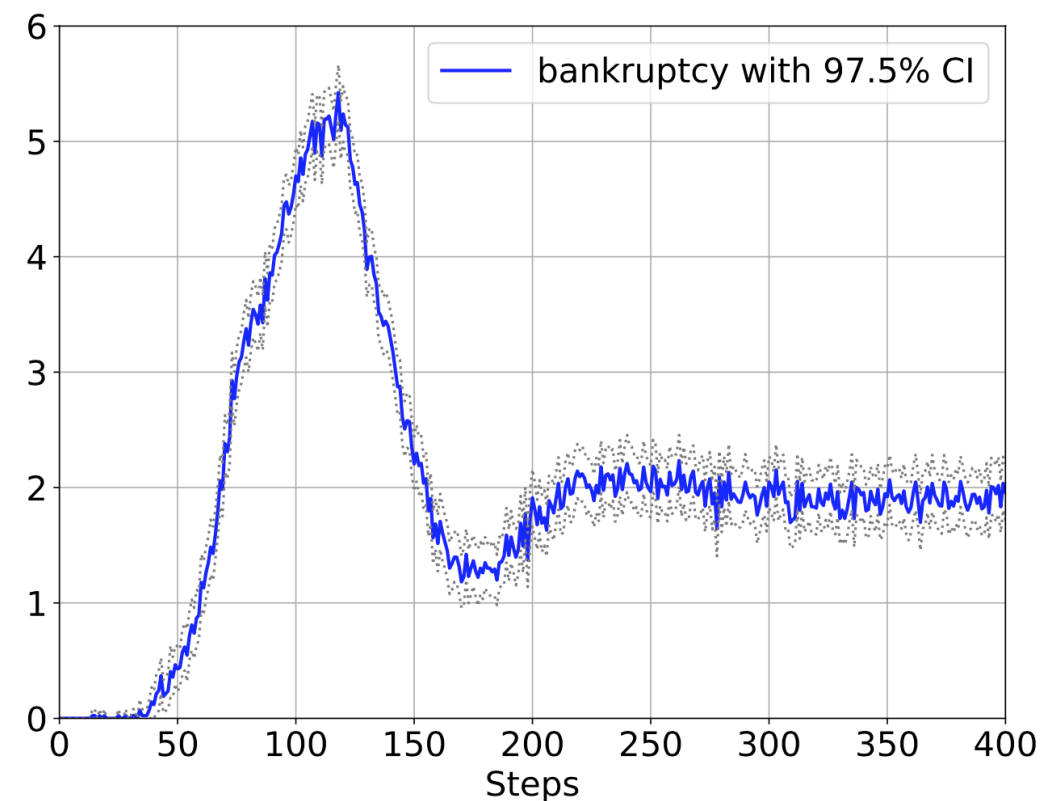
A query to study the evolution of bankruptcies **and** unemployment rate

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```

$\delta=0.005$



$\delta=0.5$



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1. Motivation and context

2. Transient analysis

3. Very briefly: steady state analysis and ergodicity diagnostics

3. PM meets SMC: Process-oriented Behavioral White-box Validation of ABMs

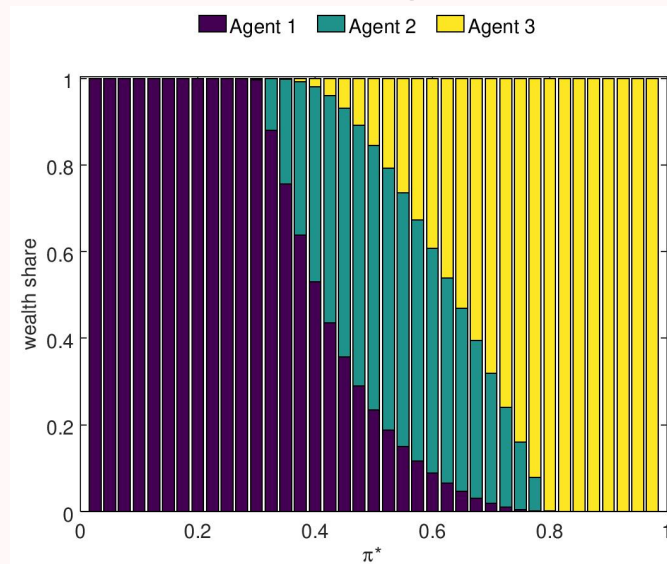
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Arbitrary choice of

- Number of sims
- Warmup period
- Time horizon

from [Kets et al2014]

Agents wealth at steady state

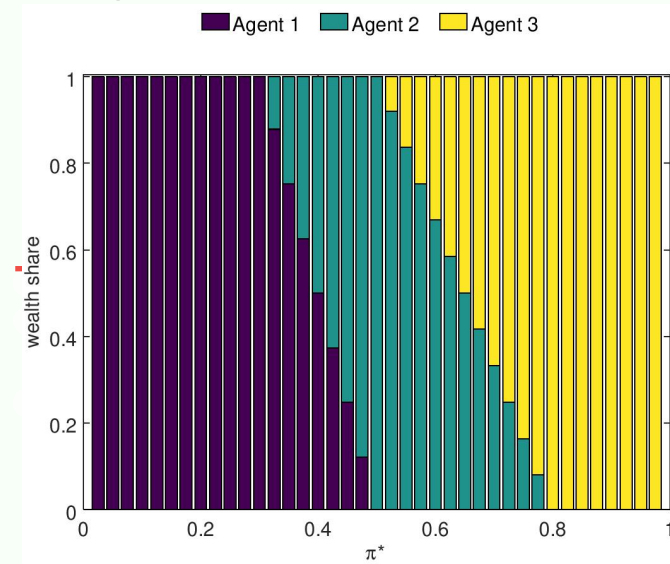


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MultiVeStA

Same as analytical solution
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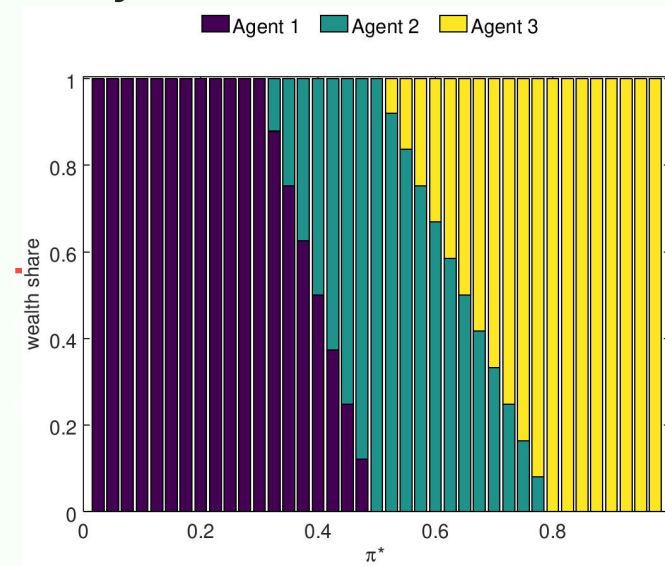
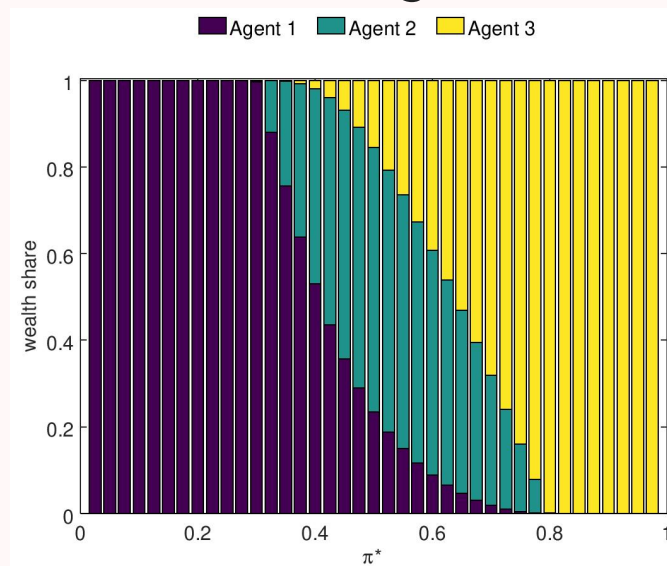
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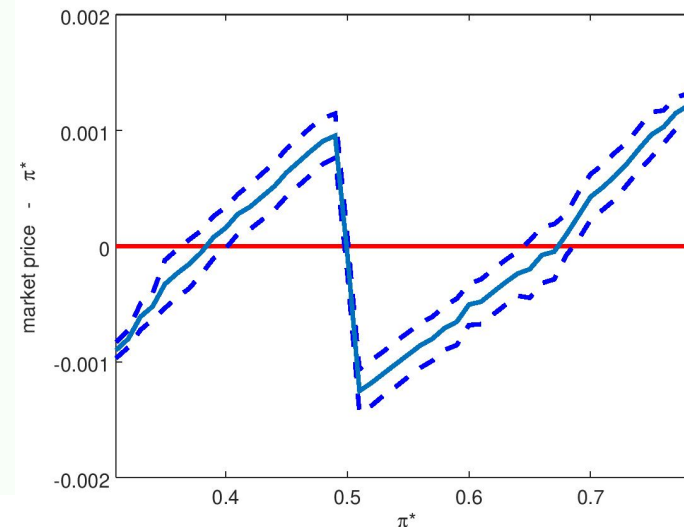
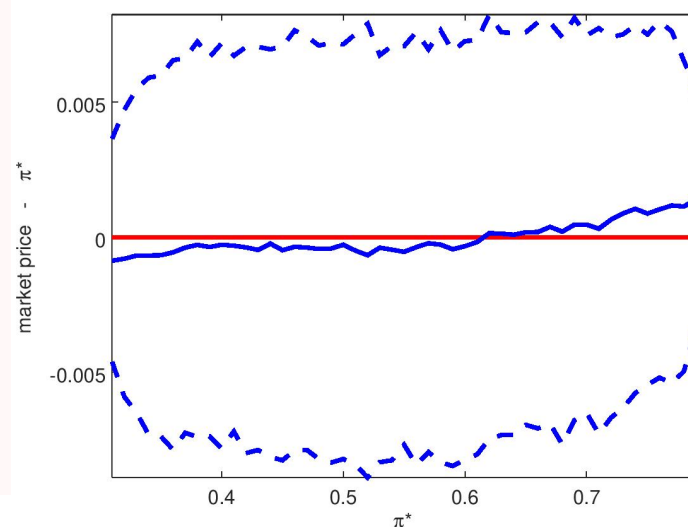
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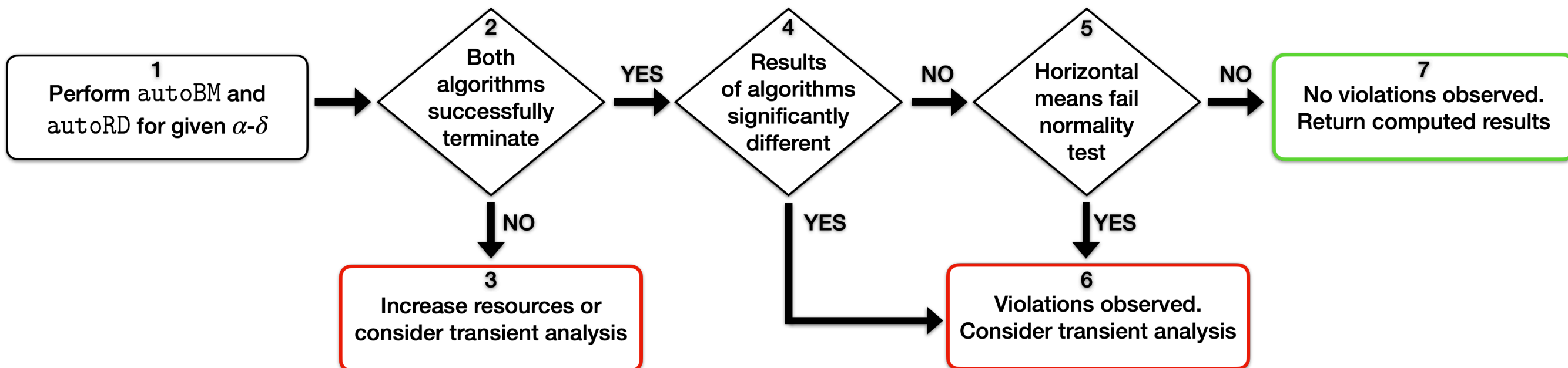
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Does the market price match π^* ?



MV in Short: a Methodology for Ergodicity Diagnostics

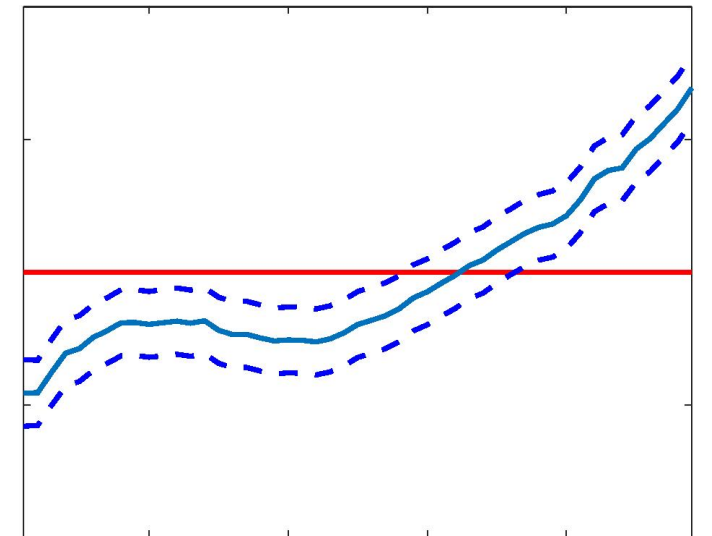
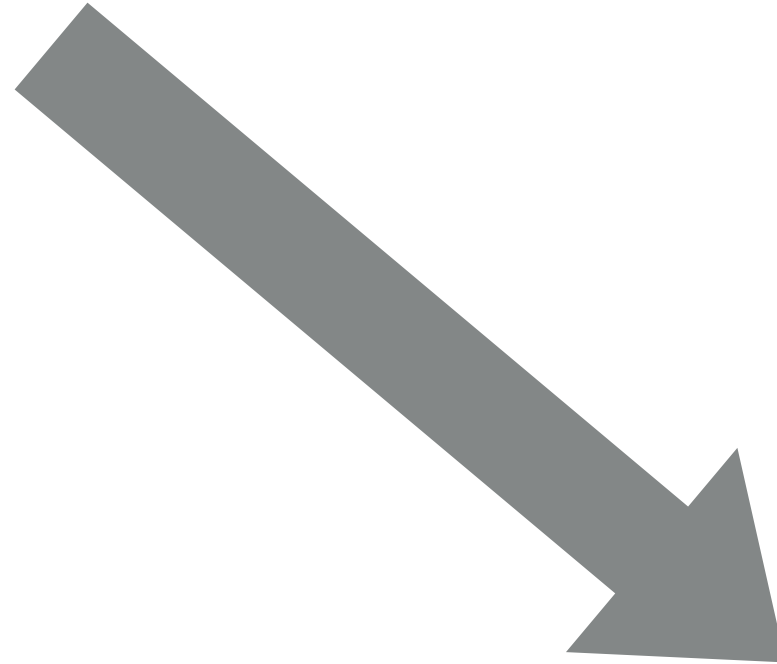


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Our Proposed Approach to Simulation-Based Analysis



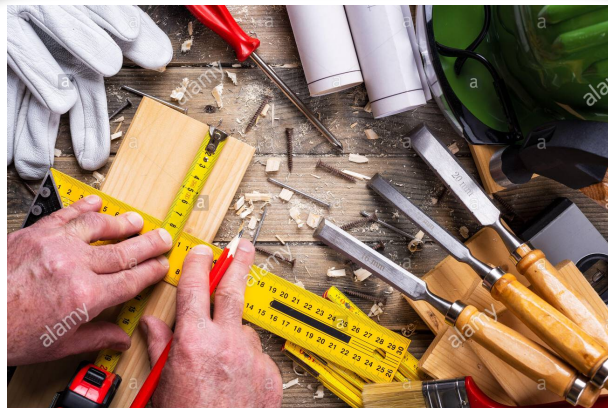
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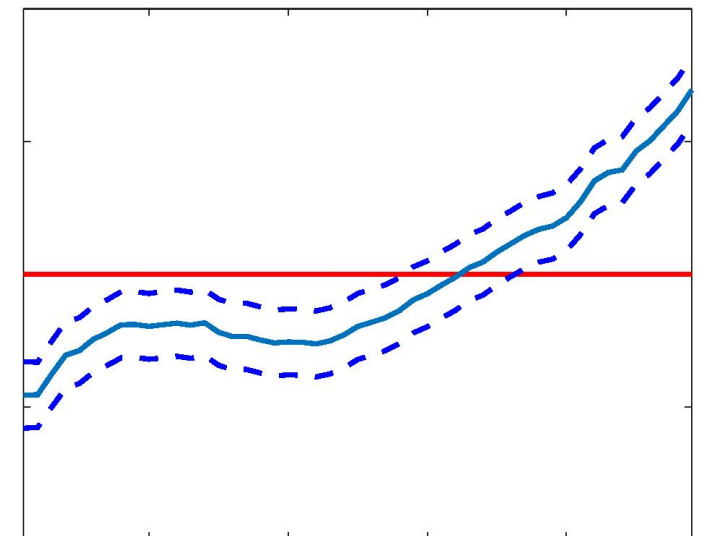
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<https://www.alamy.com/>

Handcrafted

- ▶ Mainly manual process
 - ▶ Time-consuming
 - ▶ Problems with replicability
 - ▶ Error-prone
 - ▶ Modify model, interpret CSV
- ▶ Ad-hoc implementations
 - ▶ Reliability? Efficiency?



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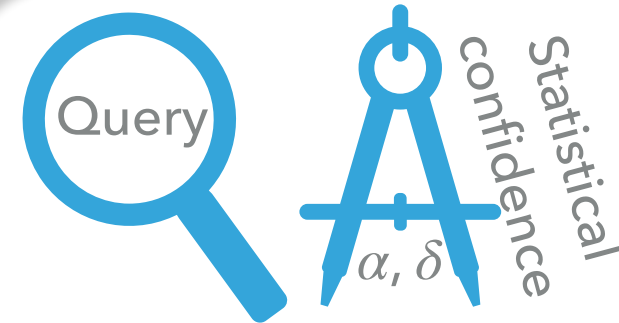
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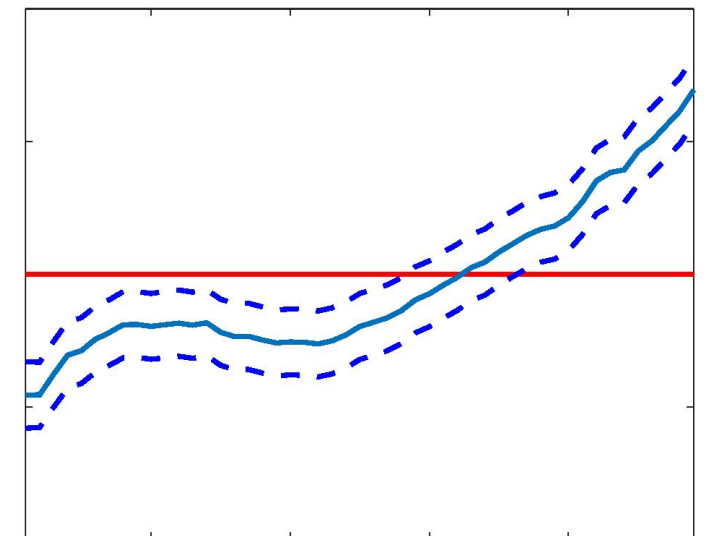
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Statistical Model Checking

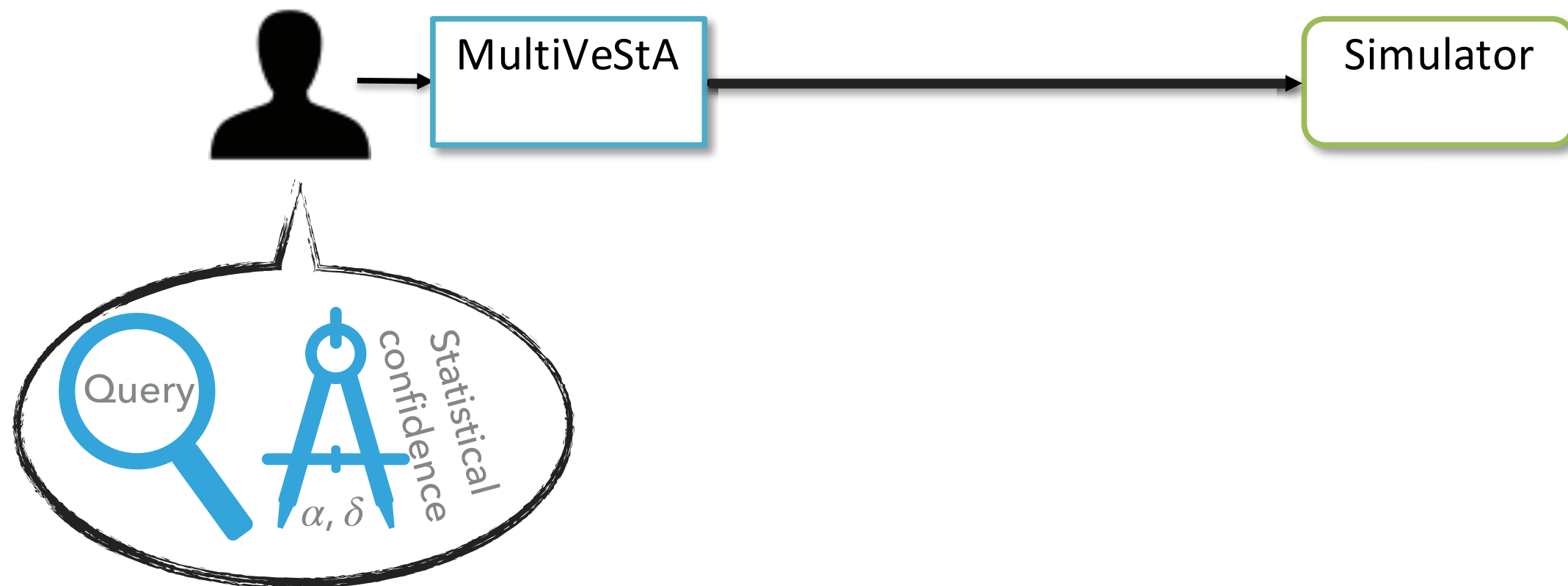
- ▶ Automatic
 - ▶ Time-saving and Reproducible
 - ▶ Promotes use of *standard* analysis
- ▶ Reference implementation
 - ▶ Reliable and Efficient



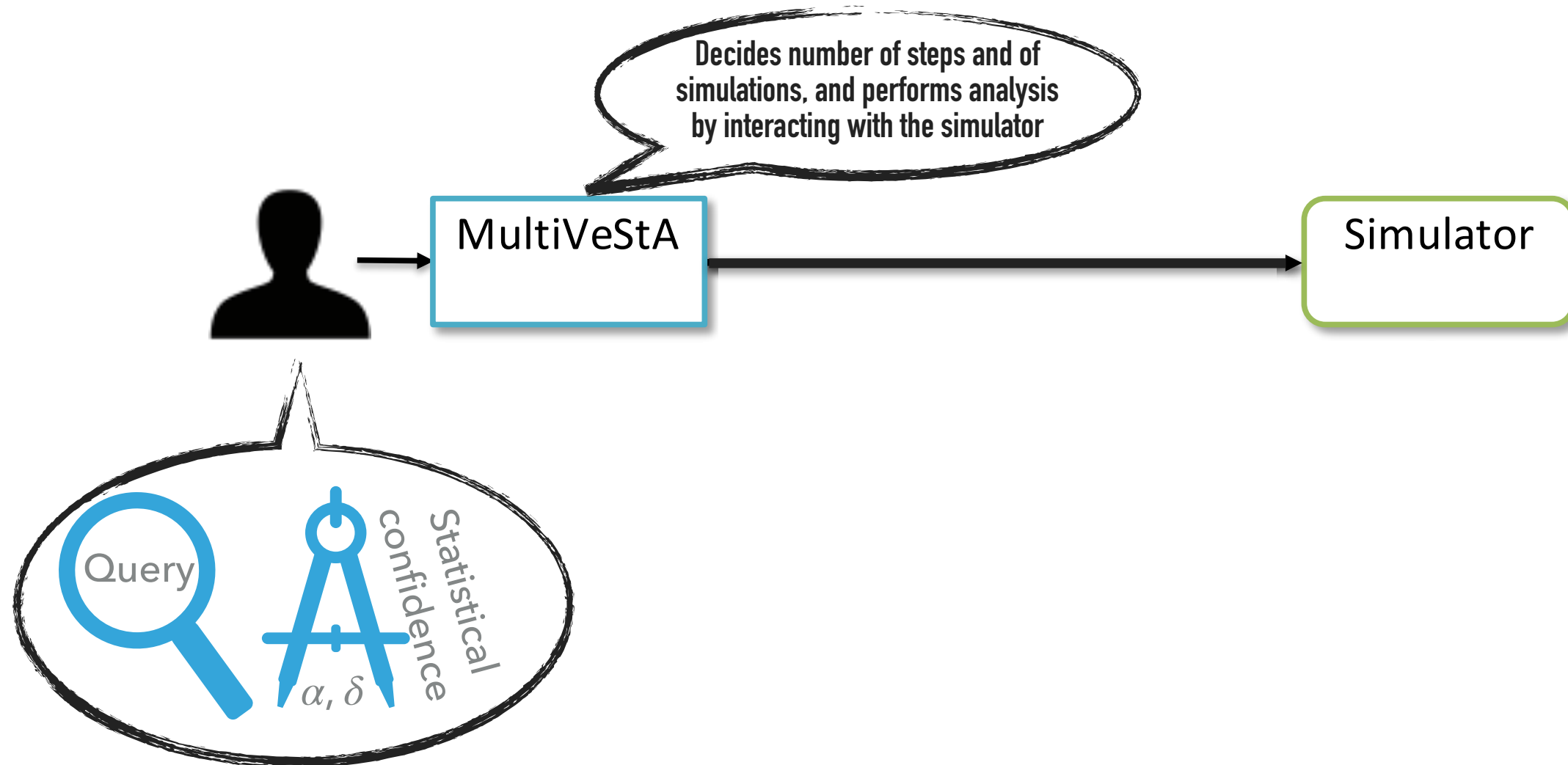
MultiVeStA: SMC For Discrete-Event Simulators



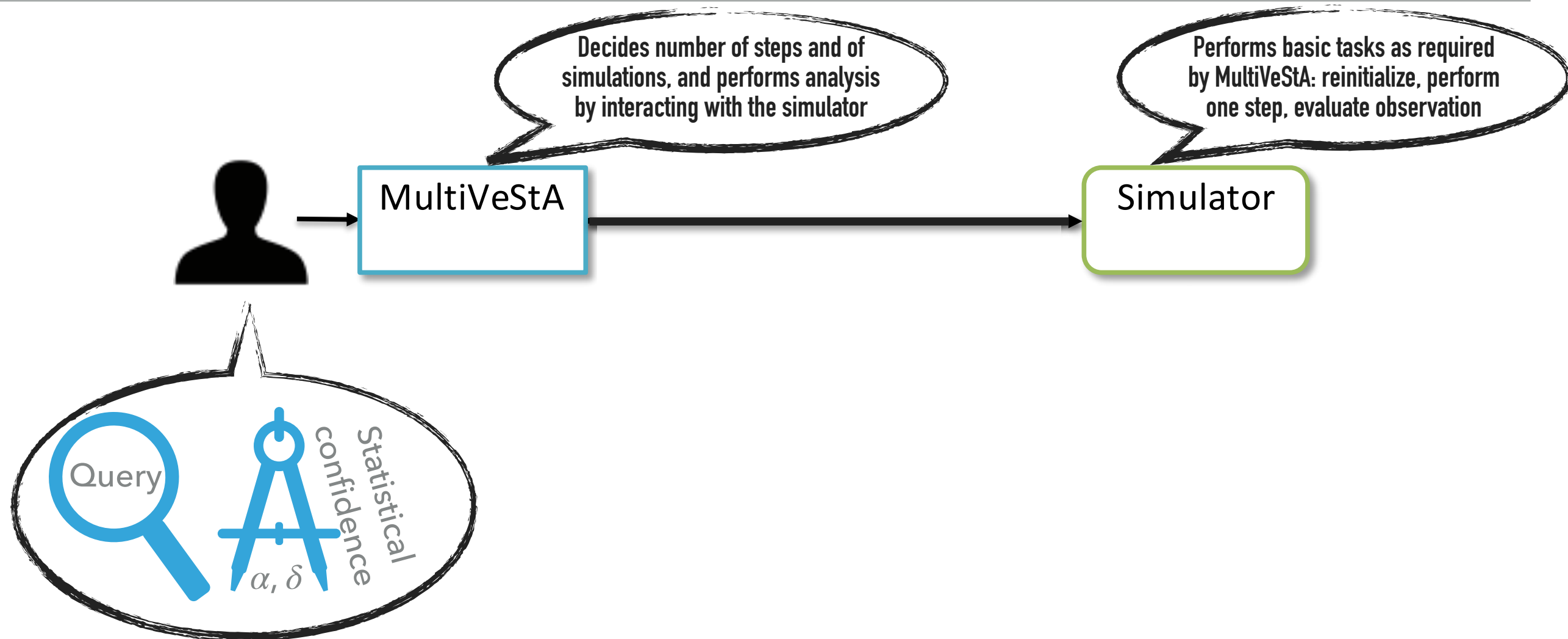
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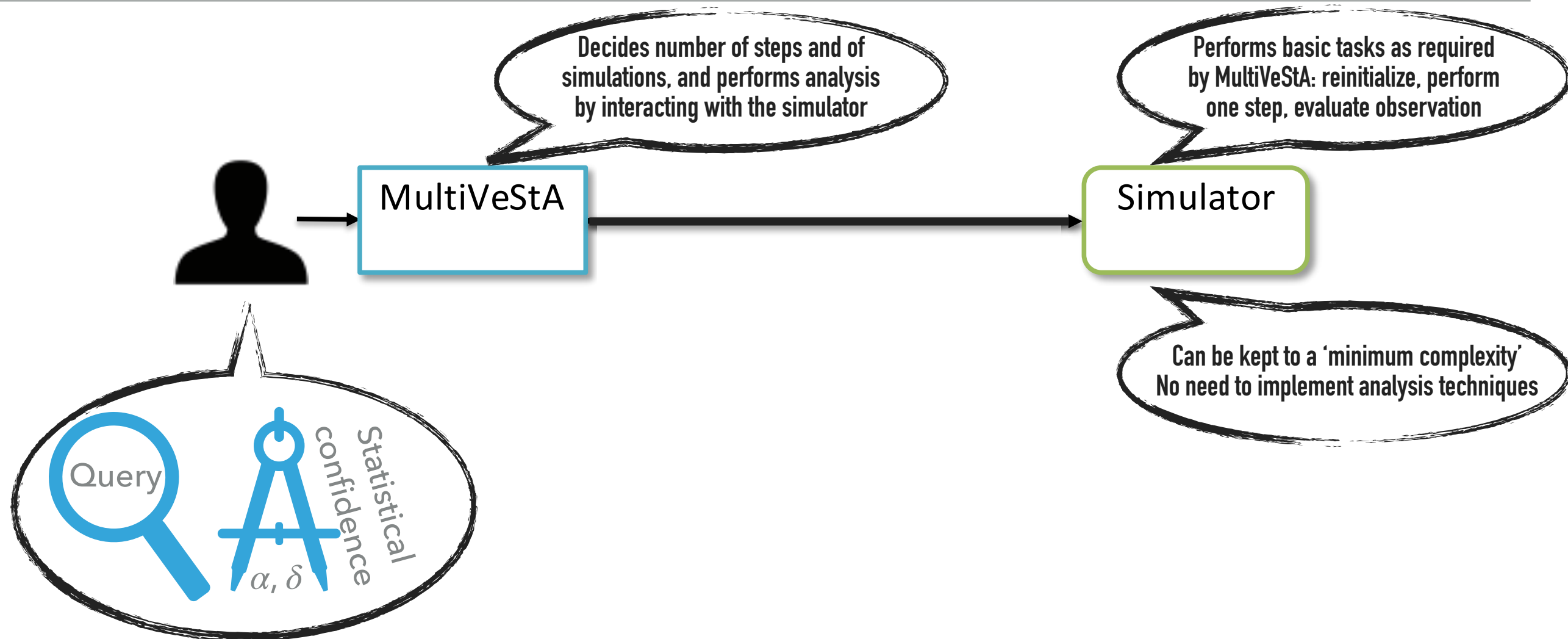
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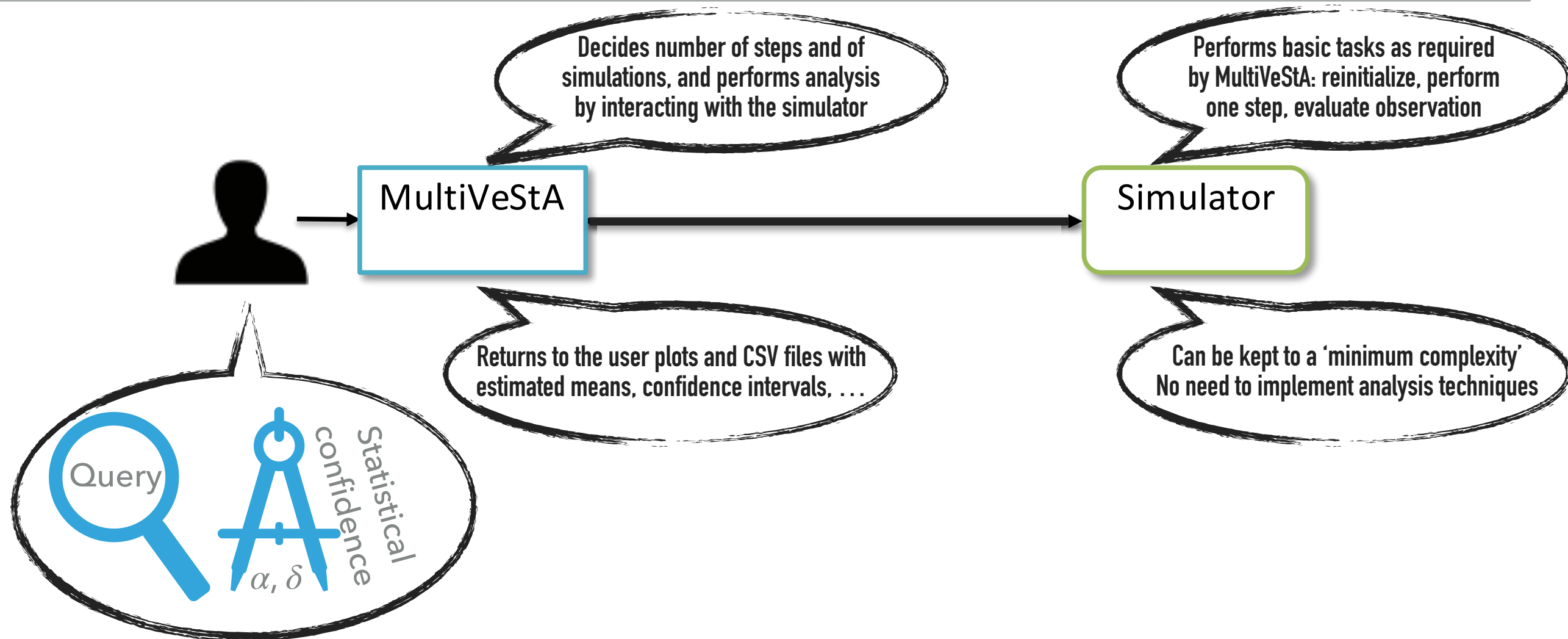
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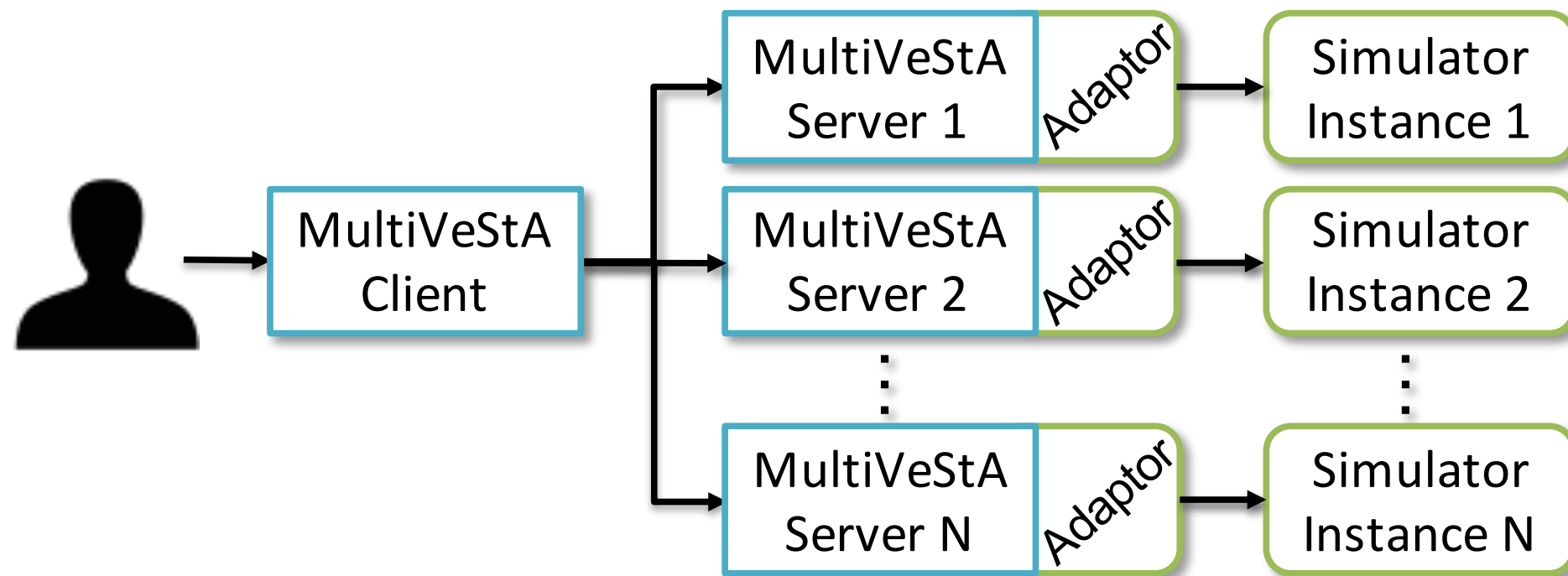
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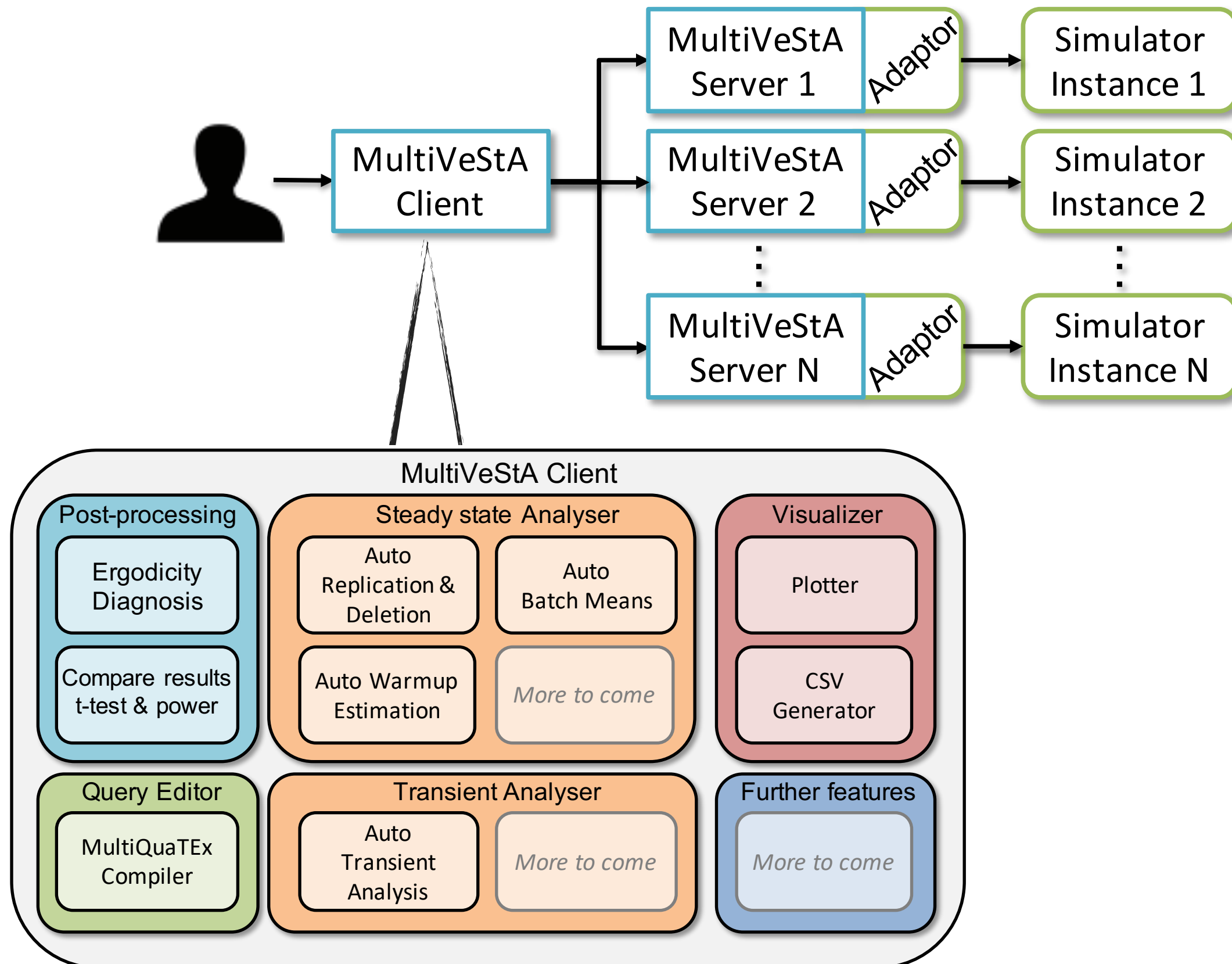
MultiVeStA: SMC For Discrete-Event Simulators



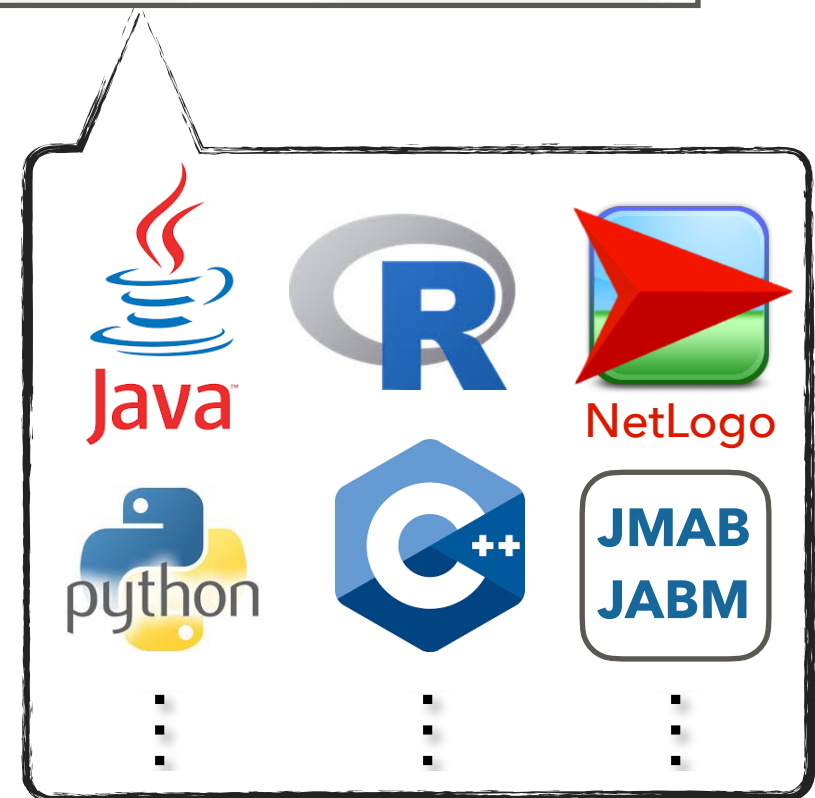
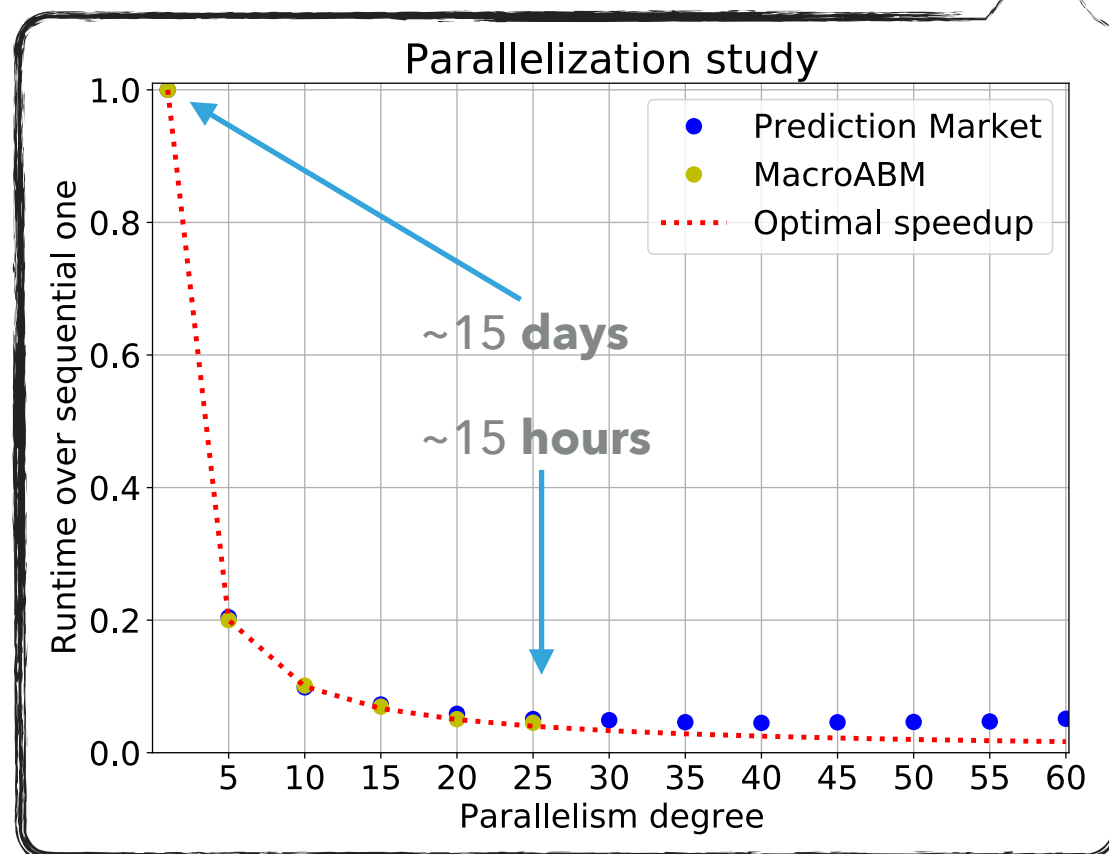
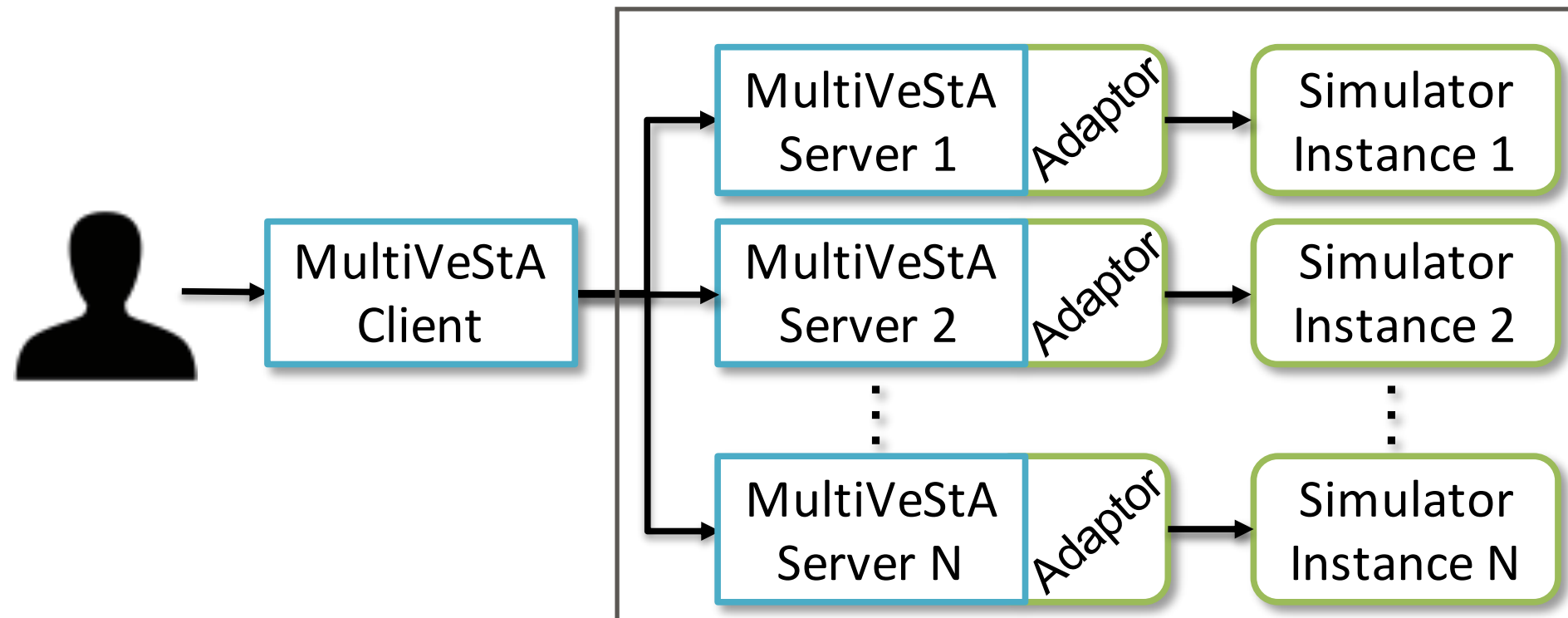
MultiVeStA: SMC For Discrete-Event Simulators



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MultiVeStA: SMC For Discrete-Event Simulators



MultiVeStA: SMC For Discrete-Event Simulators

- ▶ I aim at bridging the simulation, formal methods and ABM communities
- ▶ Future work:
 - ▶ Developing further techniques to help in the simulation-based analysis task
 - ▶ but also validation, calibration, modelling...
 - ▶ Make the techniques accessible
 - ▶ Integrate MultiVeStA with more simulation environments
 - ▶ Improve interface/GUI, ...

1. About me & EMbeDS & Institute of Economics
2. Automated Statistical Analysis of Economic Agent-Based Models by Statistical Model Checking
3. PM meets SMC: Process-oriented Behavioral White-box Validation of ABMs

PROCESS MINING MEETS STATISTICAL MODEL CHECKING TOWARDS A NOVEL APPROACH TO MODEL VALIDATION AND ENHANCEMENT

Joint work with

Roberto Casaluce, Andrea Burattin, Francesca Chiaromonte

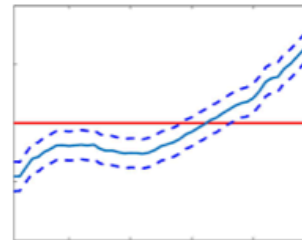
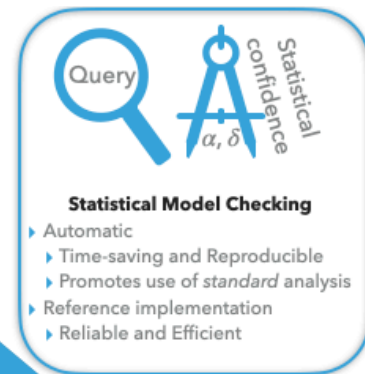
Recently published at DEC2H

zenodo.org/record/6623377

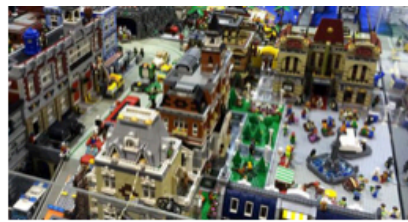
OUR METHODOLOGY WHITE-BOX BEHAVIOURAL VALIDATION



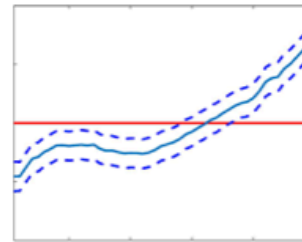
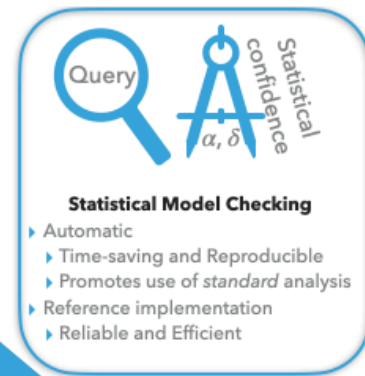
newstalkzb.co.nz/news/education/modern-lego-sets-more-complex-less-inspiring/



OUR METHODOLOGY WHITE-BOX BEHAVIOURAL VALIDATION



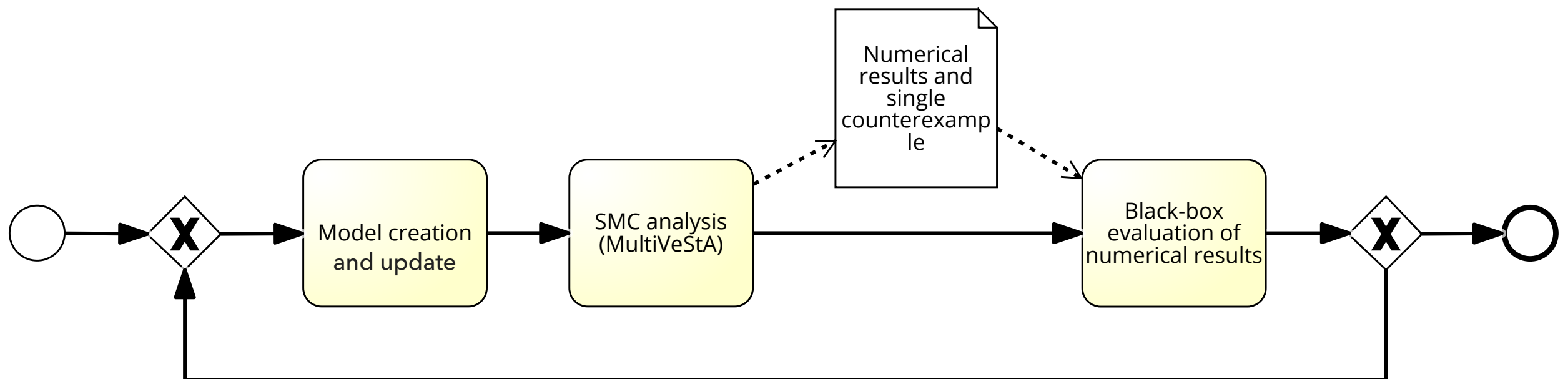
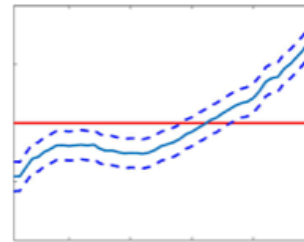
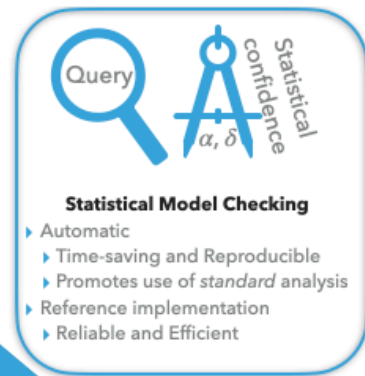
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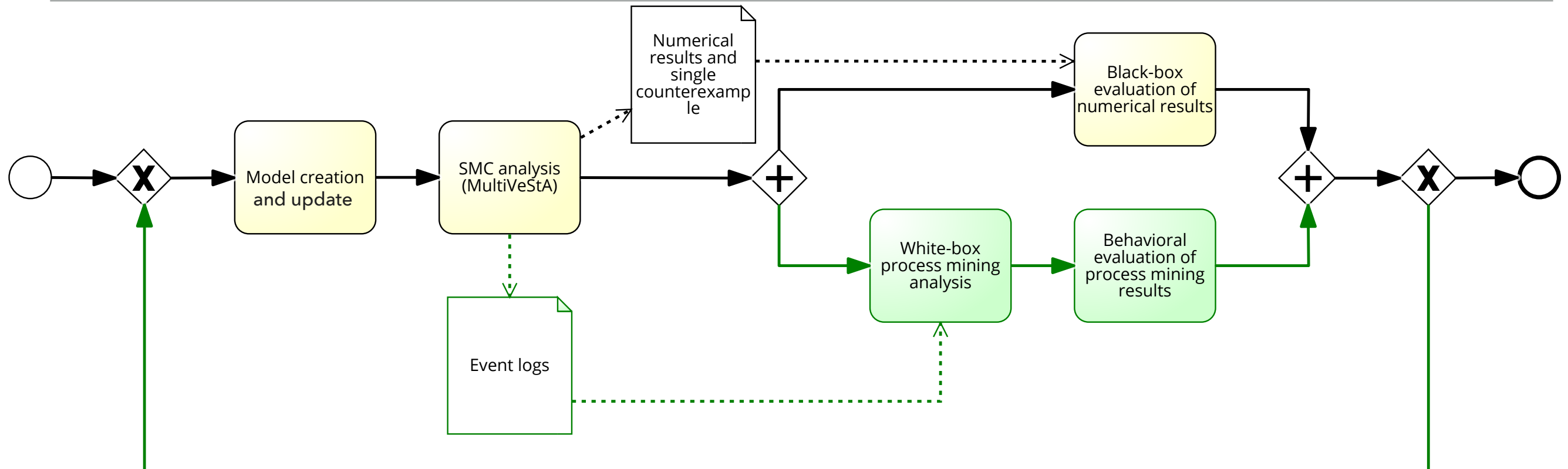
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Informed guess driven by numerical results

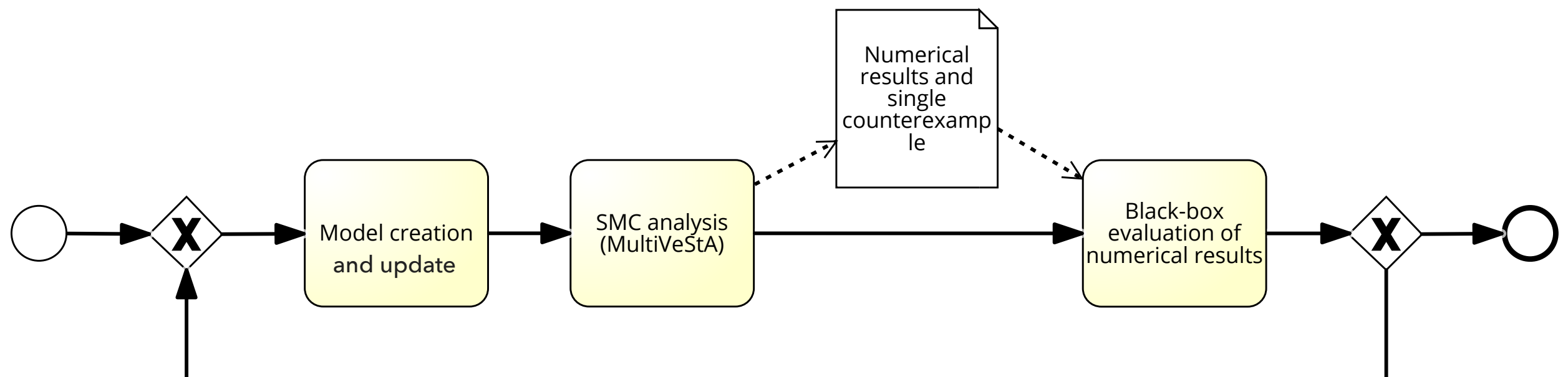
State-of-the-art life-cycle of SMC-analysed simulation

OUR METHODOLOGY WHITE-BOX BEHAVIOURAL VALIDATION



Unexpected behavior discovered with process mining and numerical results

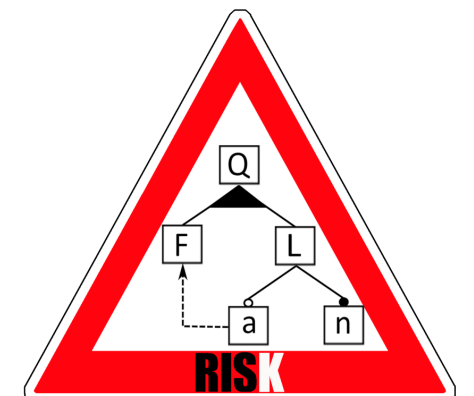
Our novel SMC- and PM-guided methodology for white-box behavioural model validation



Informed guess driven by numerical results

State-of-the-art life-cycle of SMC-analysed simulation

APPLICATION ON SIMPLE THREAT ANALYSIS EXAMPLE



APPLICATION ON SIMPLE THREAT ANALYSIS EXAMPLE

The screenshot displays the RisQFLan application interface with several panels:

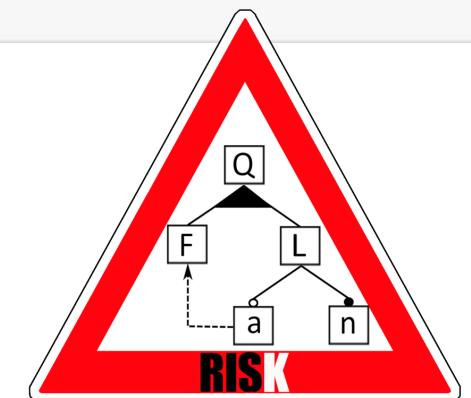
- Project Explorer:** Shows a tree structure of projects including BGP, Bypassing802, CatBurglar, company_damage, COMPANY_DAMAGE, JuvenileDelinquent, RobBank, RobBank2, and SMC. The SMC folder is expanded, showing MultiVeStA_OUTPUT and src-gen.
- RisQFLan Editor:** Contains code for defining a model, variables, attack nodes, defense nodes, countermeasure nodes, attack diagrams, and attackers. The code is for a model named RobBank_one_child.
- Console View:** Displays the output of the simulation, including the MultiVeStA client status and various parameters like alpha, delta list, and class name.
- Model Visualizer:** Shows a hierarchical diagram of the model structure, including nodes like RobBank, OpenVault, BlowUp, and LockDown.
- Outline View:** Provides a summary of the model's components, such as variables, attack nodes, defense nodes, countermeasure nodes, attack diagrams, and attackers.
- Plot View:** Displays a line graph showing the results of the simulation, with a green line representing the defense effectiveness over time.

Computers & Security
Volume 109, October 2021, 102381

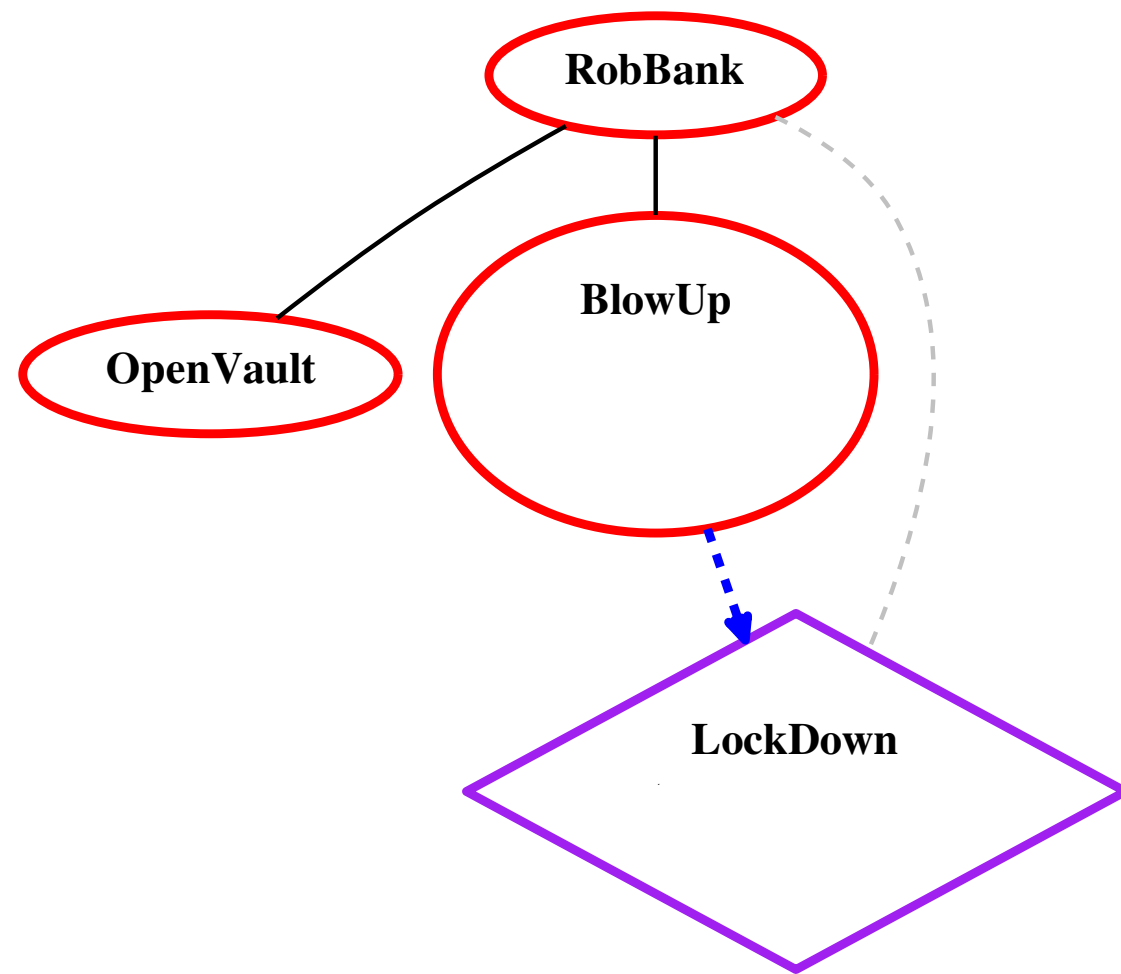
TC 11 Briefing Papers

Quantitative Security Risk Modeling and Analysis with RisQFLan

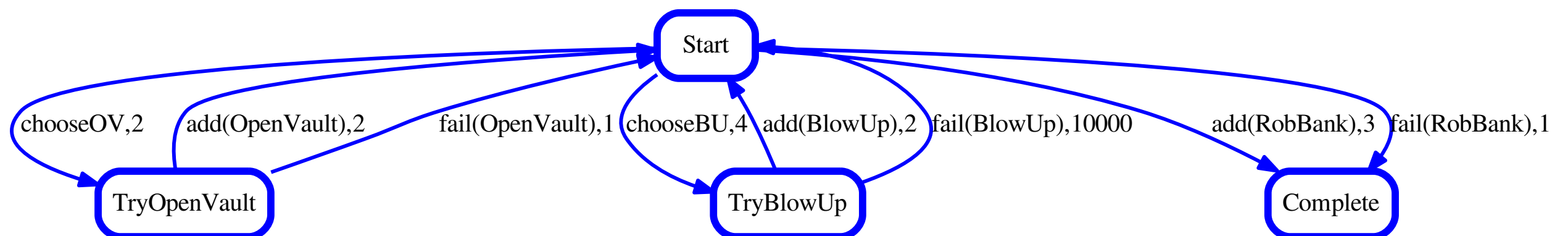
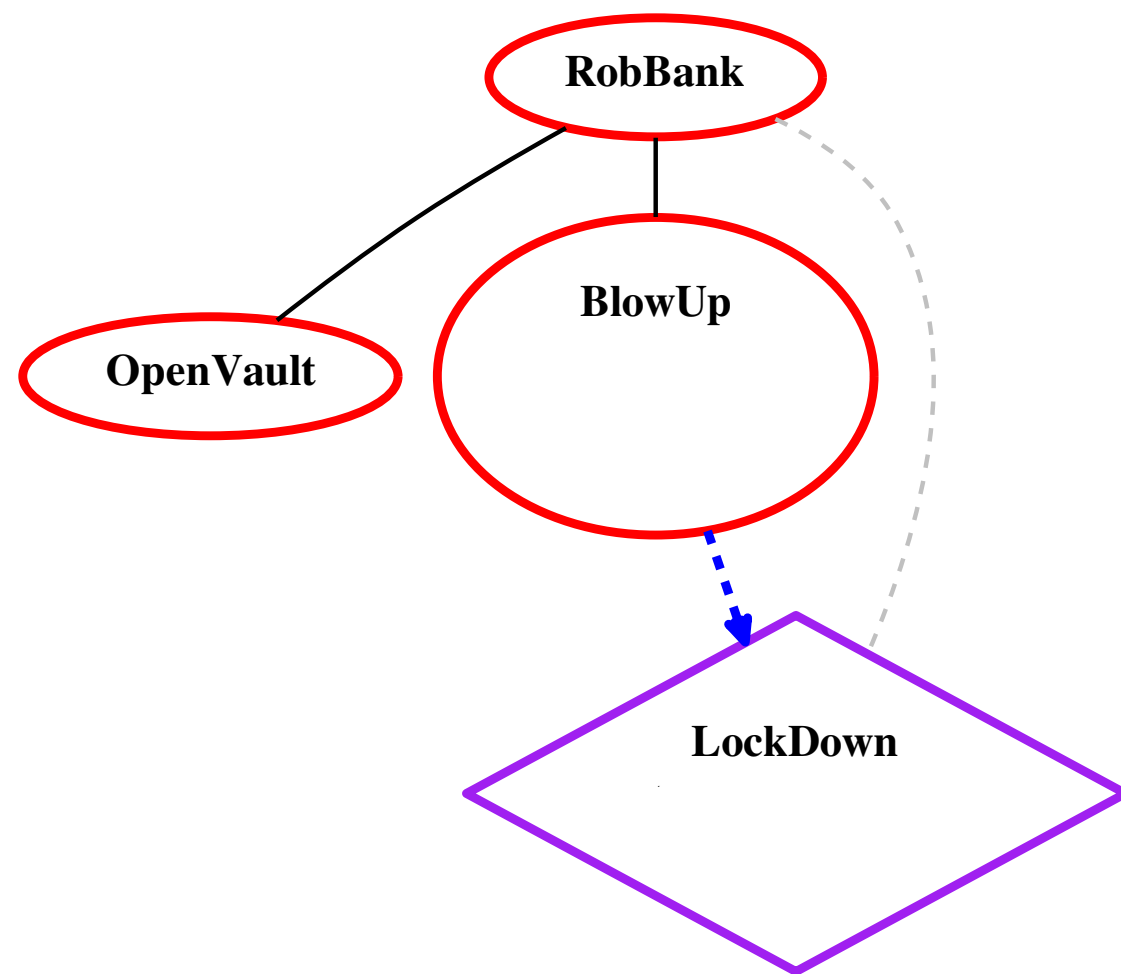
Maurice H. ter Beek^a, Axel Legay^b, Alberto Lluch Lafuente^c, Andrea Vandin^{c, d}



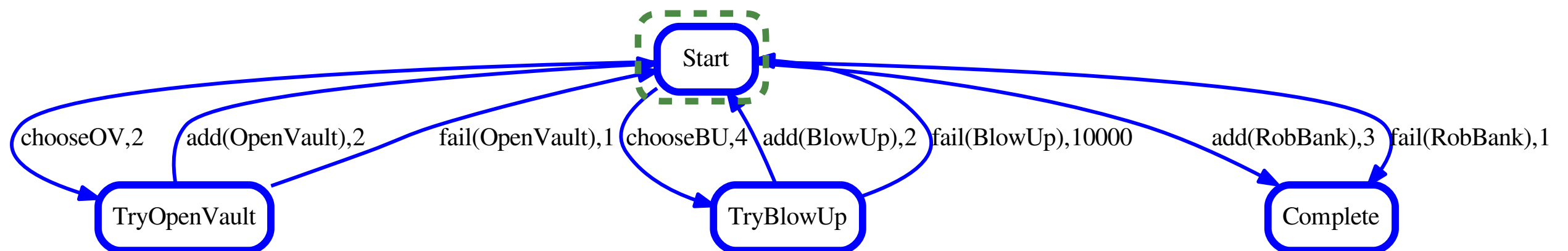
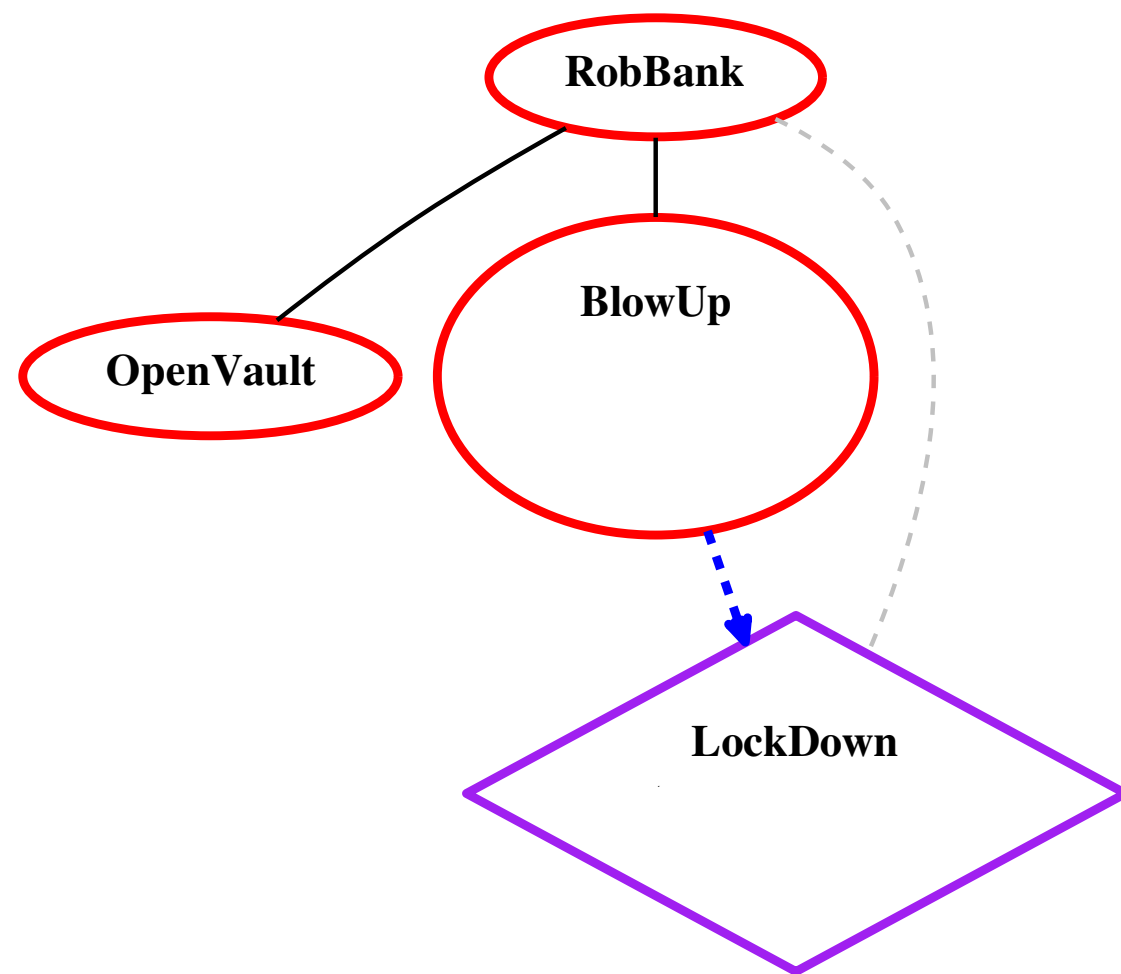
APPLICATION ON SIMPLE THREAT ANALYSIS EXAMPLE



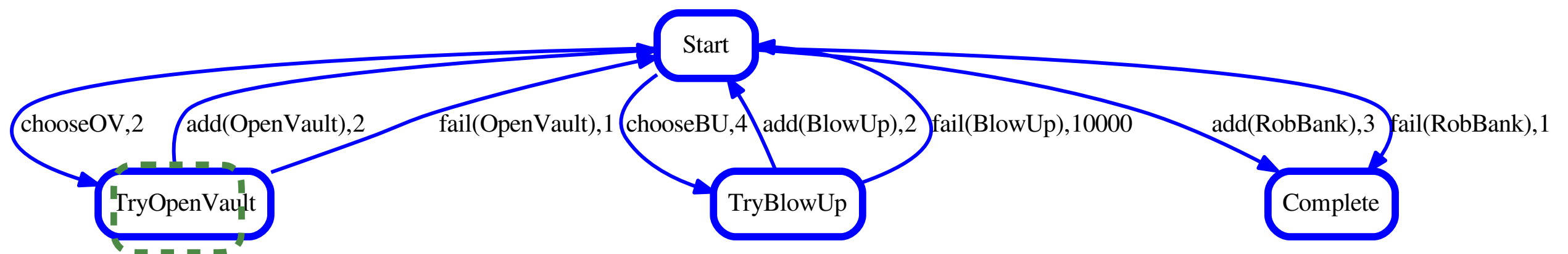
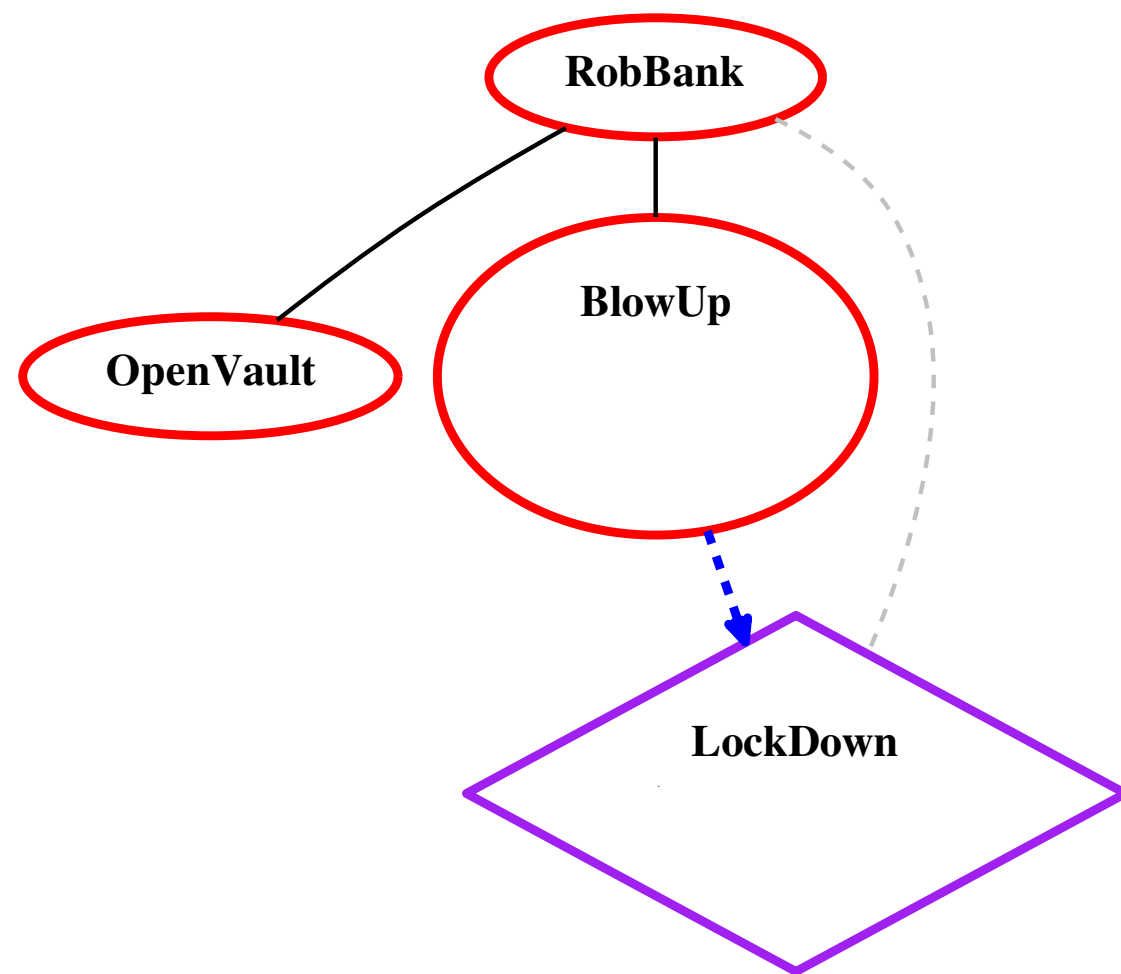
APPLICATION ON SIMPLE THREAT ANALYSIS EXAMPLE



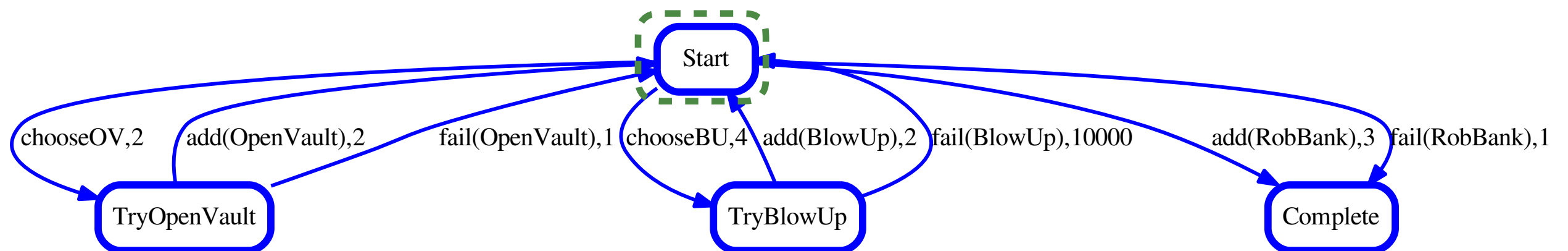
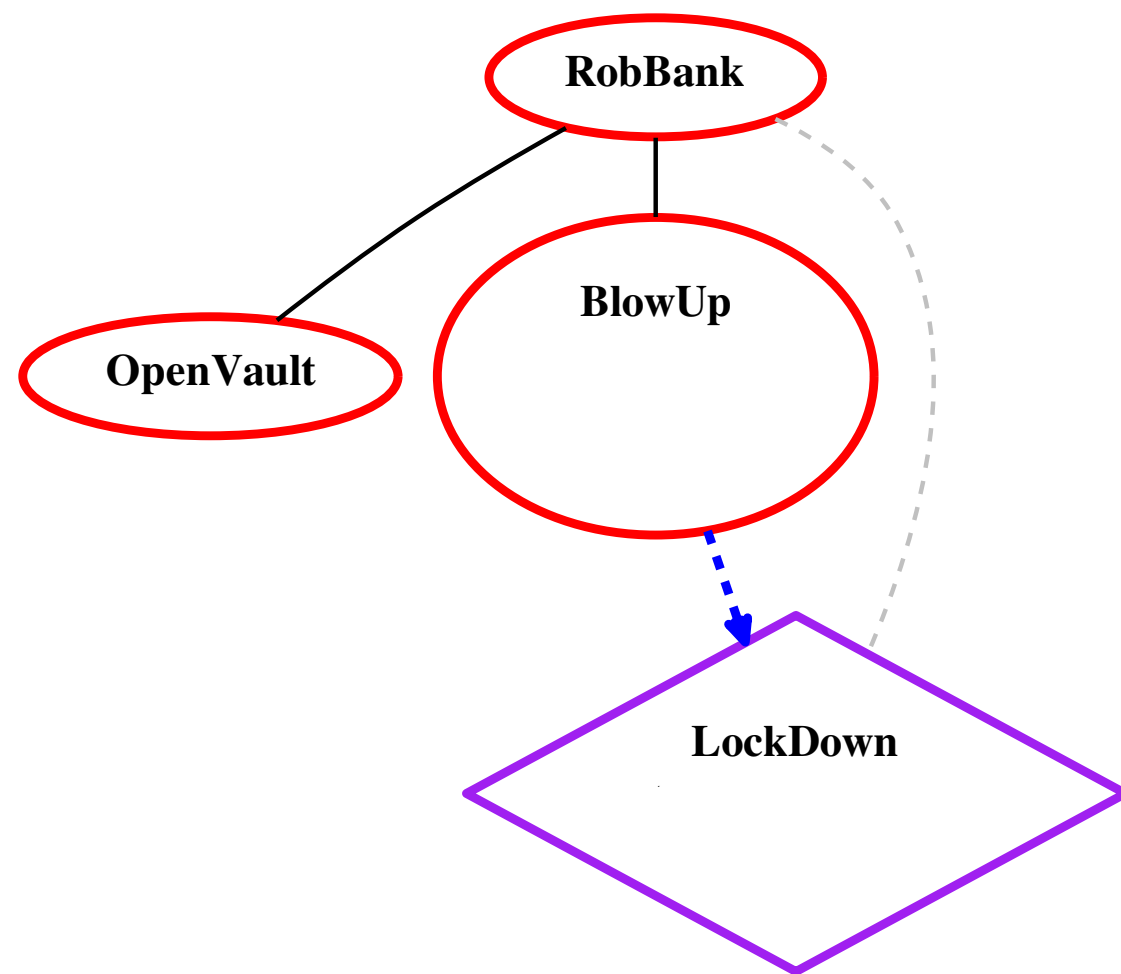
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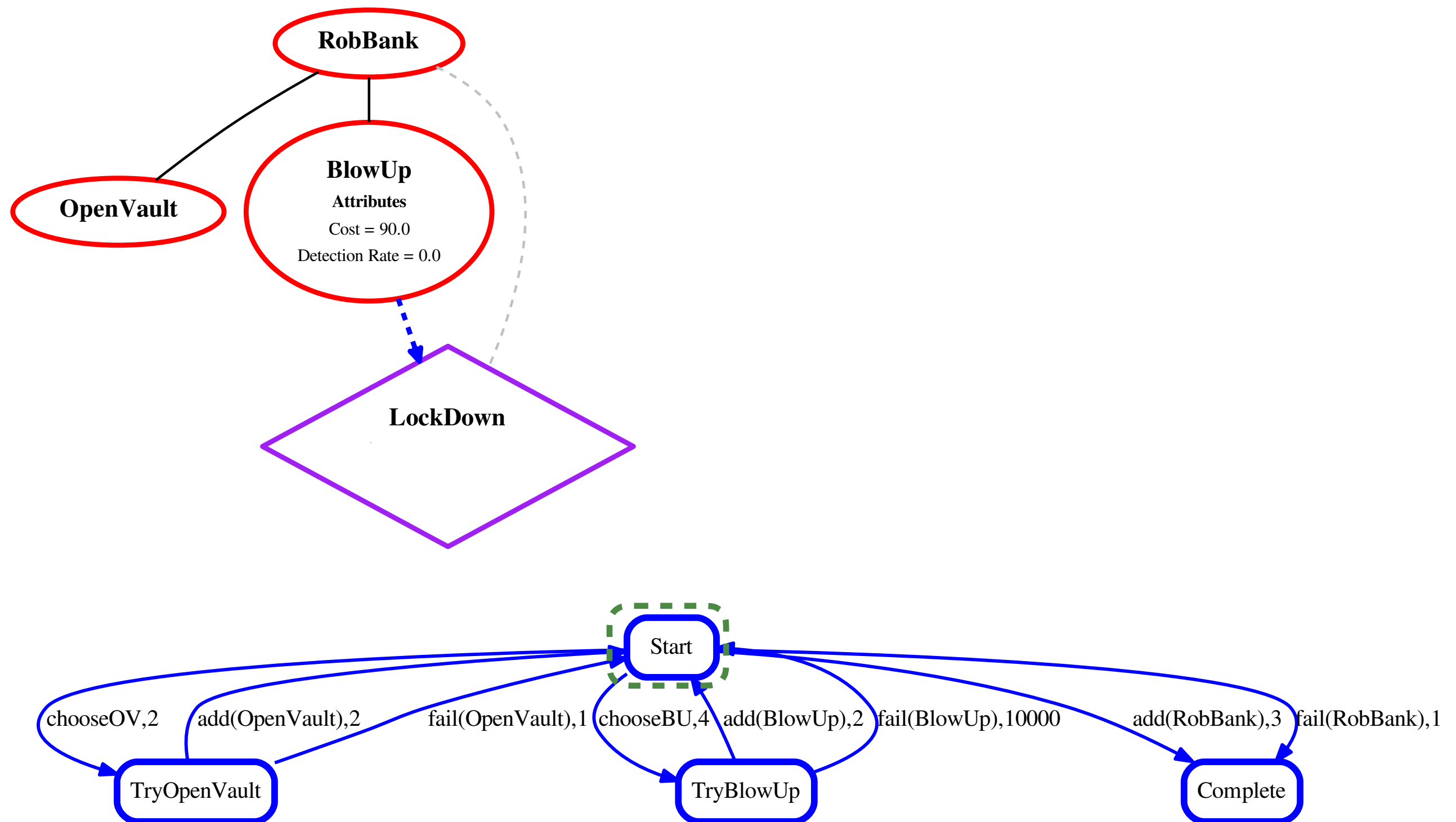
APPLICATION ON SIMPLE THREAT ANALYSIS EXAMPLE



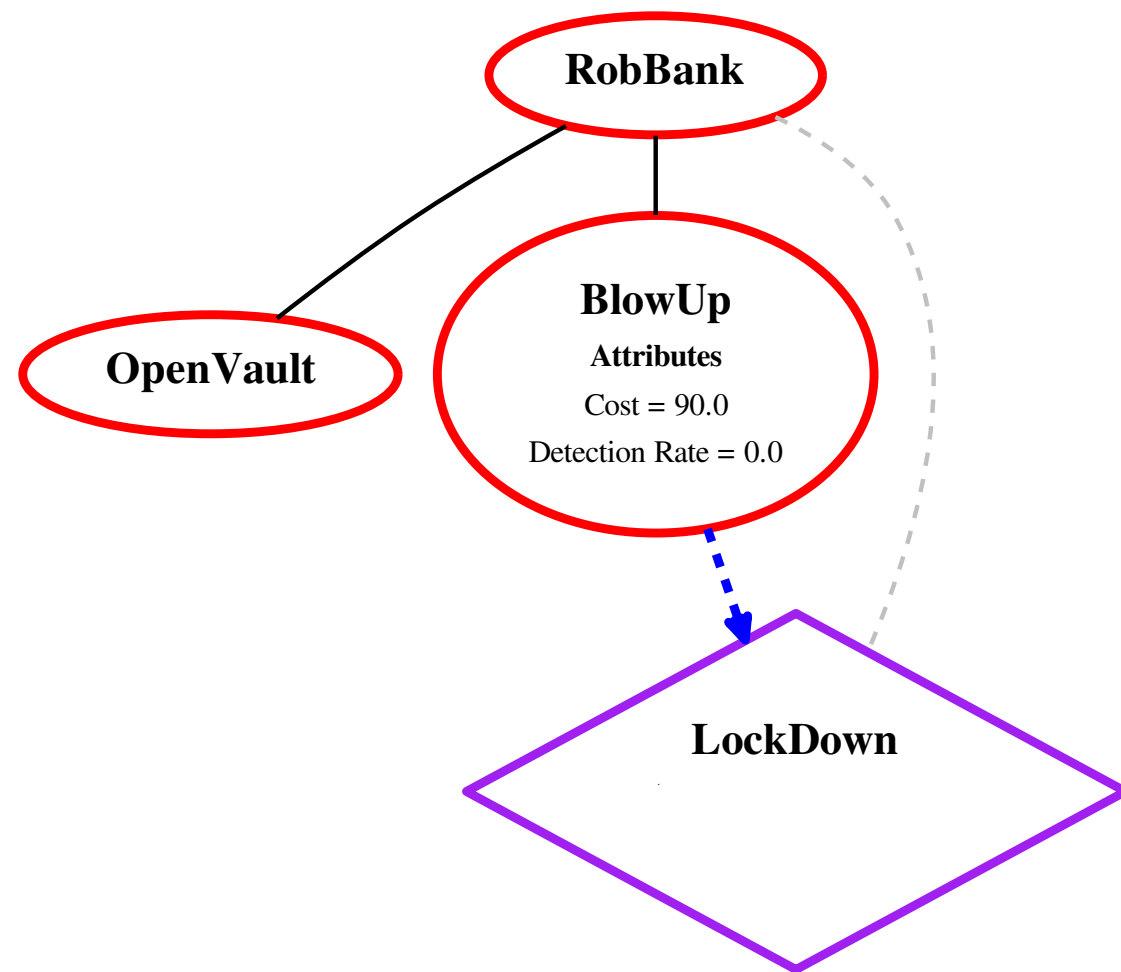
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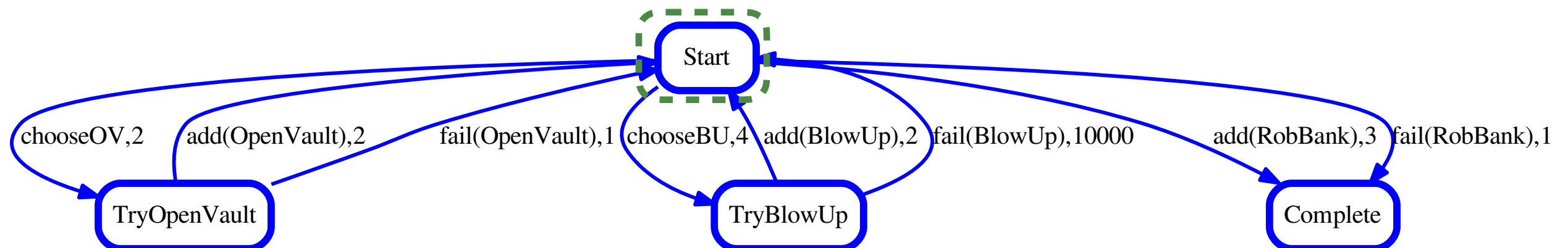
APPLICATION ON SIMPLE THREAT ANALYSIS EXAMPLE



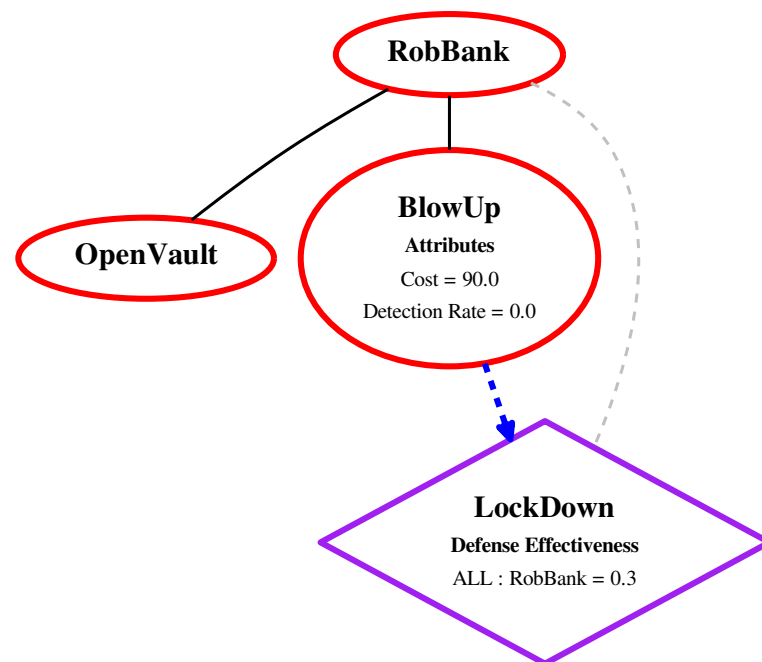
APPLICATION ON SIMPLE THREAT ANALYSIS EXAMPLE



```
1 begin attributes
2   Cost = {BlowUp = 90, OpenVault = 0}
3 end attributes
4
5 begin quantitative constraints
6   { value(Cost) <= 100 }
7 end quantitative constraints
```



APPLICATION ON SIMPLE THREAT ANALYSIS EXAMPLE

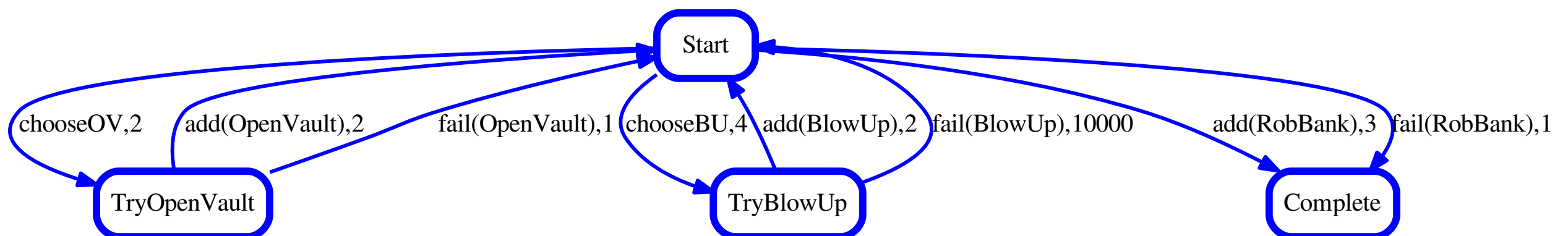


```

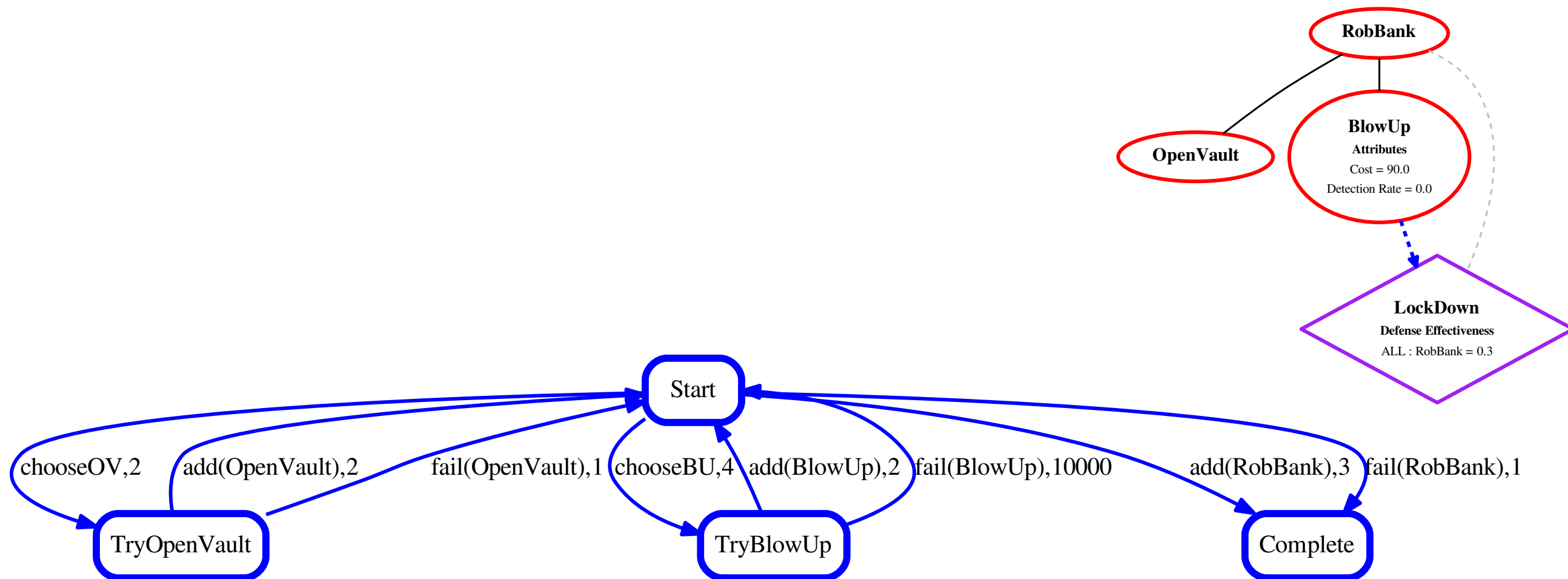
1  begin actions
2      chooseOV
3      chooseBU
4  end actions
    
```

```

1  begin attacker behavior
2      begin attack
3          attacker = Thief
4          states = Start, TryOpenVault, TryBlowUp, Complete
5          transitions =
6              // If vault open or blown up, attacker can rob
7              Start - (succ(RobBank), 3, allowed(RobBank)) -> Complete,
8              Start - (fail(RobBank), 1, allowed(RobBank)) -> Complete,
9              // Strategy where the attacker can open the vault
10             Start -(chooseOV, 2) -> TryOpenVault,
11                 TryOpenVault -(succ(OpenVault), 2) -> Start,
12                 TryOpenVault -(fail(OpenVault), 1) -> Start,
13             // Strategy where the attacker tries to blow up the vault
14             Start -(chooseBU, 4) -> TryBlowUp,
15                 TryBlowUp -(succ(BlowUp), 2) -> Start,
16                 TryBlowUp -(fail(BlowUp), 10000) -> Start
17         end attack
18     end attacker behavior
    
```



ANALYSIS OF ORIGINAL MODEL

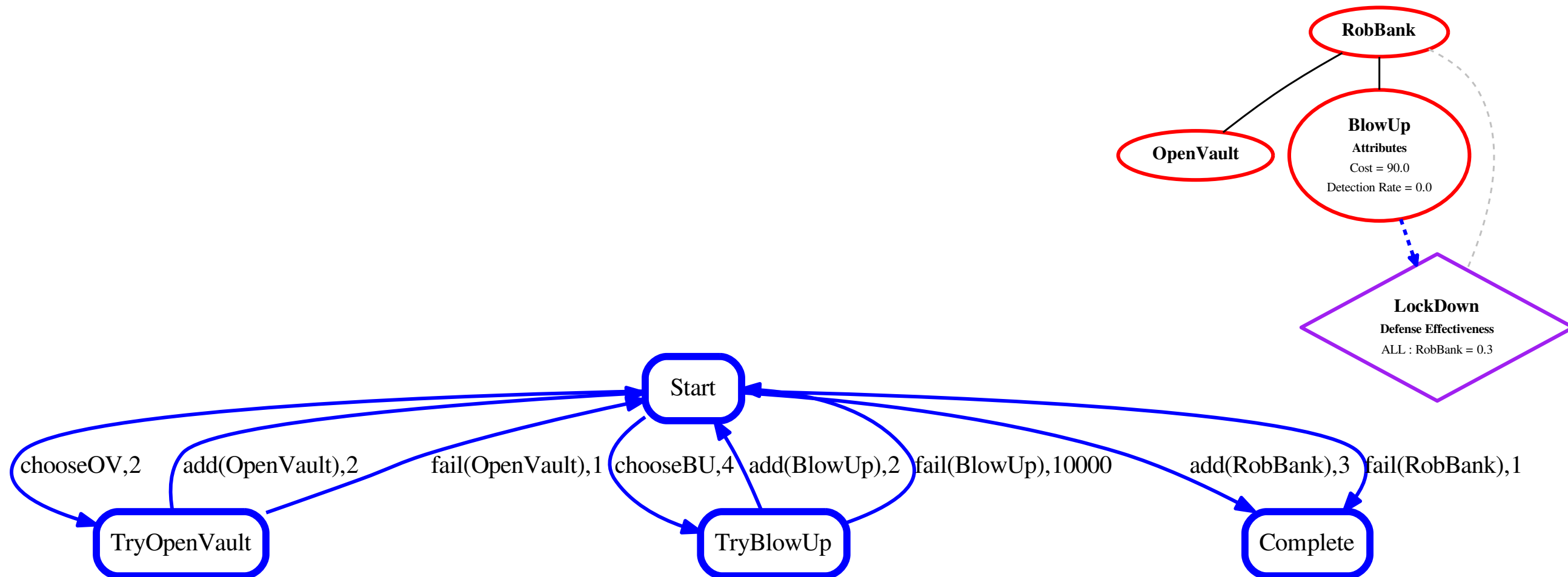


ANALYSIS OF ORIGINAL MODEL

Probability of successful bank robbery!? **0.17**

Why?

- 1) My defences are *good*
- 2) The attacker is *bad*
- 3) **Or my model is bad!?**



ANALYSIS OF ORIGINAL MODEL

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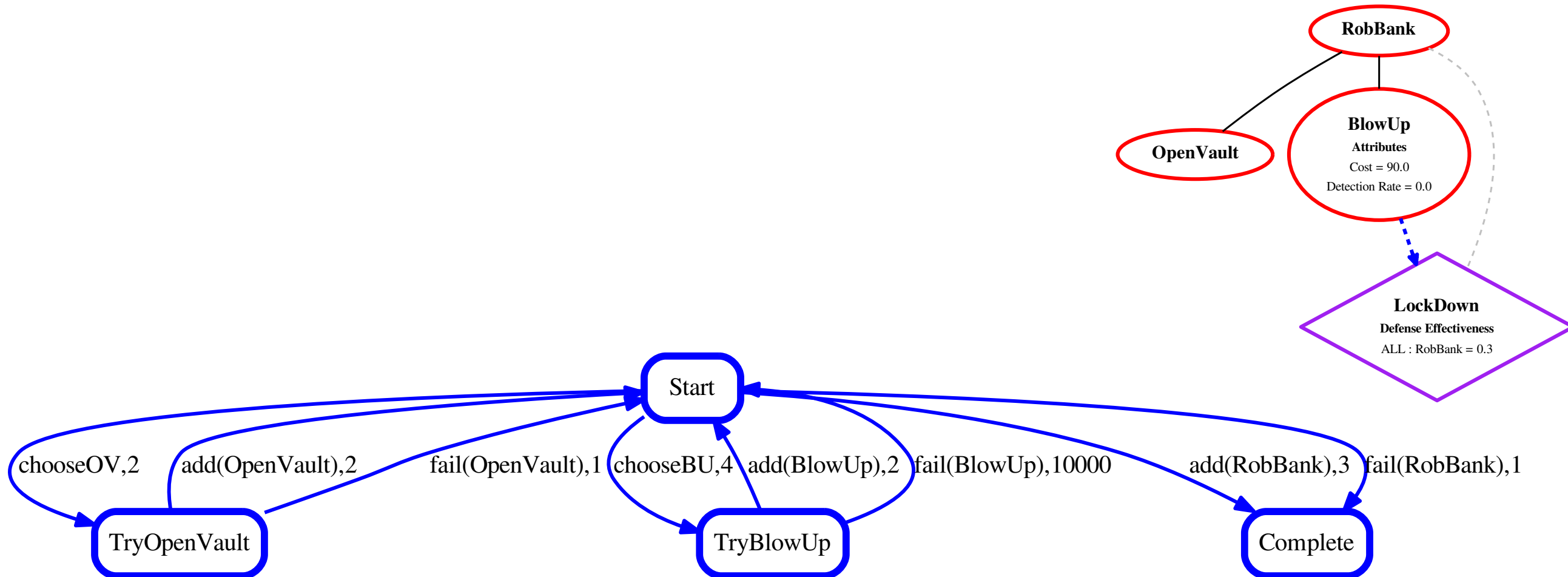
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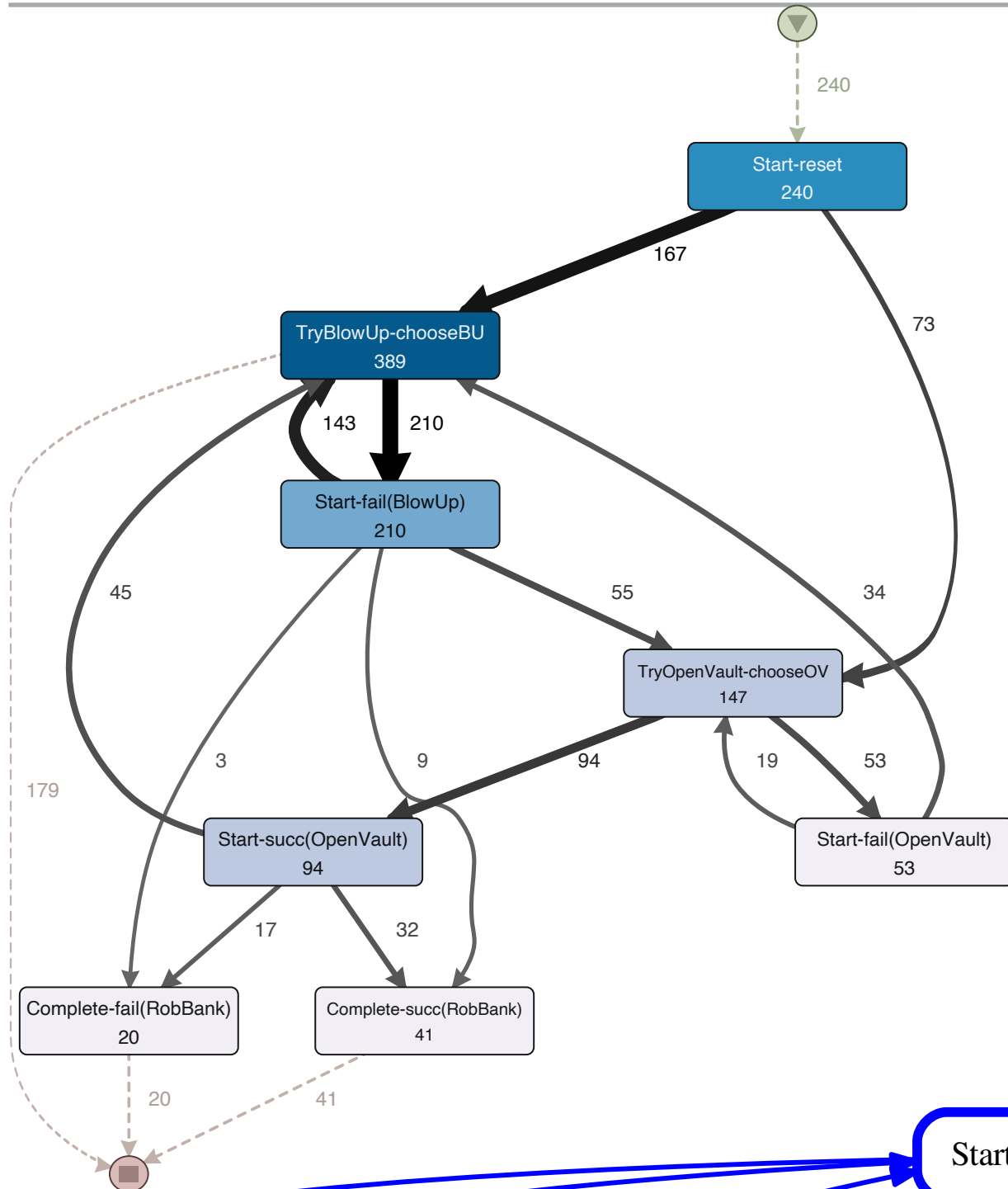
We set $\alpha=0.1$, $\delta=0.1$

MultiVeStA performs **240 simulations**

- We generate **logs** for each simulation
- We ask **Fluxicon Disco** mine these logs
- Can we spot **any issue in the model?**



ANALYSIS OF ORIGINAL MODEL



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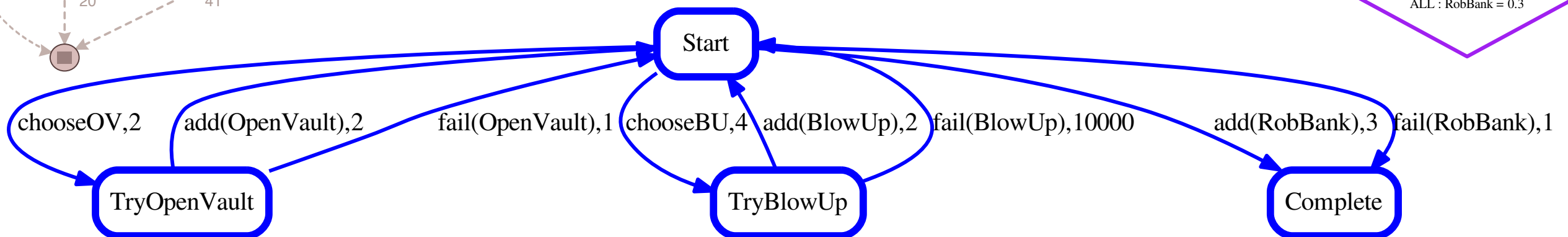
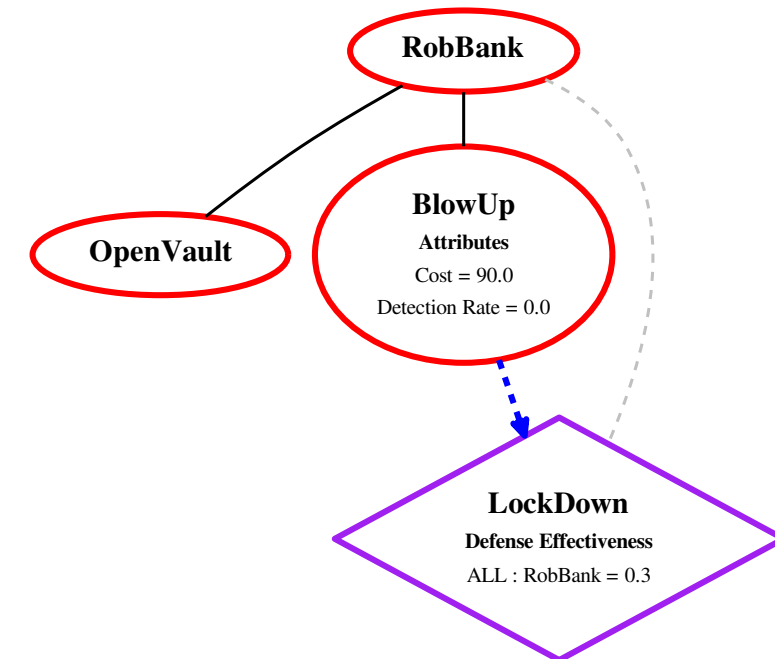
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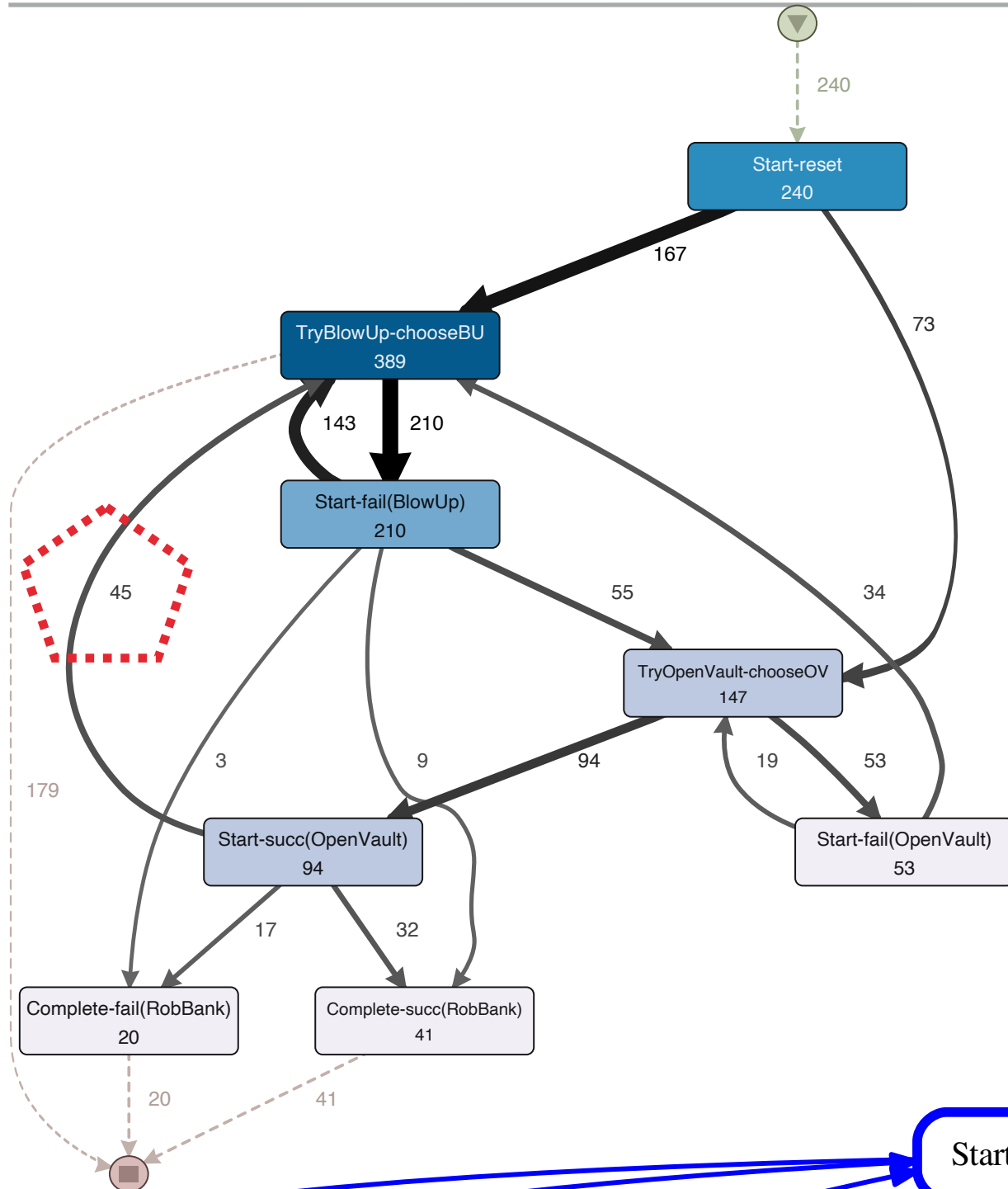
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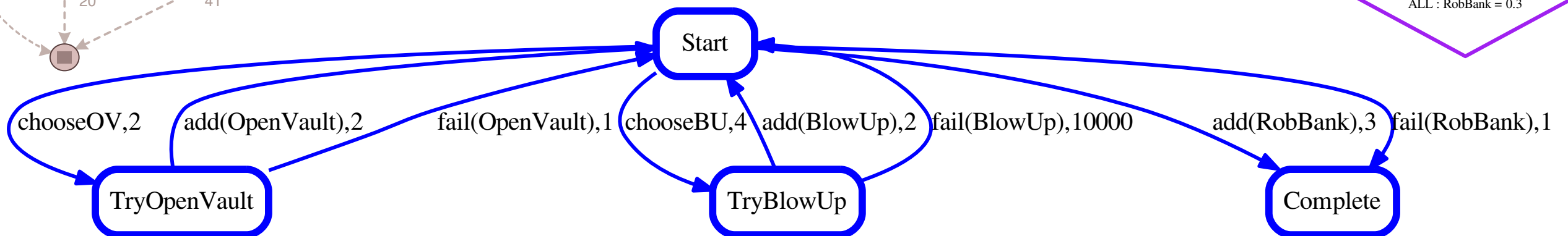
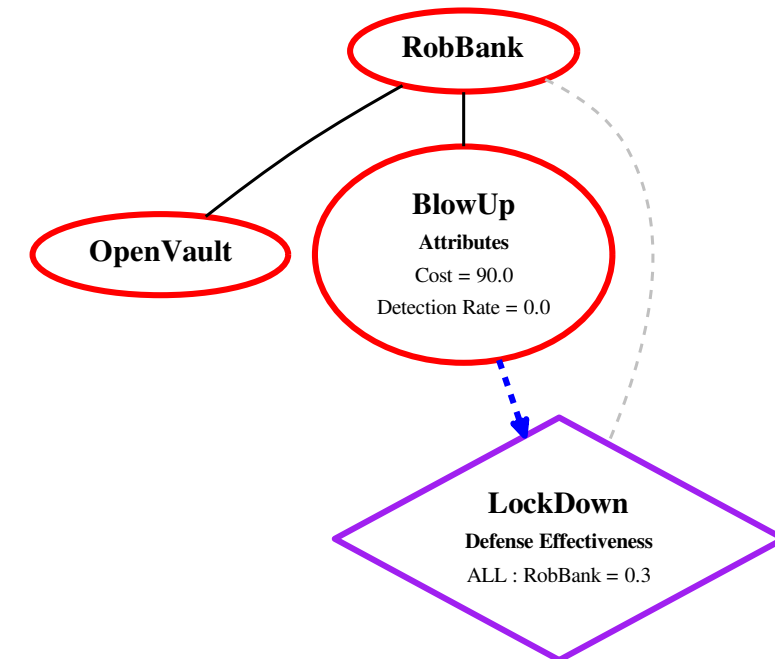
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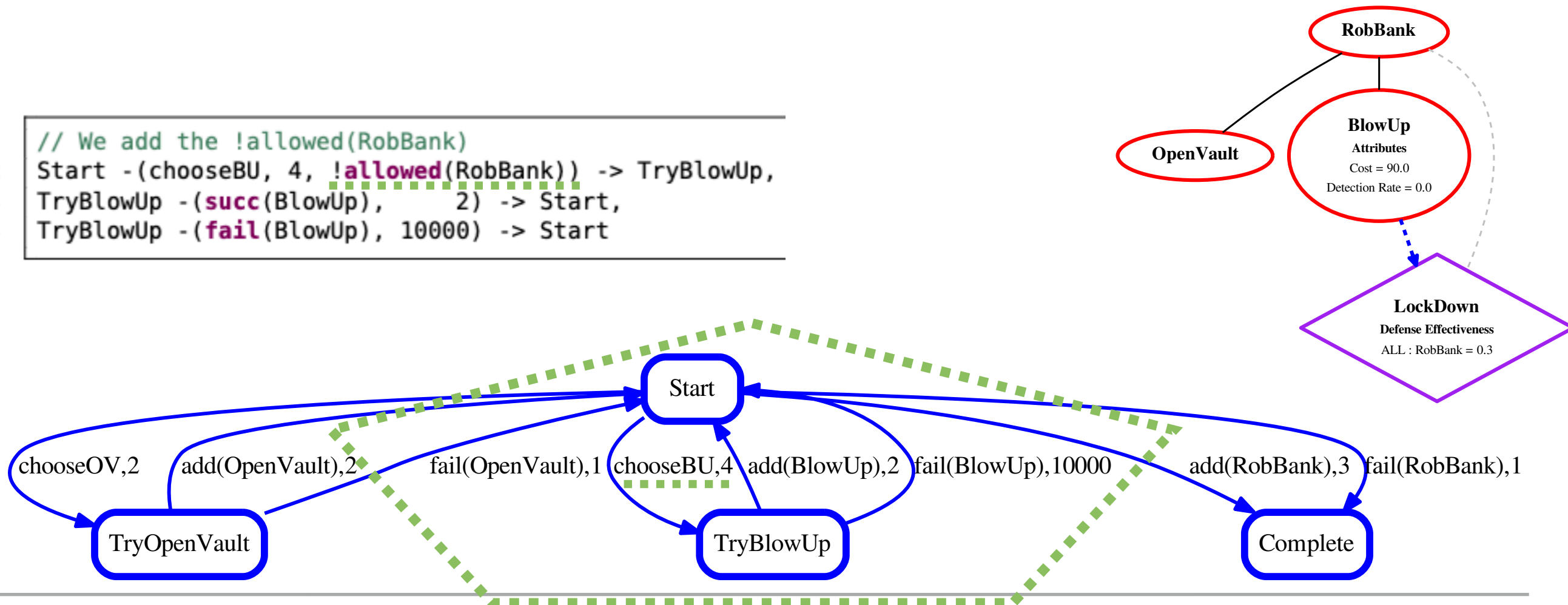
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FIRST REFINEMENT: PARSIMONIOUS ATTACKER

```
1 // We add the !allowed(RobBank)
2 Start -(chooseBU, 4, !allowed(RobBank)) -> TryBlowUp,
3 TryBlowUp -(succ(BlowUp), 2) -> Start,
4 TryBlowUp -(fail(BlowUp), 10000) -> Start
```



FIRST REFINEMENT: PARSIMONIOUS ATTACKER – ANALYSIS

Probability of successful bank robbery!? ~~0.17~~ 0.31

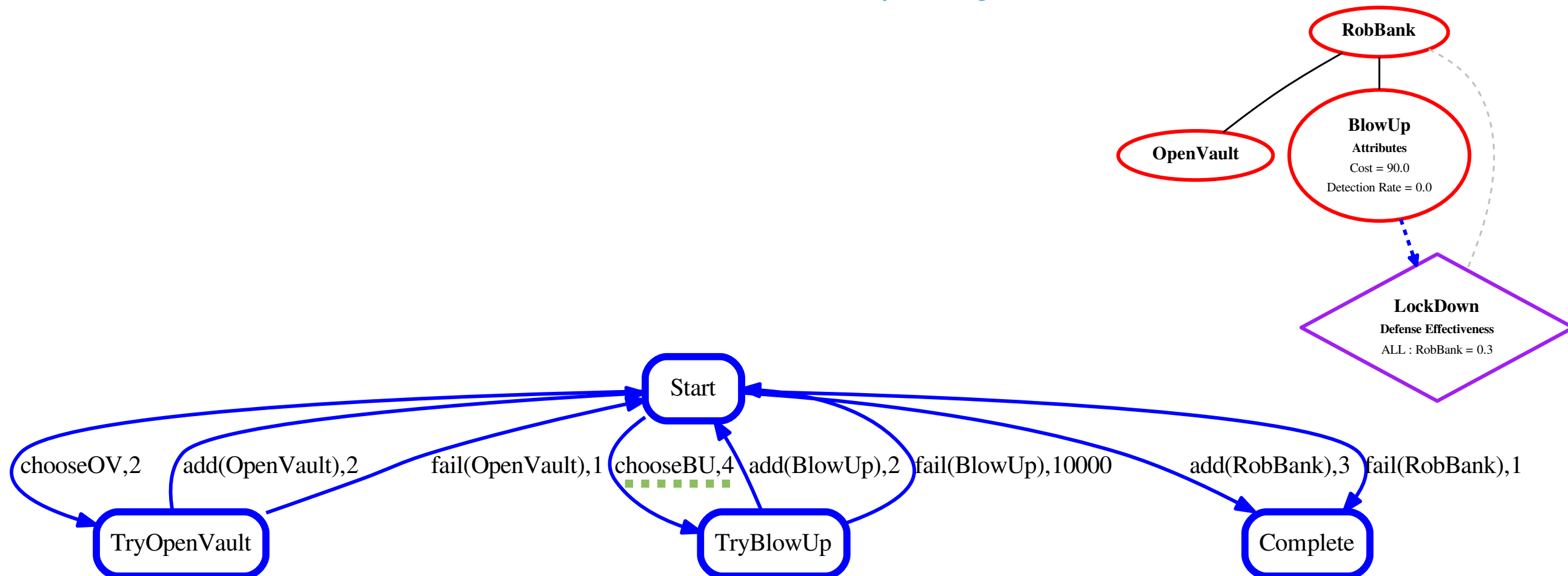
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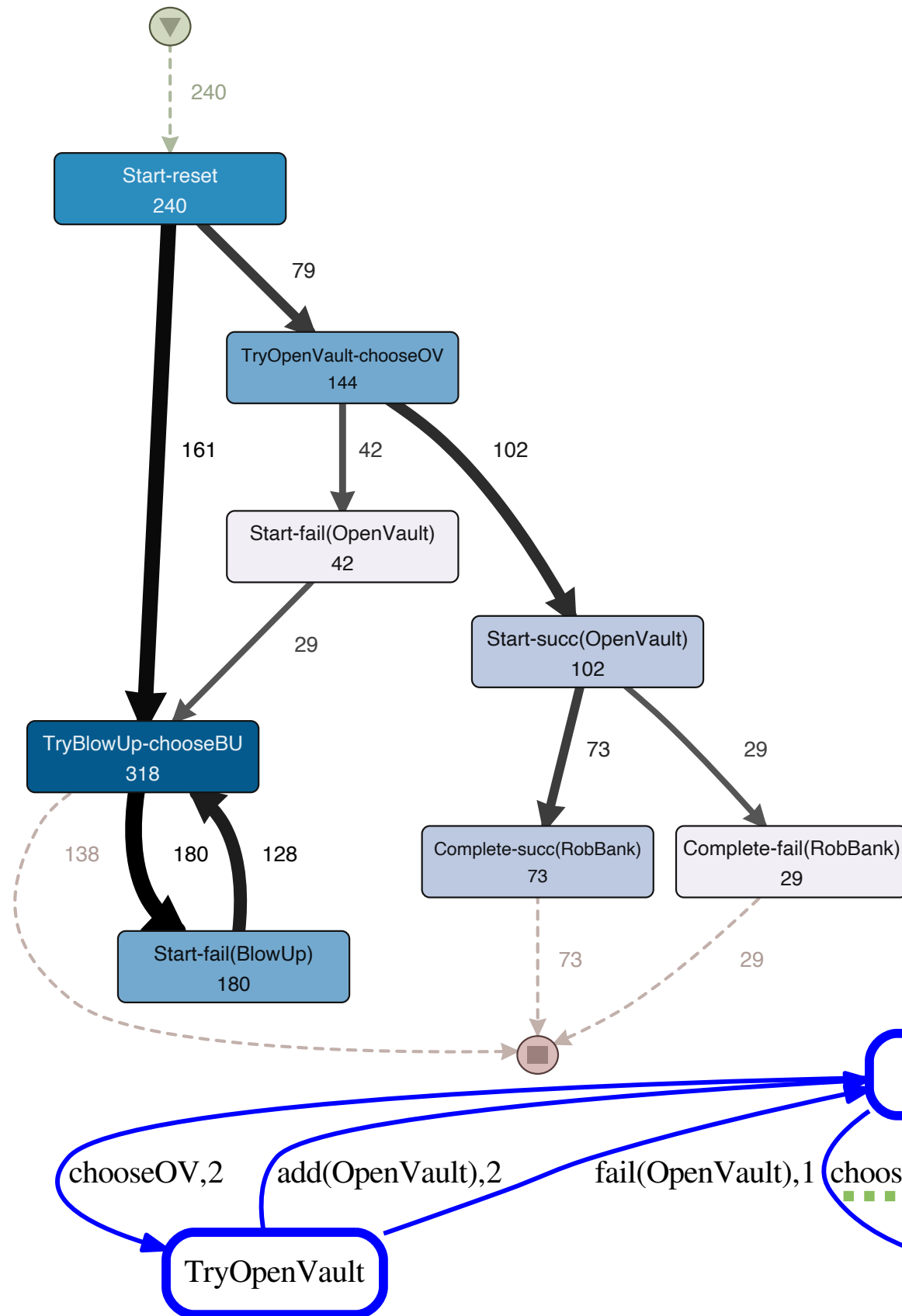
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FIRST REFINEMENT: PARSIMONIOUS ATTACKER – ANALYSIS



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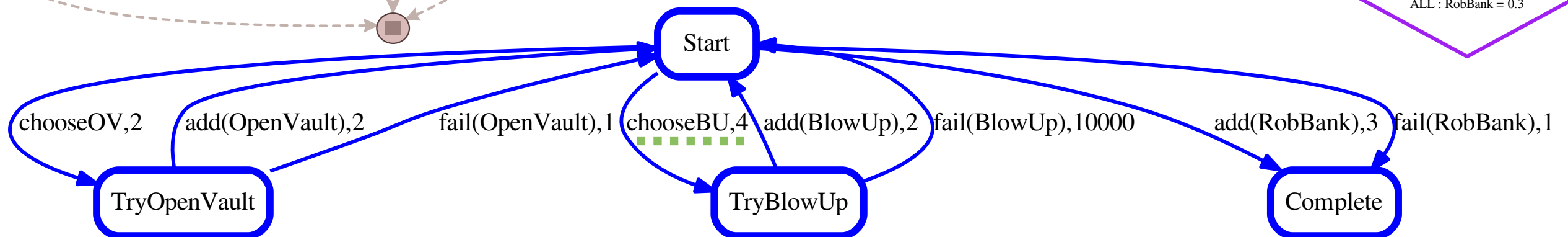
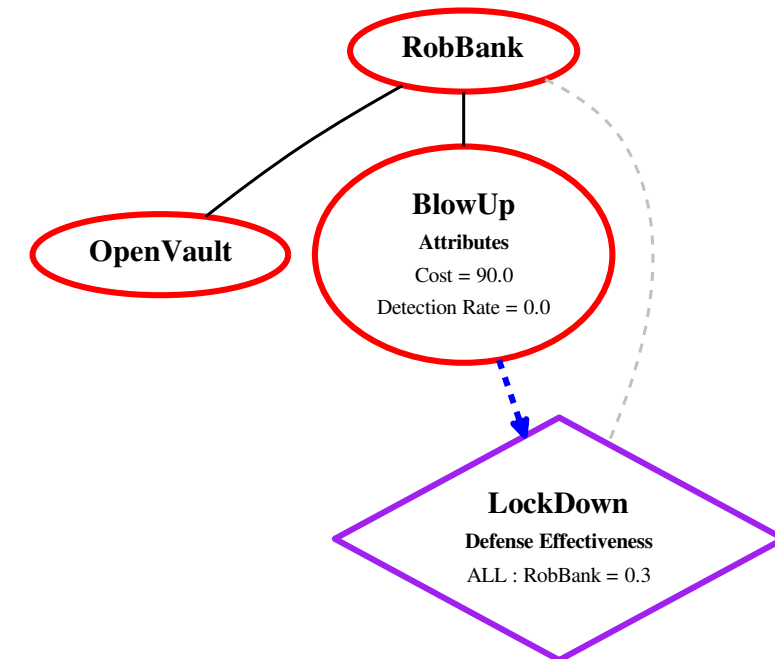
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Probability of successful bank robbery!? ~~0.17~~ **0.31**

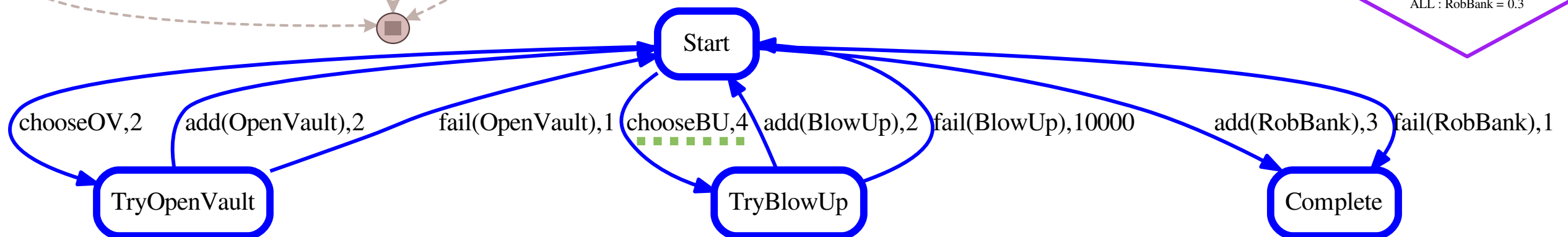
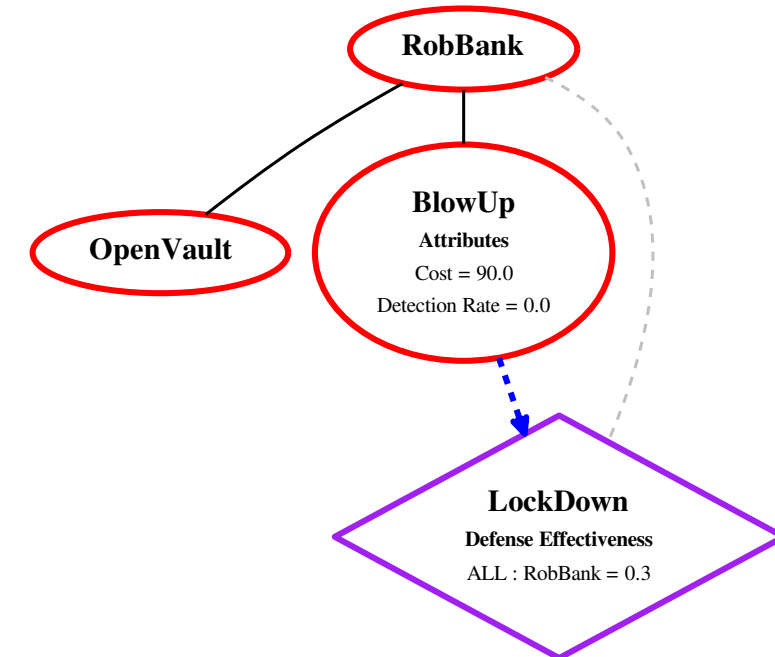
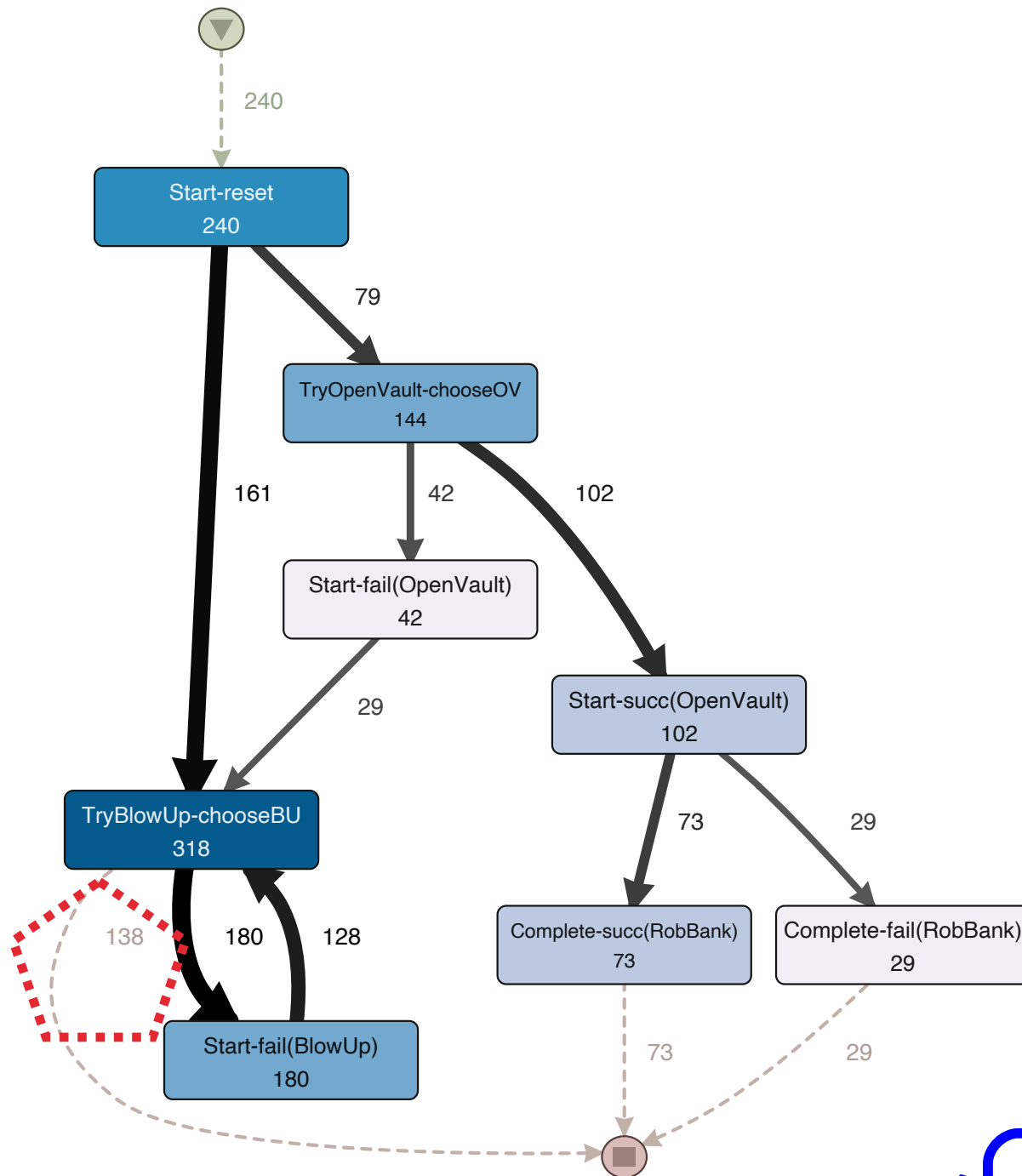
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FIRST REFINEMENT: PARSIMONIOUS ATTACKER – ANALYSIS

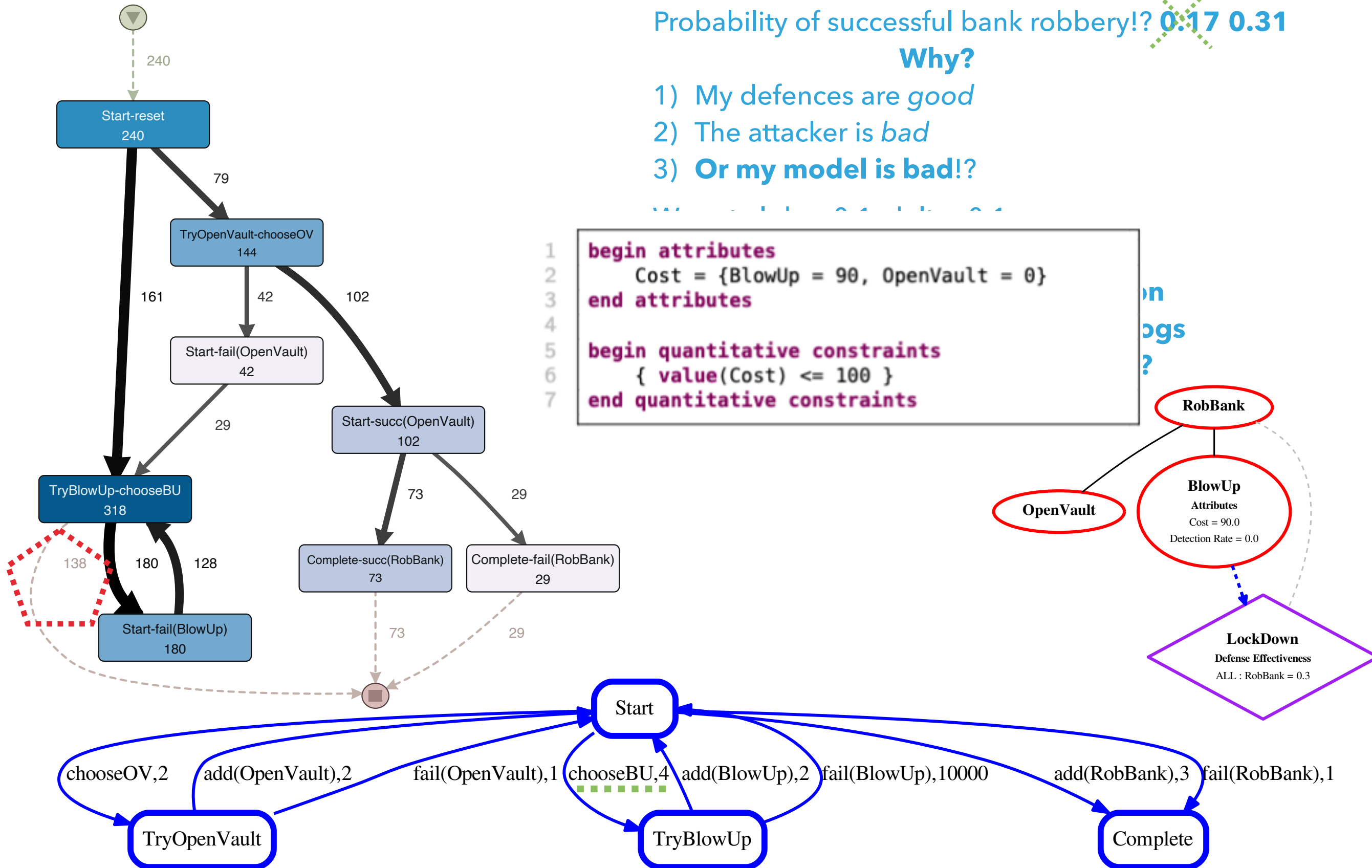
Probability of successful bank robbery!? ~~0.17~~ 0.31

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```

1 begin attributes
2   Cost = {BlowUp = 90, OpenVault = 0}
3 end attributes
4
5 begin quantitative constraints
6   { value(Cost) <= 100 }
7 end quantitative constraints
    
```

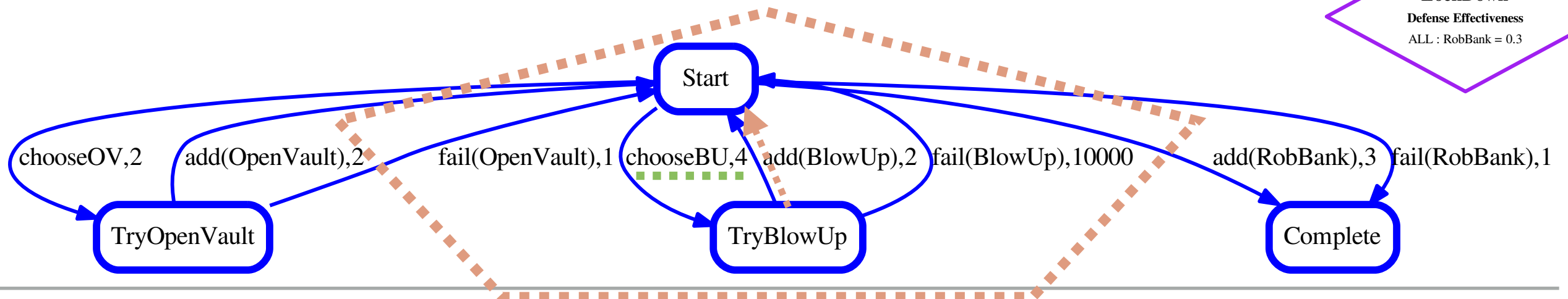
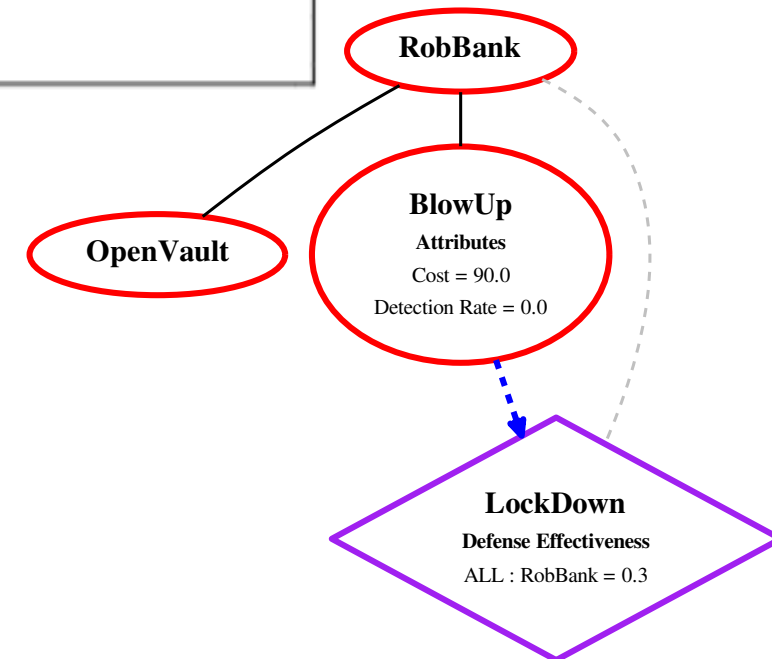


SECOND REFINEMENT

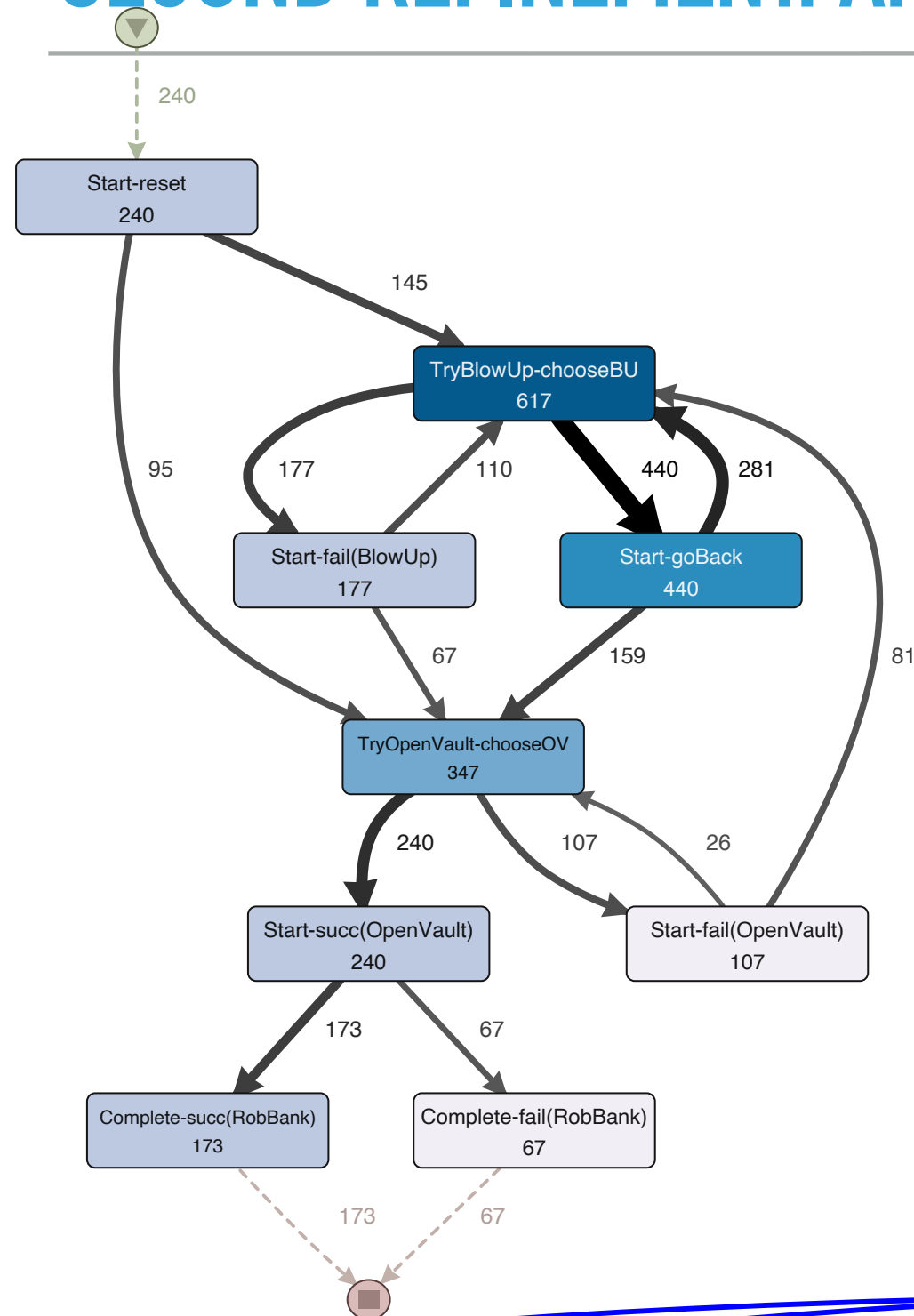
```
1 begin actions
2   chooseOV
3   chooseBU
4   goBack
5 end actions
```

```
1 begin attributes
2   Cost = {BlowUp = 90, OpenVault = 0}
3 end attributes
4
5 begin quantitative constraints
6   { value(Cost) <= 100 }
7 end quantitative constraints
```

```
1 // Strategy where the attacker tries to blow up the vault
2 Start -(chooseBU, 4, !allowed(RobBank)) -> TryBlowUp,
3 TryBlowUp -(succ(BlowUp), 2) -> Start,
4 TryBlowUp -(fail(BlowUp), 10000) -> Start,
5 TryBlowUp -(goBack, 0.00001) -> Start
```



SECOND REFINEMENT: ANALYSIS

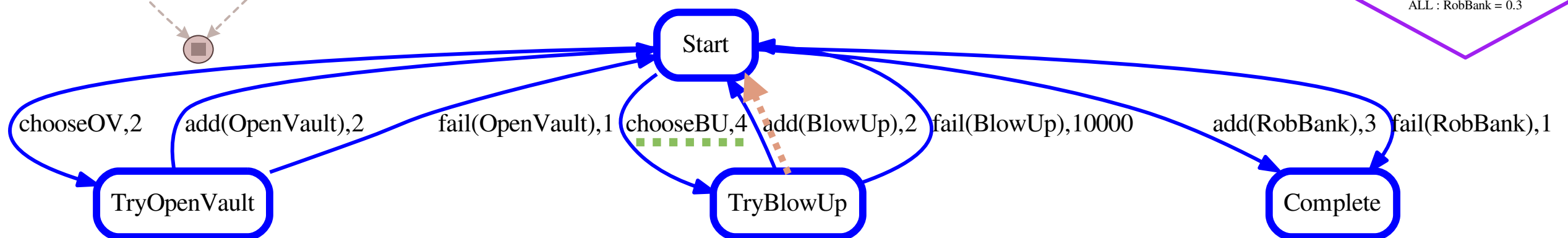
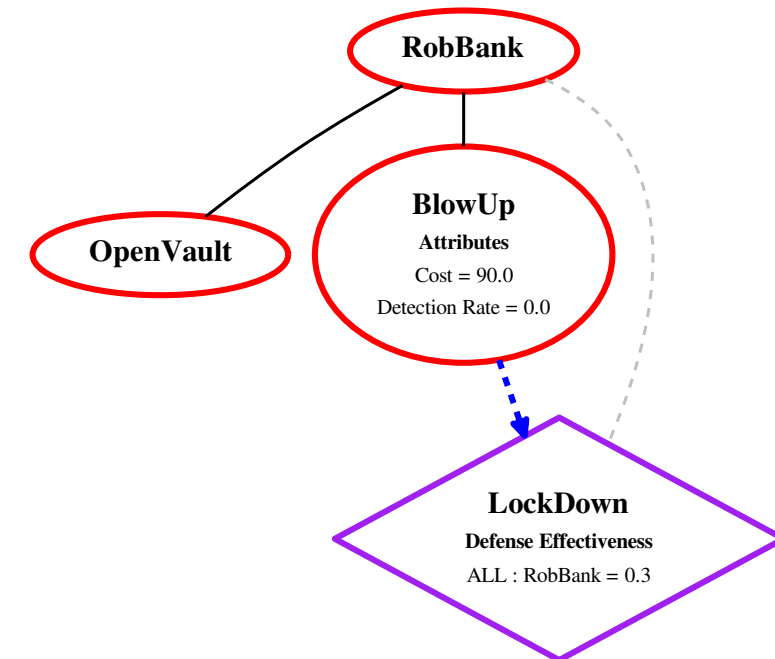


Probability of successful bank robbery!? ~~0.17~~ ~~0.31~~ **0.72**

We set $\alpha=0.1$, $\delta=0.1$

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PM MEETS SMC: CONCLUSIONS & FUTURE WORKS

- ▶ We proposed a novel methodology for validating and enhancing simulation models to make them more reliable
 - ▶ We obtained: **SMC- and PM-guided white-box behavioral model validation** and enhancement
- ▶ **Future works**
 - ▶ More realistic models, from more domains (e.g., ABM from social sciences)
 - ▶ Conformance checking might help our white-box analysis
 - ▶ Currently, we use **PM *after* SMC**:
 - ▶ **Using PM *during* SMC**: streaming PM might help improving SMC analysis
 - ▶ **Using PM *before* SMC**: discovery algorithms might be applied to real data to
 - ▶ synthesize attack-defense trees and/or attacker behaviors
 - ▶ or parts of simulation models in general

THANK YOU FOR
YOUR ATTENTION!

QUESTIONS?
FEEDBACK?

Tool, models, and draft available at:

<https://bit.ly/MultiVeStATool>

Andrea Vandin

andrea.vandin@santannapisa.it

www.santannapisa.it/en/andrea-vandin